

# **ADVANTEST<sup>®</sup>**

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## **ADVANTEST CORPORATION**

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*ATLworks*

*Tool User's Manual*

**MANUAL NUMBER 8311174-14**

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# ***Preface***

## **About This Document**

This document describes how to use ATLworks.

## **Related Manuals**

*ATL-51 Test Plan Program Reference Manual*(No. : 8431498)

*ATL-51 Pattern Program Reference Manual*(No. : 8256292)

*ASX/U-51 Hybrid Controller User's Manual*(No. : 8409598)

*T5500 Series Workbench-M2 Tool User's Manual*(No. : 8350412)

*ADVANsite Test System Software Installation/System Start User's Manual VOL2 Tester Environment Software Installation*(No. : 8227729)

*Online Manual Reference Manual*(No. : 8431523)

*T5500 Series Wave Tracer2 Reference Manual*(No. : 8262615)

*Online Manual (HTML) Reference Manual*(No. : 8700036)

# 1. *Outline and Basic Usage*

This chapter explains the outline and basic usage of ATLworks.

## 1. 1 **ATLworks Outline**

ATLworks is a software program provided for the T5500 series memory test systems to reduce the test program debugging time and enhance device evaluation efficiency.

ATLworks offers software environment which integrates diverse tester utilities with a graphical user interface (GUI) and allows you to effectively debug test plan programs and evaluate devices.

---

 **Tip**

- *If the color setting is not suitable for ATLworks, windows do not display correct colors except for the active window.*
-

## 1. 2 How to Start and Exit ATLworks

To use ATLworks, Solaris 2.5.1 released in November 1997 or later is necessary. ATLworks can be started using either of the following two methods.

- Starting from the ATLworks Startup window
- Starting from the Command Line

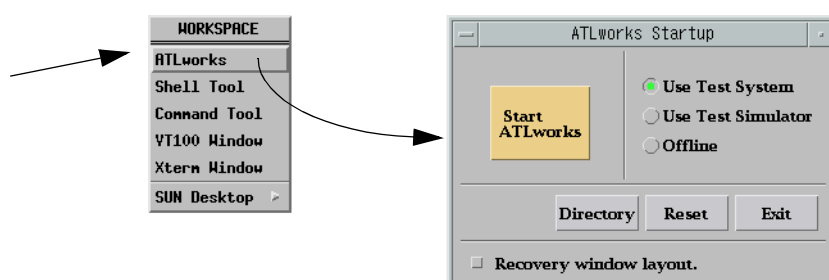
### 1. 2. 1 Starting ATLworks Startup

#### ◆ Solaris 2.5.1 + MWM (Motif Window Manager)

Place the mouse pointer on the root window, and press the left mouse button.

When the WORKSPACE menu is displayed, keep holding the left mouse button and move the mouse pointer to ATLworks, then release the button. The ATLworks Startup window is displayed shortly thereafter.

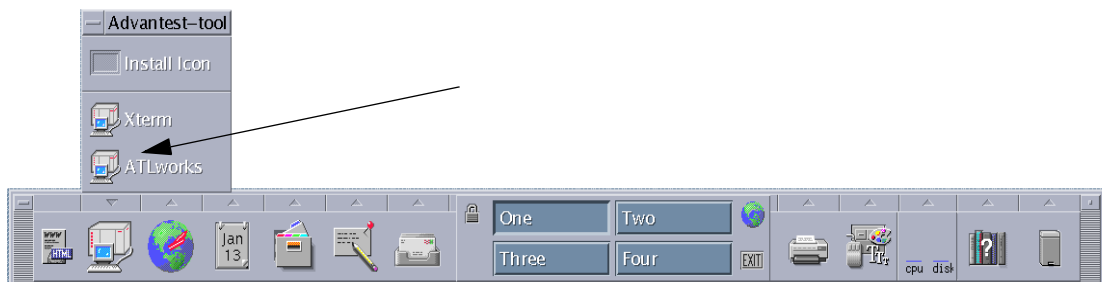
**Figure 1-1 Starting from the WORKSPACE Menu**



◆ Solaris 7 or Solaris 8 + CDE (Common Desktop Environment)

Click the left mouse button on the Advantest tool menu upper arrow in the front panel, and a sub-menu appears. Select ATLworks by clicking the left mouse button. The ATLworks Startup window is displayed shortly thereafter.

**Figure 1-2 Starting from the CDE Front Panel**



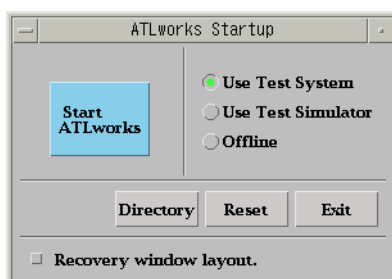
After the ATLworks installation, executing `atlworks_setup` adds the WORKSPACE and Advantest tool menus.

➔ For more information, refer to [ADVANsite Test System Software Installation/System Start User's Manual VOL2 Tester Environment Software Installation\(No. : 8227729\)](#).

### 1. 2. 1. 1 ATLworks Startup Window

This window allows you to start ATLworks, select an operation mode, specify the working directory, and stop ATLworks.

**Figure 1-3 ATLworks Startup Window**



#### Selecting the start mode

- Use Test System toggle button  
Selecting this button makes the test system online.  
(This button is disabled unless SYSTEST51SU is installed.)
- Use Test Simulator toggle button  
Selecting this button uses ATLworks with the tester simulator (TESIM). (This button is disabled unless TESIM51SU is installed.)
- Offline toggle button  
Selecting this button makes the test system offline.

#### Start ATLworks button

Selecting this button clears the ATLworks Startup window, and starts ATLworks in the mode specified in ["Selecting the start mode" on page 1-4](#).

If you exit ATLworks, the ATLworks Startup window is redisplayed.

#### Directory button

Selecting this button displays the Change Directory window. This window allows you to specify the directory where ATLworks operates.

#### Reset button

Selecting this button stops all ATLworks currently operating in the same account.

#### Exit button

Selecting this button exits the ATLworks Startup window.

#### Recovery window layout button

When this button is selected, ATLworks is started with the saved window layout.

To save the window layout, select Save & Quit in Section [1. 2. 5 "Closing ATLworks" on page 1-12](#).

Note that the Recovery window layout toggle button is disabled when the Offline toggle button is selected in the ATLworks Startup window.

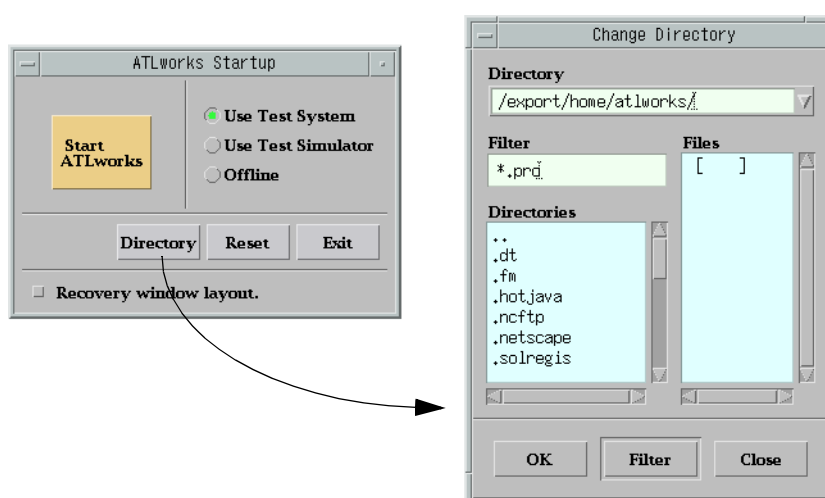


## 1. 2. 1. 2 Change Directory Window

Click the Directory button in the ATLworks Startup window, then the Change Directory window shown below appears.

This window allows you to change the working directory where ATLworks operates.

**Figure 1-4 Change Directory Window**



### OK button

Specifies a different directory and closes the Change Directory window.

### Filter button

Opens the directory selected from the Directories list box and displays files in the Files list box.

### Close button

Closes the Change Directory window.

### Tip

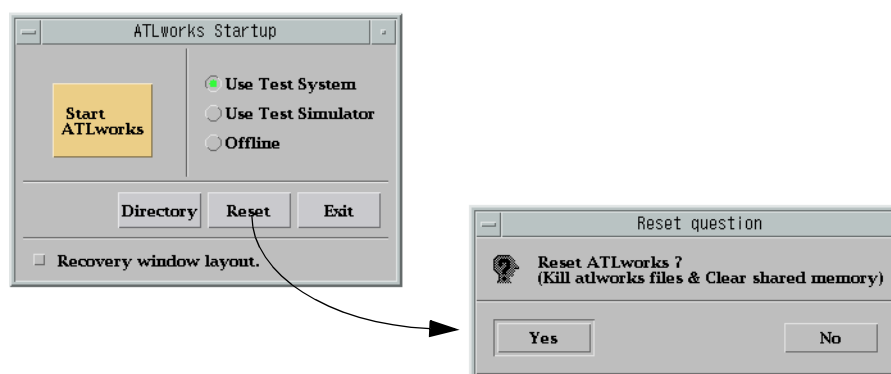
*The most recent directory setting is saved in the .startaw.dir file located below the current directory. Next time, ATLworks will start with the same directory setting.*

### 1. 2. 1. 3 Reset Question Window

Clicking the Reset button on the ATLworks Startup window displays the Reset Question window shown below.

This window prompts you to confirm if you want to reset ATLworks.

**Figure 1-5 Reset Question Window**



**Yes button**

Resets ATLworks and closes the Reset Question window.

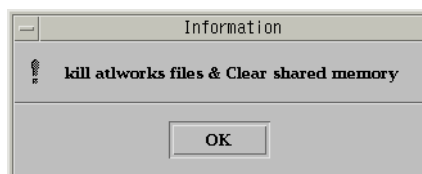
**No button**

Cancels the operation and closes the Reset Question window.

### 1. 2. 1. 4 Information Window

The Information window displays messages about the reset ATLworks.

**Figure 1-6 Information Window**



**OK button**

Closes the Information window.

## 1. 2. 2 Starting from the Command Line

To start the ATLworks Startup window from the command line, execute the command shown below.

```
% startATLworks_↓
```

To start ATLworks from the command line, move to the ATLworks working directory and enter the command shown below.

```
% atlworks_↓
```

The above command is used for ATLworks to operate with an online test system. To use ATLworks with an offline test system, execute the command shown below.

```
% atlworks -n_↓
```

To use ATLworks with the tester simulator (TESIM), execute the command shown below.

```
% atlworks -tesim_↓
```

Figure 1-7 Online State

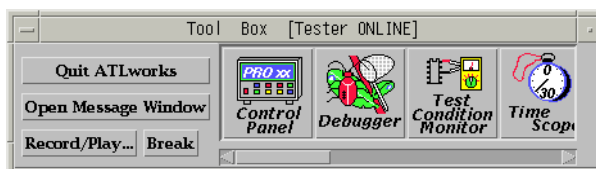


Figure 1-8 Offline State

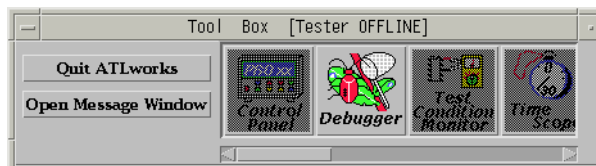
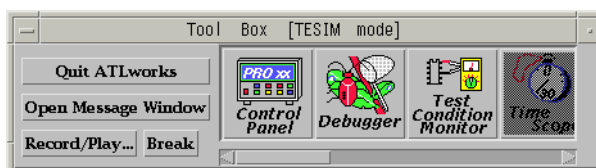


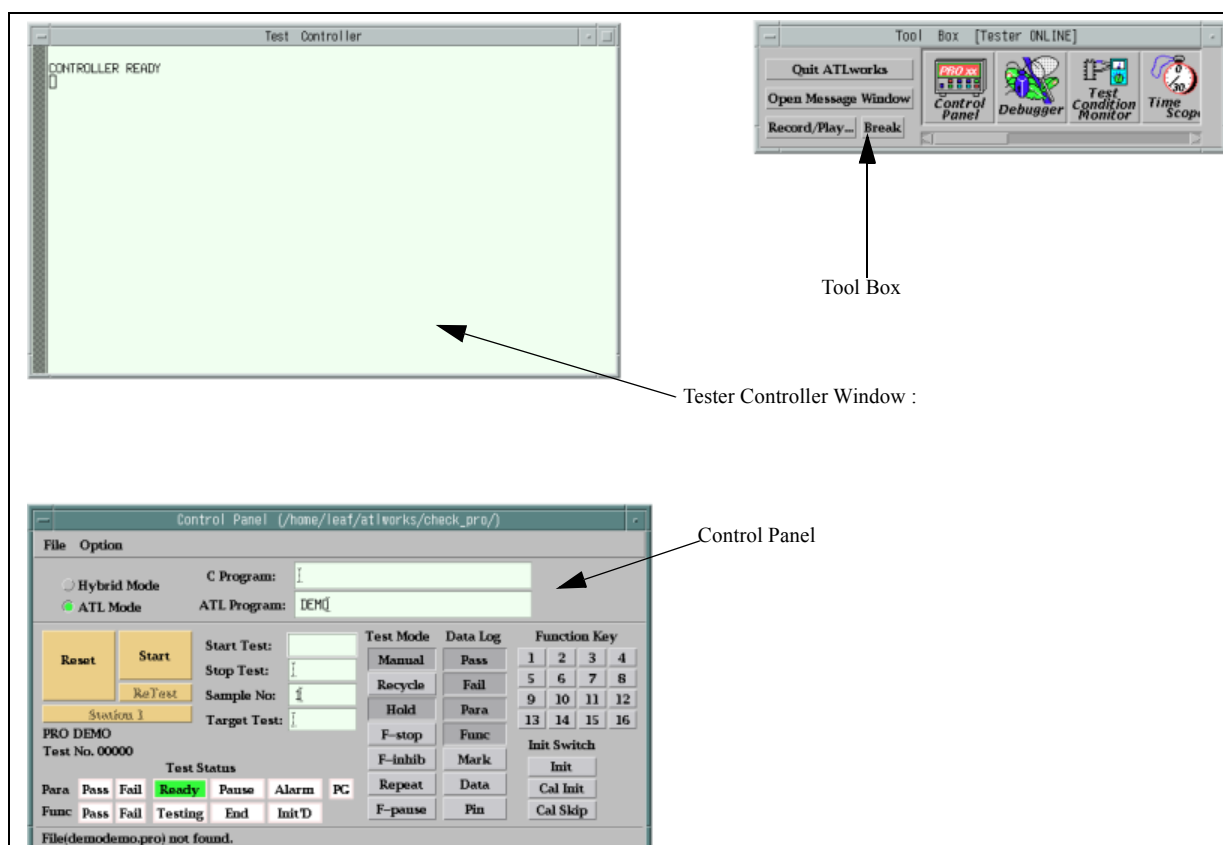
Figure 1-9 TESIM Mode



## 1. 2. 3 Initial Screen

If ATLworks is started with an online test system or the tester simulator (TESIM), the initial screen shown in [Figure 1-10](#) is displayed.

**Figure 1-10 Initial Screen**



### Tool Box

This is used to start various graphical tools and stop ATLworks operations.

### Test Controller window

This window displays the conventional MOST environment and can be used to execute key commands, character-based utilities, etc.

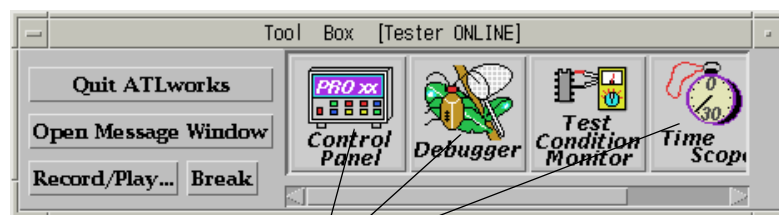
### Control Panel

This is used to execute the function of the control box provided for each test station. You can set the program name to start or reset the program.

## 1. 2. 4 Activating Various Tools

Clicking a tool button in the Tool Box activates the related tool.

**Figure 1-11 Activating Tools**



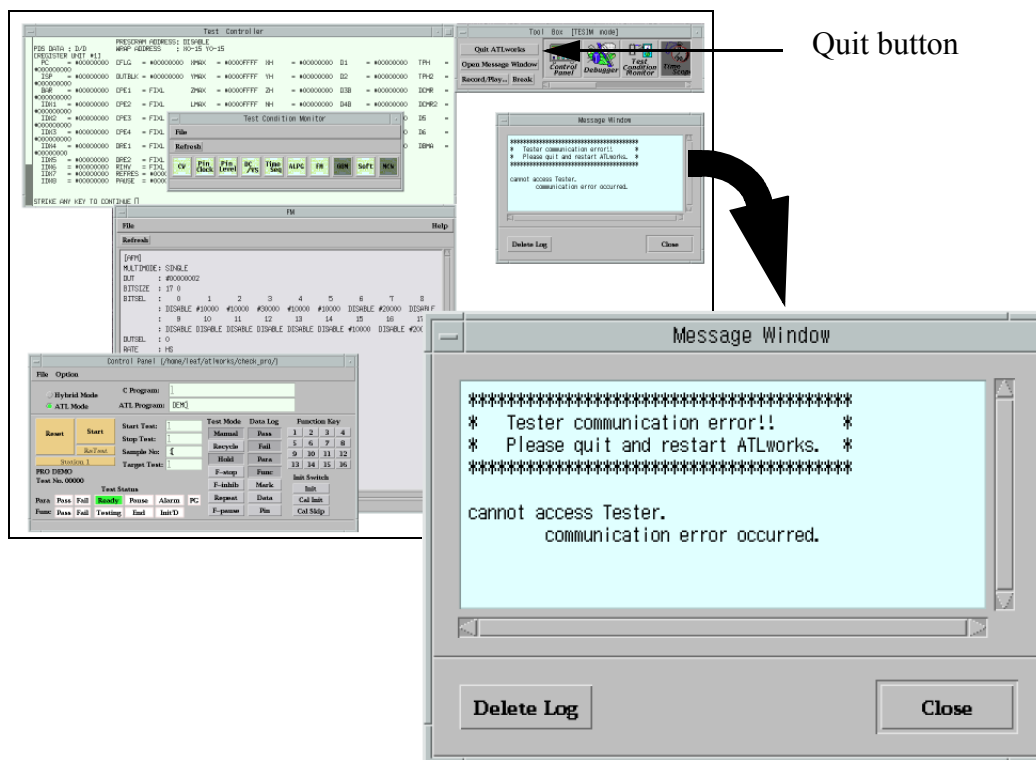
Tool Buttons

### Tip

- Tool buttons displayed in gray indicate that the related tools are not installed or cannot be used for the specified operating condition (off-line or TESIM).
- When C or C++ programs are running on the hybrid controller, ATLworks tools cannot be used (except for the help window).

## 1. 2 How to Start and Exit ATLworks

- While other utilities (/EPM etc.) or statements (such as SEND DBM, or SEND FM) which require long data transfer time are running on the test controller window, attempting to start ATLworks displays the message window shown below. When this message is displayed, ATLworks tools becomes disabled. Click the Quit button in the tool box to terminate ATLworks.

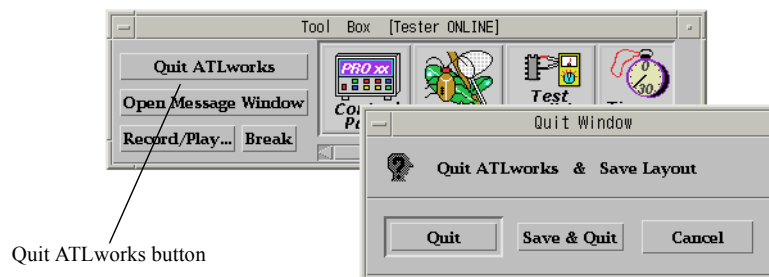


## 1. 2. 5 Closing ATLworks

Clicking the Quit ATLworks button in the Tool Box closes ATLworks.

Then, the Quit Window appears to prompt you to confirm if you want to close ATLworks.

**Figure 1-12 Closing ATLworks**



### ◆ Button Description

#### Quit button

Closes ATLworks.

#### Save & Quit button

Saves the window layout for the tools currently being used and exits ATLworks.

➔ For more information, refer to [1. 8 "Saving and Recovering the Window Layout" on page 1-28.](#)

#### Cancel button

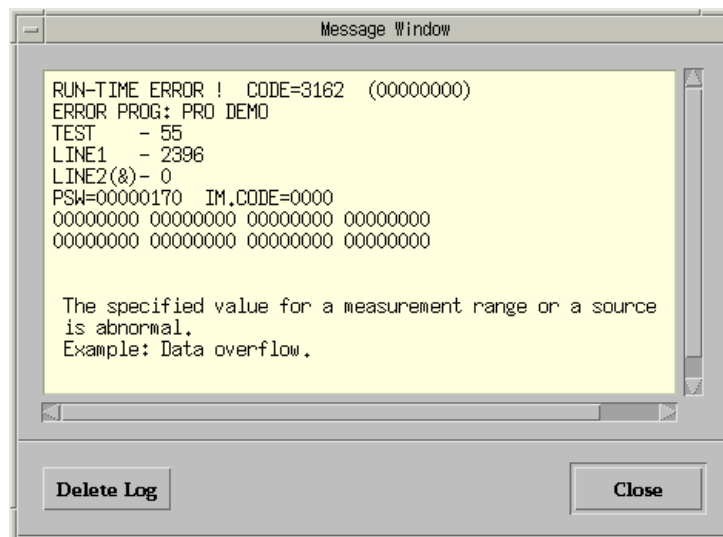
Keeps the current operation and closes the Quit window.

## 1. 3 Message Window

The message window can display a variety of messages.

The message window opens automatically if an error occurs.

**Figure 1-13 Message Window**



**Delete Log button**

Deletes the log displayed on the Message Window.

**Close button**

Closes the Message Window.



## 1. 4 Communication Error

If an error occurs during the ATLworks online operation, the message shown in [Figure 1-14](#) is sometimes displayed in the message window shown in [Figure 1-13](#). If this message is displayed, the online tester functions become disabled. Close and restart ATLworks.

**Figure 1-14 Error Message**

```
*****
*      Tester Communication error!!      *
*      Please quit and restart ATLworks.  *
*****
```

## 1. 5 Basic ATLworks Operation

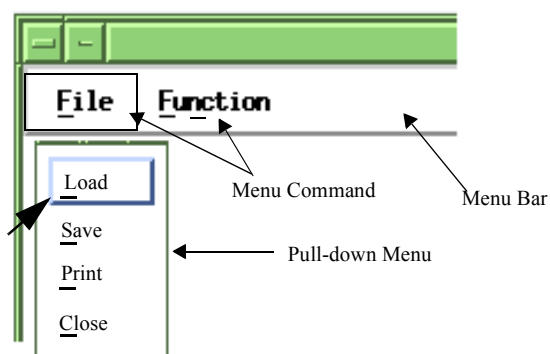
### 1. 5. 1 Menu Bar And Pull-Down Menu

The menu bar is used to operate various ATLworks functions.

Each menu command in the menu bar provides a pull-down menu which lists more detailed functions or commands.

To use the menu, move the mouse cursor onto a desired menu command, and press the mouse left button. A pull-down menu appears under the menu command button. While holding down the mouse button, move the mouse cursor to the desired menu and then release the button.

**Figure 1-15 Menu Bar and Pull-Down Menu Examples**



Place the cursor on File and click the mouse left button to display the menu.

Hold down the mouse button, move the cursor downward, and select a menu item.



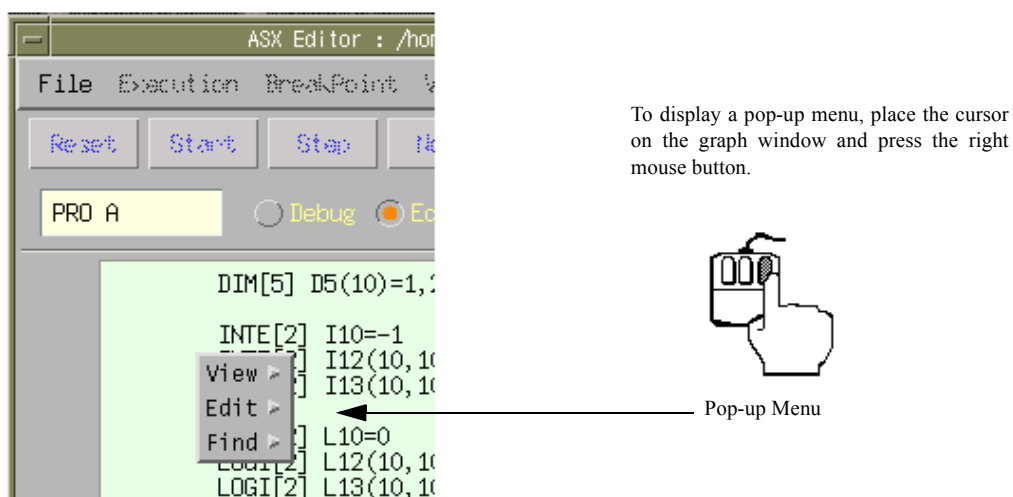
## 1. 5. 2 Pop-Up Menu

A pop-up menu appears when you press the right mouse button on the graph window. To select a command from the pop-up menu, press and hold the left mouse button, move the cursor to a desired command, then release the button.

Items in the pop-up menu are grayed out if the items are unavailable. Grayed-out items cannot be selected.

The menu items are automatically enabled depending on the current setting.

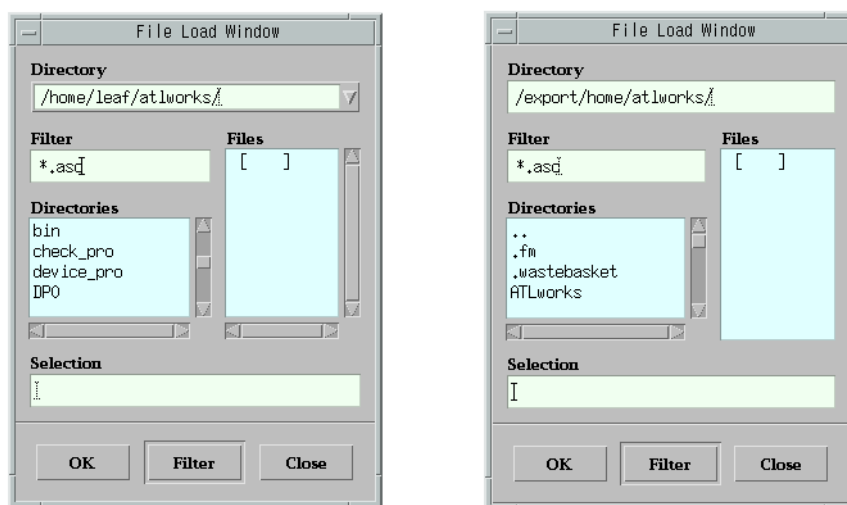
**Figure 1-16 Pop-Up Menu Example**



### 1. 5. 3 File Selection Dialog Box

The Directory field in the File Selection dialog box is different, depending on the operating system, Solaris 7, Solaris 8, or Solaris 2.5.1.

**Figure 1-17 File Selection Dialog Box**



a) Using Solaris 7 or Solaris 8

b) Using Solaris 2.5.1

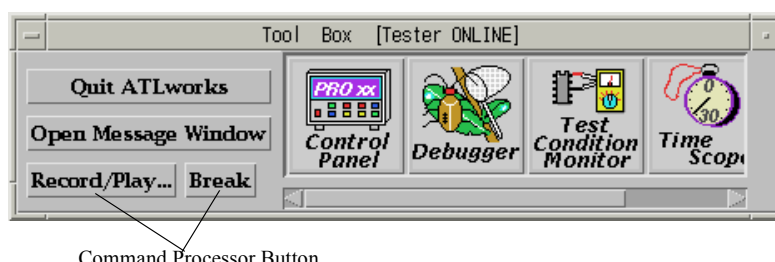
When Solaris 7 or Solaris 8 is used, an arrow button is displayed on the right of the Directory field. This enables you to quickly access to the home directory, etc. Both File Selection dialog boxes have the same function regardless of the arrow button. Therefore, this manual uses the File Selection dialog box without the arrow button to describe ATLworks.

## 1. 6 Executing Command Processor

Command Processor uses a file to execute commands.

➔ For more information, refer to [1. 7 "Command Processor" on page 1-19](#).

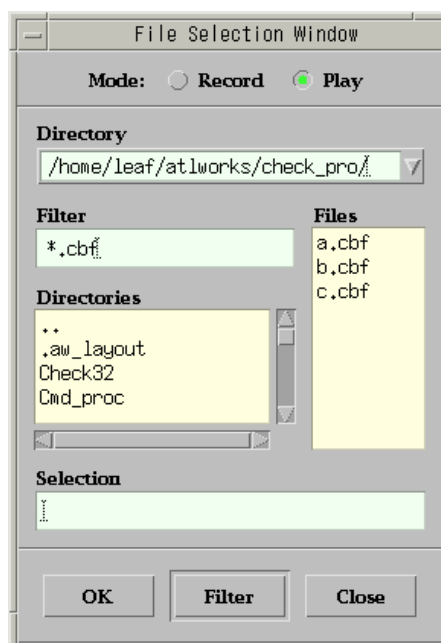
**Figure 1-18 Command Processor Button**



**Record/Play...**

Clicking this button displays the window shown below and allows you to select a file to record command selections or to execute the recorded commands. When a file is selected, selected commands are recorded in the file or the previously recorded commands are executed.

**Figure 1-19 File Selection Window**



**Break**

Clicking this button stops executing recorded commands.

## 1. 7 Command Processor

### 1. 7. 1 Overview

Command Processor is a tool used to execute commands from a file.

ASCII format commands can be saved in a command buffer file and executed in batches.

Command Processor is aimed at building an integrated tester application environment.

Application programs are independent of each other and also of Command Processor. These programs run independently.

Command Processor is a master process to activate other applications. It resembles a server/client model for X servers. Each application can automatically log in to Command Processor and registers its name, commands identified by Command Processor, the type and number of command parameters.

### 1. 7. 2 Functional Description

#### General functions

Command processor is a master process that interprets the information entered to applications. The information is the mouse entry or the ASCII text entered from the command buffer file (in batch processing mode).

Command Processor directs applications to go into record mode and records data from various sources. In record mode, all correct operations you execute on applications are recorded as the corresponding command statements.

#### Startup

Command Processor is automatically activated when ATLworks is started. All applications linked to Command Processor are also started with Command Processor.

When Command Processor attempts to communicate with an application, which has not been activated yet or has already ended, Command Processor sends a message to the tool manager and starts the application for communication. When Control Panel has not been activated yet, entering the CPNL Start command causes Command Processor to send an appropriate message to Tool Manager and to activate Control Panel. When the application is activated, commands are sent to the application and automatically processed.

When the application is terminated correctly or abnormally, it is detached from Command Processor. No matter how the application is terminated, the application can be linked to Command Processor again. This link process is performed when the application is restarted.

#### Record session

In the record session, the ATLworks operation is recorded. The mouse operation and commands included in the command buffer file can be recorded. A single command operation or the entire operation can be recorded. In record session, Command Processor checks all commands based on the registered list. Syntactically correct commands are saved as a command buffer script.

### Play session

The play operation is to execute commands recorded in the command buffer. ATLworks loads the command buffer file data and operates in batch mode until all commands in the command buffer file are executed or the Break button is clicked. If an error is detected, an error message is displayed and allows you to quickly identify the error and modify the command setting. If the record session is used to create the command buffer file, no error message is displayed.

Most ATLworks widgets are set to the insensitive mode during the play session. While the command buffer file is being used, accessing to ATLworks applications is not allowed. However, clicking the Break button can stop the play session and allows you to access applications.

### Command type

Commands whose names are basically created using the abbreviation of application names, require the keyword and arguments. Keywords are used to specify a correct command path so that the command can be executed through the graphical user interface. The keywords represent a label, cascade or button.

For example, the command, CPNL Data Log Pass, takes the same action as the Data Log Pass button of the Control Panel application. (For more information on the command language syntax, refer to each application.)

### Case sensitivity

Command Processor is not case-sensitive. Both uppercase and lowercase characters can be used to enter all application names and commands with a keyword.

### Character string variable

Character string variables are used to specify a file name, etc. Character string variables are case-sensitive and can be expressed using up to 70 characters.

For example, CPNL Station Program <string>, etc.

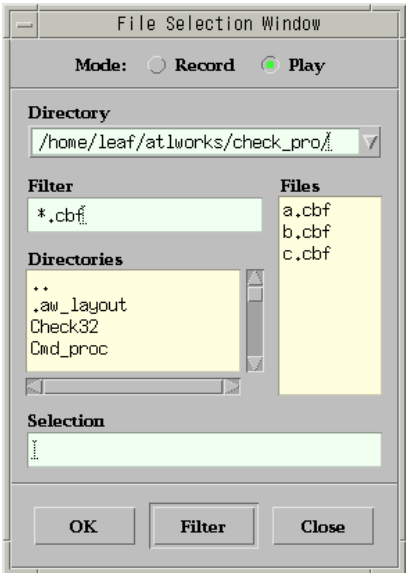
Character string variables have to be enclosed by double quotes. If a character string variable contains more than one space, Command Processor treats the words delimited by spaces as individual elements, which are arguments, in syntax analysis.

## 1. 7. 3 Operating Instructions

### Record/Play button

Clicking the Record/Play button displays the File Selection dialog box. This dialog box allows you to scroll through the directory list and select a file. When a directory is selected, command buffer file names with the .cbf extension are listed.

Figure 1-20 File Selection Window



- Record session
 

When Record is selected from the Mode menu and a file name is entered in the Selection text box, the record session starts.

Then, the Record Session dialog box is displayed to notify you that the record session starts. The Record Session dialog box does not disappear until the record session is stopped or completed. In the record session, all correct operations are recorded.

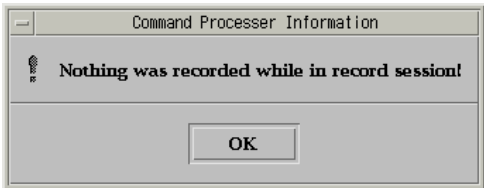
To stop the record session, click the Stop button in the dialog box shown in [Figure 1-21](#).

Figure 1-21 Record Session Dialog Box



When the record session is stopped without recording commands, the message is displayed in the Command Processor Information window as shown below.

Figure 1-22 Record Information Message





- Play session

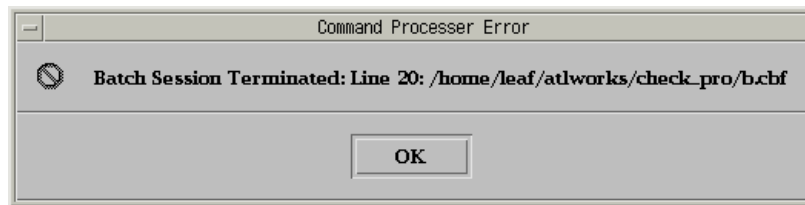
When Play is selected from the Mode menu of the dialogue box shown in [Figure 1-20](#) and a command buffer file is selected, Command Processor is set to the batch mode in the play session. When an error occurs, the play session stops. Errors are categorized as syntax errors or operation errors.

#### Break button

Clicking the Break button causes the play session to stop. The Break button can also be used to stop CPNL Start and other command operations.

While the batch file is being processed, clicking the Break button displays a message dialog box that shows the batch file name, location and command line number where the batch process was stopped.

**Figure 1-23 Batch Session Break Message**



## 1. 7. 4 Command Buffer File

A command buffer file is an ASCII text file that is created through the Command Processor record session or using the offline text editor.

### File name rules

The ATLworks command buffer file name extension is .cbf.

### Backup file

Command Processor saves the command buffer file as a backup file before starting the record session. The backup file is created only when an already-existing file is used for the record session. The backup file is left in the system.

A backup file name consists of the selected file name used for record and ~. For example, the backup file name of Dev8086.cbf is Dev8086.cbf~.

### File Nesting

To nest a command buffer file, use the CPROC include command or add the CPROC Run <file-name> command to the command buffer file. There is no restriction on the nesting depth. The nesting depth is determined by the workstation type, the memory size and the command buffer file size. However, the best way is to make the nesting depth smaller. A large nesting depth may cause a stack overflow.

### Comment

/Characters between /\* and \*/ are processed as a comment, and ignored. These characters can freely be used to make it easier to read the batch file. Comments can be put anywhere, however, the comments cannot be nested.

### Header information

A command buffer file consists of a header and command lines. The header includes the following information.

- Name  
Command buffer file name
- Created By  
File owner name. It is also a login name.
- Date  
Date when the file was created is displayed.

## 1. 7. 5 Command Buffer File Example

This section shows a command buffer file example.

```
/**
**
**Name:      /home/usr/kevin/integration.cbf
**
** Created By: Kevin H. Perot
**
** Date:      31-Aug-1992 4:47 PM
**
**/

/* Setup for TestMode */
CPNL TestMode Manual Enable
CPNL TestMode Recycle Enable
CPNL TestMode Hold Enable

/* Setup for DataLog */
CPNL DataLog Pass Enable
CPNL DataLog Fail Enable
CPNL DataLog Parametric Enable
CPNL DataLog Functional Enable

/* Setup for Function Key */
CPNL Function_keys 1 Enable
CPNL Function_keys 2 Enable
CPNL Function_keys 3 Enable
CPNL Function_keys 4 Disable

CPNL Station Test_No 100

CPNL Start
```

## 1. 7. 6 Error Messages

Error messages are displayed in the Command Processor Error dialog box.

Command Processor displays two types of error message: system error messages, which the UNIX system displays to show that the library function failed to start, and general error messages displayed by the command processor.

CPRC: Failed to initiate communication channel

CPRC: Failed to receive recorded file from <Application>

CPRC: Failed to broadcast to <Application>

CPRC: Cannot connect to additional applications

CPRC: Failed to communicate to the <Application>

The above error messages are displayed when a communication error occurs between Command Processor and applications.

Communication channel to the <Application> is lost

The above error message is displayed when an internal error occurs in some applications.

Bad Vector Table Expected function name: <Vector Syntax>

Invalid vector table

Unexpected command is received from <Application>

The above error messages are displayed when something is wrong with the vector table.

Error: Failed to send command to the <Application>

The above error message is displayed when an application is not ready to receive the command.

Expecting a file name

Failed to open "<File-Name>"

Cannot read "<File-Name>"

Cannot write to "<File-Name>"

Invalid file type <File-Name>

Improper file permission <File-Name>

Failed to open <File-Name> to record

Error: "/" is a read only file system.

Error: <Path> is a directory.

Error: Invalid directory <dir-name>.

Error: Failed to change directory path.

The above error messages are displayed when Command Processor cannot open the command buffer file or when the specified path is invalid.

Error: Provide a directory name.

The above error message is displayed when the directory name is not specified.

Error: Mismatch at or near: "<Keyword>"

The above error message is displayed when the syntax is incorrect.

Error: Ambiguous Keyword: "<Keyword>"

The above error message is displayed when the keyword is ambiguous. For example, st is an ambiguous keyword because st is applicable to Stop and Start.

Error: Incomplete command, expects one or more Keywords.

The above error message is displayed when the command is incorrect.

Expected Keyword(s): <Followed by a list of the Keywords>

When an error message including "Mismatch," "Ambiguous" or "Incomplete" is displayed, clicking the help button displays the above error message.

Error: Decimal number <number> is too large

The above error message is displayed when a too large integer was entered.

Invalid time period <value>

The above error message is displayed when an invalid value was entered with the wait command.

Record session <File-Name> is already enabled

The above error messages are displayed when the start symbol "/" or the stop symbol "\*" is not written in the command buffer file.

Record session <File-Name> is already enabled

The above error message is displayed when you attempt to start the record session that has already been started.

Record Session is already disabled!

The above error message is displayed when you attempt to stop the record session that has already been stopped.

Incorrect file nesting "<File-Name>" inside <File-Name>

The above error message is displayed when the command buffer file is used to access another command buffer file that has already been used.

Batch Session Terminated: Line <Line>: <File-Name>

The above message is displayed when the batch session is terminated.

<Function-Name> Failed to allocate memory

The above error message is displayed when sufficient memory space is not allocated to the system.

## 1. 7. 7 Restrictions

- Only the Control Panel tool can use Command Processor with ATLworks.  
However, if an optional tool is installed, tools such as Wave Tracer2 or Data Scan can also be used.  
➔ For more information, refer to [T5500 Series Workbench-M2 Tool User's Manual\(No. : 8350412\)](#) or [T5500 Series Wave Tracer2 Reference Manual\(No. : 8262615\)](#).
- If ASXDebugger has already been started, the dialog box shown below is displayed and Command Processor cannot start the play session. Terminate ASXDebugger and restart it.

**Figure 1-24 Command Processor Information**



- When Command Processor executes the play session, TCM and TimeScope tools are automatically closed.

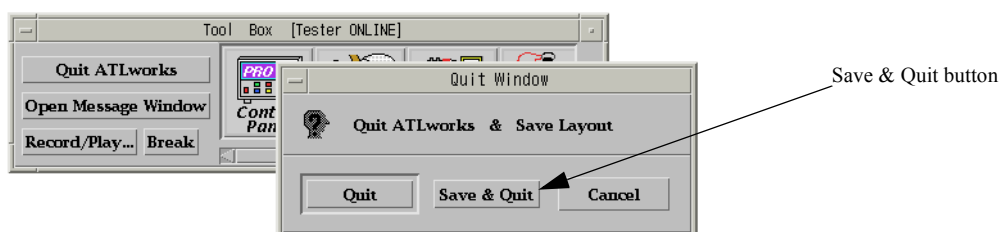
## 1. 8 Saving and Recovering the Window Layout

### 1. 8. 1 Saving the Window Layout

To save the window layout, click the Quit ATLworks button in the Tool Box. The Quit window appears. Clicking the Save & Quit button in this window saves all active tool layout information and terminates ATLworks.

If ATLworks has been started without a test system connected, the Save & Quit button is disabled.

**Figure 1-25 Saving the Window Layout**

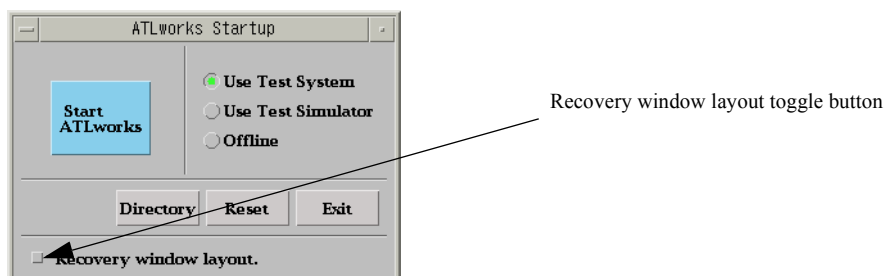


### 1. 8. 2 Recovering the Window Layout

When you want to recover the previously saved window layout, select the Recovery window layout toggle button in the ATLworks Startup window and click the Start ATLworks button.

Note that the Recovery window layout toggle button is disabled when the Offline toggle button is selected in the ATLworks Startup window. (The previously saved window layout cannot be recovered when ATLworks is not used with a test system.)

**Figure 1-26 Recovering the Window Layout**

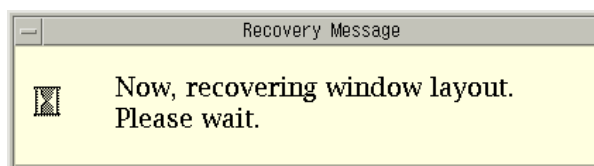


### 1. 8. 3 Window Layout Recovery Message

While the window layout is recovered, the dialog box shown below is displayed.

While the recovery operation is being performed, performing other operations may affect the recovery operation. Wait until this window disappears.

**Figure 1-27 Window Layout Recovery Message**



### 1. 8. 4 Window Layout Settings to Be Saved

Some window layout settings can be saved, depending on the individual tool.

This section lists what can be saved for each window layout.

#### Tool Box

- Location and size of the tool box

#### Control Panel

- Location and size of the main window
- Location and size of the User Area Command window
- Location and size of the Controller Command window
- Location of the DUT Information window
- Location of the Category window

➔ For more information on each window, refer to [2. "Control Panel" on page 2-1](#).

#### ASXDebugger

- Location and size of the ASXDebugger main window, the current program name, and the line number
- Toggle button and text box settings in the Compilation switch window
- Toggle button setting and shell file name in the Option window
- Location and size of the Variable List and Variable windows, and the displayed variable names
- Location and size of the Insert window, the number of displayed files, and the displayed file names
- Location and size of the Pattern window, and the displayed file name
- Location and size of the Socket window, and the displayed file name



- Location and size of the Scramble window, and the displayed file name

➔ For more information on each window, refer to [3. "ASXDebugger" on page 3-1.](#)

#### Test Condition Monitor

- Location of the main window
- Location of the Control Word monitor
- Location and size of the Pin Clock monitor
- Location and size of the Pin Level monitor
- Location and size of the DC/VS monitor
- Location and size of the Time Sequence monitor
- Location and size of the ALPG monitor
- Location and size of the FM monitor
- Location and size of the OBM monitor
- Location and size of the Soft monitor
- Location and size of the MCW monitor
- Location and size of the Test Head Status monitor

➔ For more information on each window, refer to [4. "Test Condition Monitor" on page 4-1.](#)

#### Time Scope

- Location of the main window
- Location and size of the All Test window
- Location and size of the Select Test window

➔ For more information on each window, refer to [6. "Execution Time Measurement Tool\(Time Scope\)" on page 6-1.](#)

#### DBMtool

- Location and size of the main window

➔ For more information on each window, refer to [7. "DBMtool" on page 7-1.](#)

## 1. 8. 5 Restrictions

- If you saved off-screen windows, the entire windows will be restored on the screen.
- Even if you save minimized windows, you cannot restore them.
- If you use CDE, only the windows displayed with Tool Box on the same screen (workspace) can be restored.

## 2. Control Panel

Control Panel is a tool which is used to load, start or stop the test program and to make data log settings for each test station. It is also used to display the test status and Pass/Fail information necessary to recognize the operating condition of the tester.

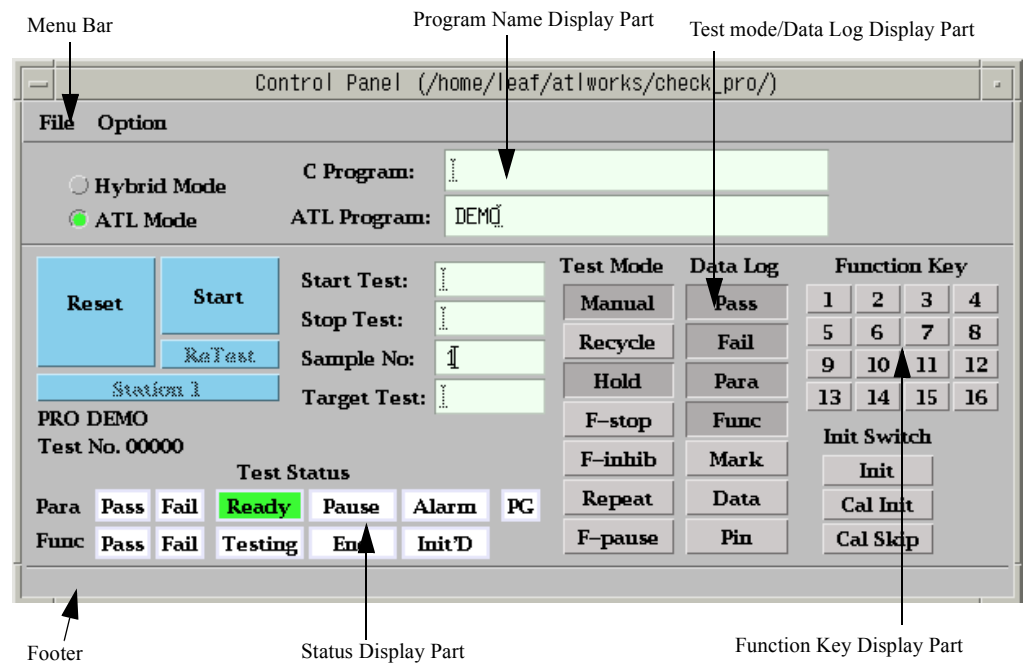
Restrictions

- Control Panel can be used with the online tester or the tester simulator (TESIM). It cannot be used when ATLworks is activated in the tester offline mode.
- While the device test is being performed in the AUTO mode, Control Panel does not display the current test number. The last test number is displayed at the end of the test.
- When the test controller is operating in the hybrid mode, the test status will not change unless the ATL\_init function is executed.

### 2. 1 Main Window

When ATLworks is connected to the online tester, Control Panel is automatically activated, and the main window shown in [Figure 2-1](#) is displayed.

Figure 2-1 Main Window



## 2. 2 Function Description

### 2. 2. 1 Controller Mode

The hybrid controller allows you to execute C or C++ language programs in addition to ATL language programs. The controller has the two operating modes described below and can indicate the main program language. The controller functions according to the selected mode.

- Hybrid mode

This operating mode uses the C language program as the main program.

➔ For more information, refer to [ASX/U-51 Hybrid Controller User's Manual\(No. : 8409598\)](#).

- ATL mode

This operating mode uses the ATL language program as the main program.

➔ For more information, refer to [ATL-51 Test Plan Program Reference Manual\(No. : 8431498\)](#).

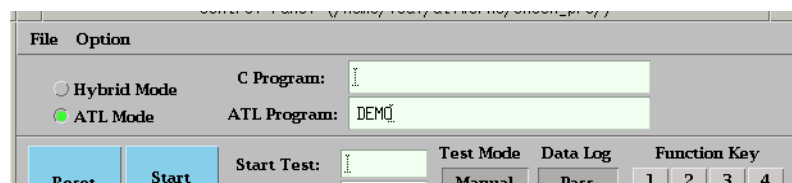


Tip

*The hybrid controller requires a system equipped with a TP5 or TP6 tester processor with 256 MB memory, and the license.*

### 2. 2. 2 Controller Mode Selection

Figure 2-2 Controller Mode Selection



- Hybrid Mode toggle button

When this toggle button is set to ON, the operating mode is set to the hybrid mode. The test program specified in the C program text box will be executed.

- ATL Mode toggle button

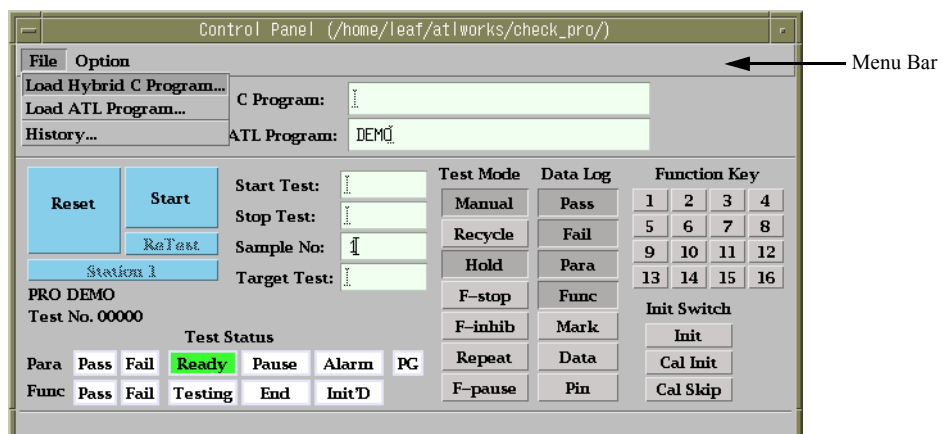
When this toggle button is set to ON, the operating mode is set to the ATL mode. The test program specified in the ATL Program text box will be executed.

### 2. 2. 3 Loading a Test Program

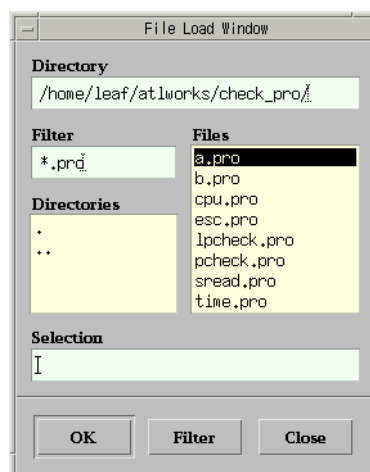
When you select Load Hybrid C Program... or Load ATL Program... from the File menu (shown in [Figure 2-3](#)), [Figure 2-4](#) is displayed. Select a test program from the window and load it. When the program has been loaded, the program name is displayed in the C Program or ATL Program text box.

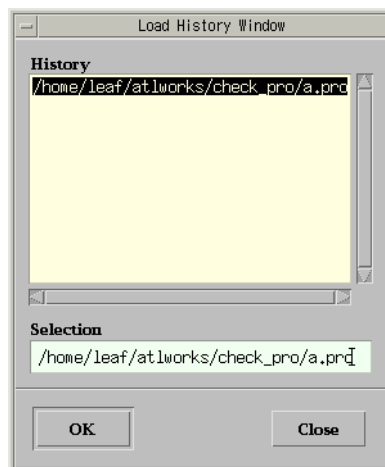
The loaded test program is registered and displayed in the window shown in [Figure 2-5](#). (To display the window, select History... from File.) When you select a previously loaded test program from the window and click on the OK button, you can reload it.

**Figure 2-3 Selecting Load**



**Figure 2-4 File Load Window**



**Figure 2-5 Load History Window**

## 2. 2. 4 File Load Window

→ See [Figure 2-4](#)

When you click the OK button, the file selected from the Files list is loaded and then set in the Program text box.

When you click the Filter button, the directory selected from the Directories list is displayed in the Files list.

When you click the Close button, the File Load window closes.

## 2. 2. 5 Load History Window

When you click the Load button, the file selected from the Files list is set in the Program text box and then the program is loaded for debugging.

When you click the Close button, the Load History Window closes.

## 2. 2. 6 Setting a C Program Name

The test program name currently set is displayed in the C Program text box.

When you use the keyboard to change the program name displayed in this text box and press the Return key, you can set a new test program name. A maximum of 128 characters can be used.

When the new test program name is set, the operating mode is automatically switched to the Hybrid Mode.

To specify a file in the NFS mounted directory, include "asxFile:" before the file name.

## **2. 2. 7 Setting an ATL Program Name**

The test program name currently set is displayed in the ATL Program text box.

When you use the keyboard to change the program name displayed in this text box and press the Return key, you can set a new test program name. A maximum of 32 characters can be used.

When the new test program name is set, the operating mode is automatically switched to the ATL Mode.

## **2. 2. 8 Starting/Stopping a Test Program**

When you click the Start button, the test program is started.

When you click the Reset button, the test program currently being executed is stopped.

When you click the ReTest button, the last test is re-executed.

## **2. 2. 9 Displaying/Changing the Station Number**

A station number (1 or 2) currently monitored by Control Panel is displayed on the Station button.

To change the station monitored by Control Panel, click the Station button.

## **2. 2.10 Displaying the Current Program Name and Test Number**

The current program name and test number are displayed below the Station button.

When REMARK is set, the program name is changed to REMARK.

## **2. 2.11 Displaying Pass/Fail**

Pass or Fail is indicated as the result of the DC parametric test Para and functional test Func.

## **2. 2.12 Setting a Start/Stop Test Number**

The current start test number is displayed in the Start Test text box. The current stop test number is displayed in the Stop Test text box.

When you change the Start Test and Stop Test text box settings and press the Return key, you can set new start and stop test numbers.

## **2. 2.13 Setting a Sample Number**

The current sample number is displayed in the Sample No. text box. When you change the text box setting and press the Return key, you can set a new sample number.

## 2. 2.14 Setting a Target Test Number

When you type a desired test number in the Target Test text box and press the Return key, you can execute the specified test number only. This function can also be performed by starting the /PART utility to specify the same number as the start and stop numbers.

To cancel it, set 0.

## 2. 2.15 Displaying the Test Status

The items listed below are used to indicate the current test status.

- Ready
- Testing
- Pause
- End
- Alarm
- Init'D
- PG (displayed only when the tester operates online in Manual or Hold mode)

## 2. 2.16 Setting the Test Mode

The items listed below are referred to as Test Mode switches and used to set the current test mode. Clicking a switch turns the setting on or off.

- Manual
- Recycle
- Hold
- F-stop
- F-inhib
- Repeat
- F-pause

## 2. 2.17 Setting the Data Log Mode

The items listed below are referred to as Data Log switches and used to set the current data log mode. Clicking a switch turns the setting on or off.

- Pass
- Fail
- Para
- Func
- Mark
- Data
- Pin

## 2. 2.18 Setting Function Keys

The current function key switch setup is displayed in Function Keys 1 to16. Clicking a switch turns the setting on or off.

## 2. 2.19 Setting Init Switches

The Init, Cal Init (Calibration Init), and Cal Skip switches are used to indicate the current settings. Clicking a switch turns the setting on or off.



## 2. 2.20 Executing User Area Commands

When you select User Area Command... from the Option menu, the window shown in [Figure 2-6](#) is displayed. This window performs the same function as when you enter U from the controller window.

**Figure 2-6 User Area Command Window**

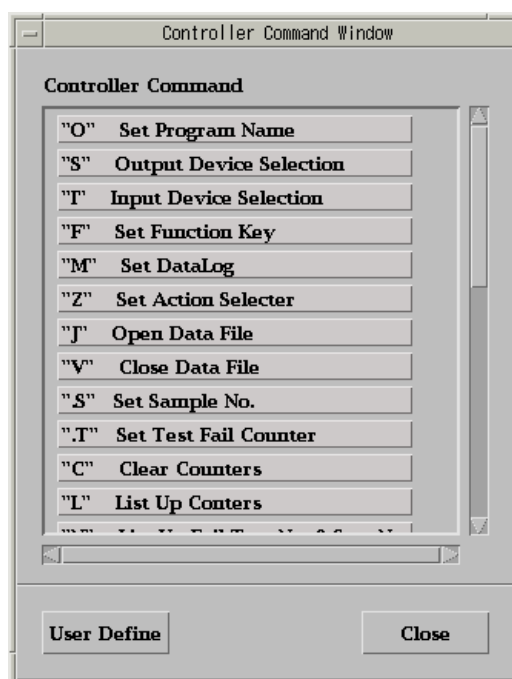


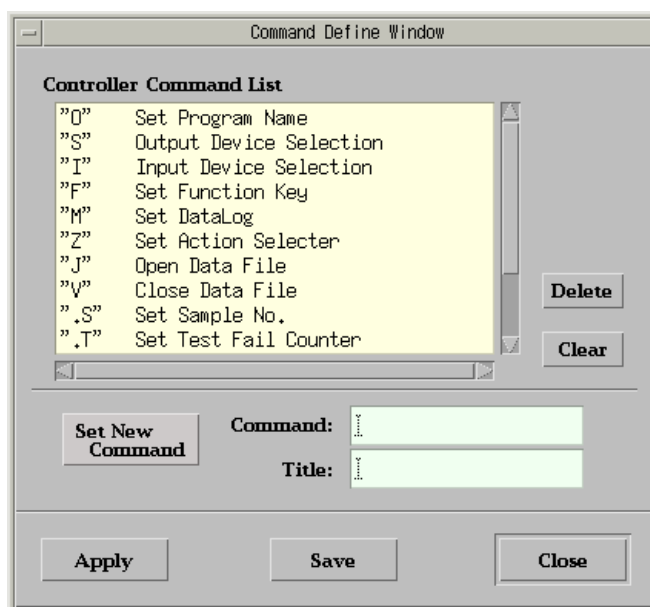
## 2. 2.21 Displaying Controller Commands

When you select Controller Command... from the Option menu, the window shown in [Figure 2-7](#) is displayed. Clicking a button on this window execute the corresponding command used for the tester controller.

When you click the User Define button, the window shown in [Figure 2-8](#) is displayed, and you can define up to 30 commands and register them.

**Figure 2-7 Controller Command Window**



**Figure 2-8 Command Define Window**

To register a command, type a command name and a comment (omissible) in the Command and Title text boxes of the Command Define Window, individually, and click the Set New Command button. When a command is registered, it is displayed in the Controller Command List. When you click the Apply button, the registered command is set a button in the Controller Command Window. In addition, you can also save the edited contents in a file by clicking on the Save button. The file is located in the current directory and named controller\_command.define.

The Delete button is used to delete the specified commands from the Controller Command List.

The Clear button is used to clear the entire Controller Command List.

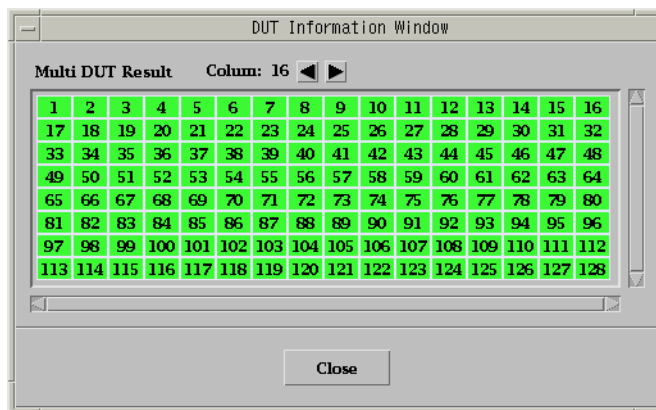
Tip

*The controller commands and the UserArea commands cannot be used while the program is operating on the tester controller.*

## 2. 2.22 Displaying Parallel Test Information

When you select DUT Pass/Fail... from the Option menu, the window shown in [Figure 2-9](#) is displayed. This window indicates parallel test information, such as the number of DUTs tested in parallel, Pass/Fail result for each DUT, and DUT numbers.

**Figure 2-9 DUT Information Window**



### Multi DUT Result

The numbers of the DUTs specified for the parallel test are displayed.

- The numbers of the DUTs excluded from the parallel test are displayed in gray.
- The numbers of the target DUTs are displayed in white.
- The numbers of the DUTs which passed the most recent test are displayed in green.
- The numbers of the DUTs which failed the most recent test are displayed in red.

When the test plan program is completed, the total pass/fail results are displayed in colors.

The numbers of the DUTs which were not tested are displayed in white.

### Column

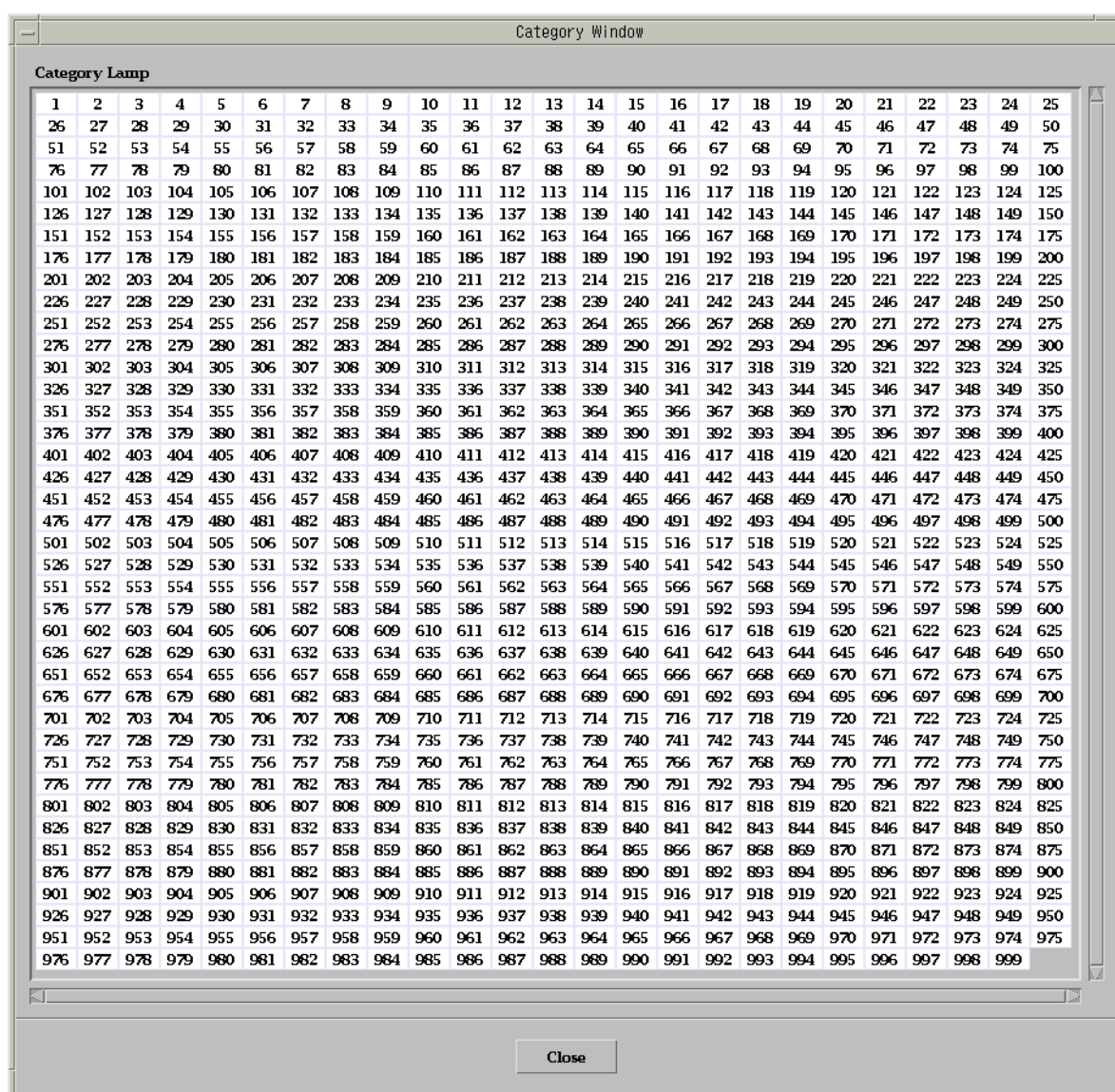
Clicking the left or right arrow allows you to change the number of columns.

The number of columns can be set from eight to a half the maximum number of DUTs to be tested in parallel.

## 2. 2.23 Displaying Category Information

When you select Category... from the Option in the menu bar, the window shown in [Figure 2-10](#) is displayed. The current specified category is displayed in red.

**Figure 2-10 Category Window**



## 2. 3 Command Language

Command Processor is used to execute a command written in a file and access Control Panel. Command Processor can interpret commands described below. The basic commands are listed below.

<application><Menu><Menu Option><Field><value>

A command is a path to access a desired field through the graphical user interface. The command setting is different, depending on the number of sub-menus or sub-windows. For example, to save the User Area, open the Option menu in the Control Panel main window and select User Area. Then, select the Save User Area button. The following command is created:

CPNL Options User\_Area save\_user\_area

The command language is not case-sensitive.

➔ For more information on effective commands, refer to [2. 3. 1 "Data Field Commands" on page 2-13](#).

Once a command is entered, applications will not accept any entry until the command is complete.

### 2. 3. 1 Data Field Commands

#### ◆ Control Panel Window

Control Panel window commands

- CPNL Start  
Starts Control Panel.
- CPNL Reset  
Resets Control Panel.
- CPNL Retest  
Retests the devices.
- CPNL Station Number <number>  
Sets a station number.
- CPNL Station Program <name>  
Sets a program name.
- CPNL Station Start\_No <number>  
Sets a start test number.
- CPNL Station Stop\_No <number>  
Sets a stop test number.
- CPNL Station Sample\_No <number>  
Sets a sample number.
- CPNL Station Target\_No <number>  
Sets a target test number.

### Test Mode Commands

- CPNL TestMode Manual Enable/Disable  
Enables or disables the Manual mode.
- CPNL TestMode Recycle Enable/Disable  
Enables or disables the Recycle mode.
- CPNL TestMode Hold Enable/Disable  
Enables or disables the Hold mode.
- CPNL TestMode Repeat Enable/Disable  
Enables or disables the Repeat mode.
- CPNL TestMode Fail\_Inhibit Enable/Disable  
Enables or disables the Fail Inhibit mode.
- CPNL TestMode Fail\_Pause Enable/Disable  
Enables or disables the Fail Pause mode.
- CPNL TestMode Fail\_Stop Enable/Disable  
Enables or disables the Fail Stop mode.

### Data Log Commands

- CPNL Datalog Pass Enable/Disable  
Enables or disables the Pass mode.
- CPNL Datalog Fail Enable/Disable  
Enables or disables the Fail mode.
- CPNL Datalog Functional Enable/Disable  
Enables or disables the Functional mode.
- CPNL Datalog Parametric Enable/Disable  
Enables or disables the Parametric mode.
- CPNL Datalog Mark Enable/Disable  
Enables or disables the Mark mode.
- CPNL Datalog Data Enable/Disable  
Enables or disables the Data mode.
- CPNL Datalog Pin Enable/Disable  
Enables or disables the Pin mode.

### Function Keys Commands

- CPNL Function\_keys <number> Enable/Disable  
Enables or disables the specified Function Key.

### Init Commands

- CPNL Station Init Enable/Disable  
Enables or disables the Init setting.
- CPNL Station CalInit Enable/Disable

Enables or disables the CalInit setting.

- CPNL Station CalSkip Enable/Disable  
Enables or disables the CalSkip setting.

#### ◆ File Load Window

- CPNL File Load <directory-name / file-name>  
Changes the directory and specifies a program name (to load the program).

#### ◆ User Area Window

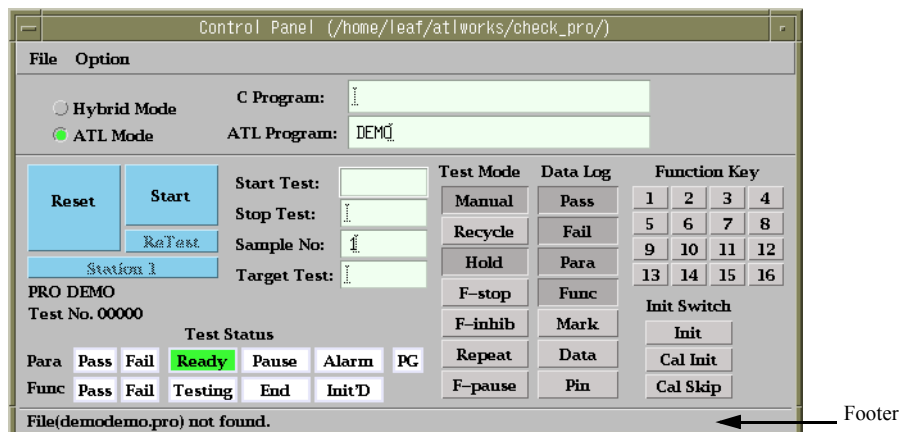
- CPNL Options User\_Area clear\_var\_prog\_area  
Clears the variable and the program area.
- CPNL Options User\_Area clear\_key\_commands  
Clears the command keys.
- CPNL Options User\_Area clear\_all\_above  
Clears all of the above.
- CPNL Options User\_Area clear\_file\_channels  
Clears the file channels.
- CPNL Options User\_Area save\_user\_area  
Saves the user area.
- CPNL Options User\_Area recover\_user\_area  
Recovers the user area.



## 2. 4 Error Messages

If an error occurs, an error message is displayed in the Control Panel window footer.

**Figure 2-11 Error Messages**



File Name Error (\*.pro).

The program name is not \*.pro.

During Testing! cannot load file.

The selected file cannot be loaded because a program is currently being executed.

Invalid TestNumber

The specified start or stop test number is not from 0 to 99999.

Invalid Sample Number.

Displayed when you enter a number other than 0 to #7FFFFFFF as a sample number.

Invalid Target TestNumber

The specified target test number is not from 0 to 99999.

File (file-name) not found.

The specified file is not found.

C Program not found.

The specified C Program is not found.

ATL Program not found.  
The specified ATL Program is not found.

### 3. ASXDebugger

ASXDebugger is a debugging tool used to modify breakpoints and variables with easy mouse operation.

#### Restrictions

1. ASXDebugger cannot be used while other utilities are operating in the controller window.
2. When a program including INSERT statements is displayed in the ASX debugger main window, the breakpoint line number is different from the line number (which was displayed in the lower part of the main window at the setup time).
3. Even if you delete the breakpoint setup information using the U command (1) in the controller window, the breakpoint information for ASXDebugger is retained. To restore the normal display, restart ASXDebugger (by clicking the ASXDebugger button in the tool box).
4. A cross compiler is necessary to compile programs. The compiler type depends on the test system as shown below.

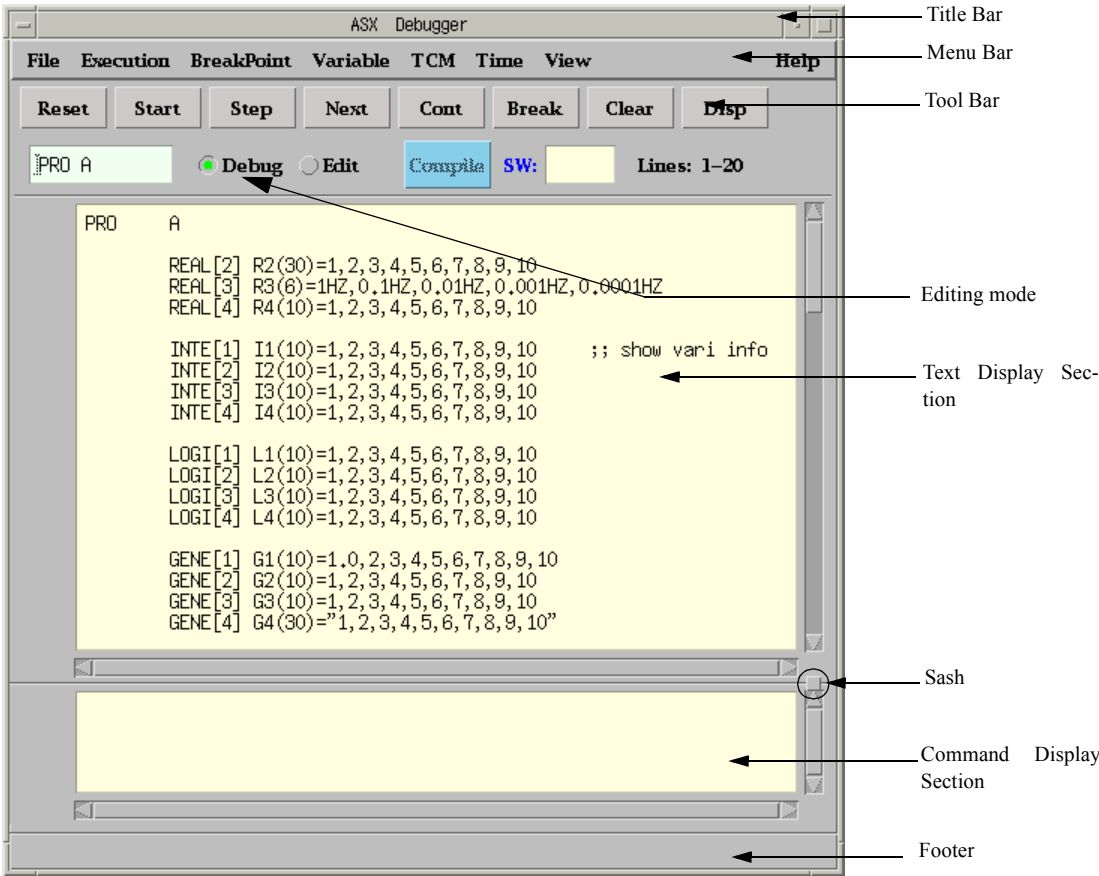
| Test System Model              | Cross Compiler |
|--------------------------------|----------------|
| T5500 series                   | PXTL51-00S     |
| T5377, T5376, T5375, and T5371 | PXTL51-36S     |

5. Restrictions on the hybrid controller
  - a. When CrossWind (which is a development tool included in Tornado) is started, ASXDebugger operates only for ATL programs. (The ASXDebugger cannot be used in a C program that includes the ATL.)
  - b. Unless ASXDebugger is set to the Hybrid debug mode (/HYBRID/S), C programs cannot be loaded, edited, debugged, or compiled. (The ASXDebugger cannot debug both ATL and C sections in a C program that includes the ATL.)
  - c. While the C program operation is broken at test\_main(), the following menus and menu items cannot be used:
    - Skip to n'th Breakpoint, Target test and RON/ROF Monitor in the Execute menu
    - Unit in the Variable menu
    - TCM Menu
    - Insert in the View menu
    - Time Menu
  - d. You can refer to variable pointers in the C program but cannot change data.
  - e. If a C program that includes the ATL is active-linked as a library module, the ATL section in the library module cannot be executed by using Step.

### 3. 1 Main Window

When you click the Debugger button in the tool box, the window shown in [Figure 3-1](#) is displayed.

Figure 3-1 ASXDebugger



**Tip**

When the editing mode is the Edit mode, disabled menus are not displayed.

When the editing mode is the Debug mode, program editing is disabled.

- Title Bar
- Displays the name of the source program file which is displayed in the text display section.
- Menu Bar

Allows you to select commands from the listed menu and to operate ASXDebugger.

The contents of the menu bar depend on the editing mode of Edit or Debug.

➔ For more information, refer to [3. 2 "Edit Mode" on page 3-5](#) and [3. 3 "Debug Mode" on page 3-11](#).

### Tool Bar

Buttons are provided for frequently used menu commands on the tool bar. Operation is the same as the corresponding menu commands. The tool bar can only be used in the Debug mode.

### Object

Displays the currently loaded object file name.

The C object file in the NFS mounted directory includes "asxFile:" before the file name.

### Editing mode

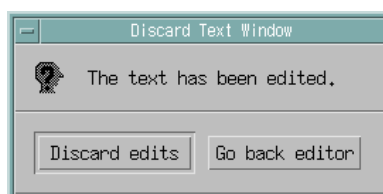
Switches between the Edit mode (text editing) and the Debug mode (program execution).

When you switch the editing mode, the menu bar changes accordingly.

In the Debug mode, you cannot modify the program. To modify the program, switch to the Edit mode.

When you edit the source program and attempt to switch to the Debug mode without saving the source program in a file, the window shown in [Figure 3-2](#) is displayed.

**Figure 3-2 Message Window**



- Discard edits button  
Discards the edited contents and changes the editing mode to the Debug mode.
- Go back editor button  
Returns the editing mode to the Edit mode without saving the edited contents.

### Lines

Displays the number of displayed program lines.

### Text Display Section

Displays the source file contents.

When the Debug mode is selected, you cannot modify the program.

### Command Display Section

#### Edit mode

Displays an error message to indicate a compilation error, etc.

#### Debug mode

Displays the executed commands (Start, Reset, Continue, etc.).

### Sash

By dragging the sash up or down, you can change the number of lines for the text and command display sections.

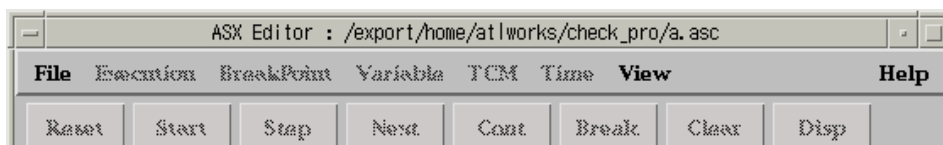
#### Footer

Displays an error message.

## 3. 2 Edit Mode

When the Edit mode is selected, the menu bar is changed as shown in [Figure 3-3](#).

**Figure 3-3 Edit Mode Menu Bar**



### 3. 2. 1 File Menu

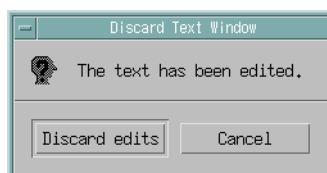
The File menu includes nine commands described below.

#### New

Clears the text display section.

When you select this menu during editing, the window shown in [Figure 3-4](#) is displayed to ask if you discard the edited contents.

**Figure 3-4 Confirmation Window**



- Discard edits button  
Discards the edited contents and clears the text display section.
- Cancel button  
Cancels clearing the text display section.

#### Load Hybrid C Program...

Loads the C program to be debugged.

➔ For more information, refer to [3. 2. 1. 1 "File Load Window" on page 3-7](#).

#### Load ATL Program...

Loads the ATL program to be debugged.

➔ For more information, refer to [3. 2. 1. 1 "File Load Window" on page 3-7](#).

#### Save

Saves the currently displayed source file.

#### Save As...

Saves the currently displayed source file as a new file name (Creates a new file).

➔ For more information, refer to [3. 2. 1. 2 "File Save Window" on page 3-8](#).

**Option...**

Sets optional ASXDebugger functions.

➔ For more information, refer to [3. 5 "Option Window" on page 3-40](#).

**ATL Compile Option...**

Sets optional ATL Compiler functions.

➔ For more information, refer to [3. 6 "ATL Compile Option Window" on page 3-44](#).

**Hybrid C Compile Option...**

Sets optional Hybrid C Compiler functions.

➔ For more information, refer to [3. 7 "C Compile Option Window" on page 3-46](#).

**Close**

Quits ASXDebugger.



### 3. 2. 1. 1 File Load Window

When you select Load Hybrid C Program... or Load ATL Program... from the File menu, the sub-window shown in [Figure 3-5](#) is displayed. This sub-window is used to select a source file for debugging.

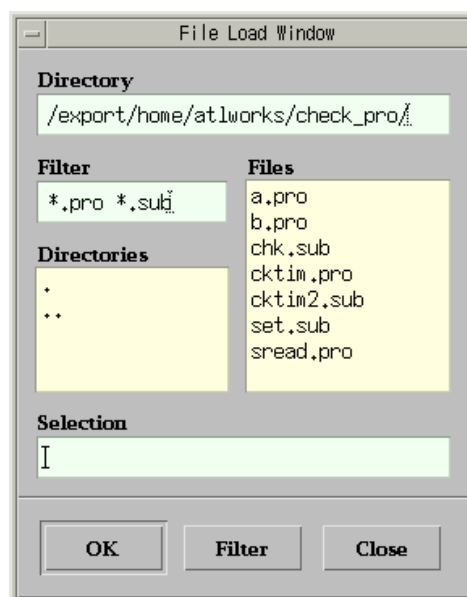
Load Hybrid C Program...

C program files are listed in the Files list.

Load ATL Program...

ATL program files are listed in the Files list.

**Figure 3-5 File Load Window**



**OK button**

Displays the file selected from the Files list in the text display section.

**Filter button**

Extracts files from the selected directory according to the filter setting and displays them in the Files list.

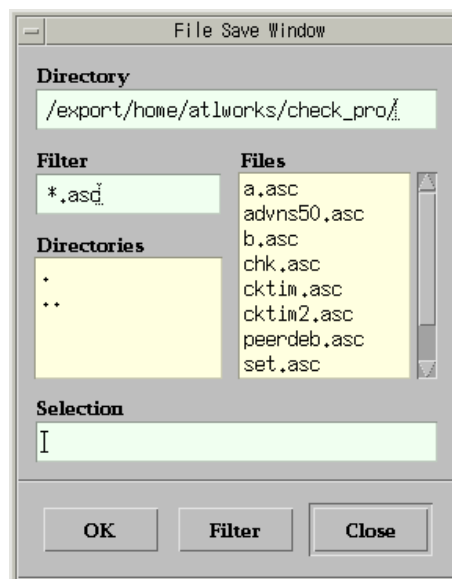
**Close button**

Closes the File Load Window.

### 3. 2. 1. 2 File Save Window

When you select Save As... from the File menu, the sub-window shown in [Figure 3-6](#) is displayed. This sub-window is used to save the currently displayed source file as a new file name.

**Figure 3-6 File Save Window**

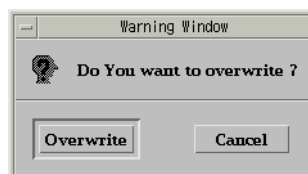


#### OK button

Stores the currently displayed program in the file specified in the File Name text box.

When you attempt to save the file as another name (or a new name) and the specified file already exists, the window shown in [Figure 3-7](#) is displayed.

**Figure 3-7 Overwrite Confirmation Window**



- Overwrite button  
Overwrites the existing file.
- Cancel button  
Cancels saving the file.

#### Filter button

Extracts files from the selected directory according to the filter setting and displays them in the Files list.

#### Close button

Closes the File Save Window.

### 3. 2. 2 Compile Button

Saves the source program displayed in the text display section in the file, and compiles it.

When the ATL program is displayed, use the cross compiler to compile it.

When the Hybrid C program is displayed, execute make according to the setting specified in the C Compile Option window.

➔ For more information, refer to [3. 7 "C Compile Option Window" on page 3-46](#).

When compilation has been successfully completed, the object object file is generated, saved, and reloaded. If a compilation error occurs, the error message and the error line number are displayed in the command display section.



Tip

*The compiler type depends on the test system as shown below.*

| Test System Model              | Cross Compiler |
|--------------------------------|----------------|
| T5500 series                   | PXTL51-00S     |
| T5377, T5376, T5375, and T5371 | PXTL51-36S     |

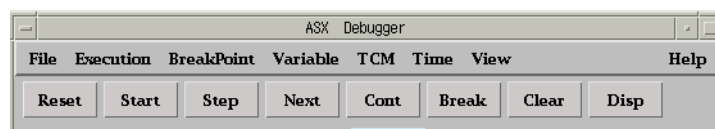
### 3. 2. 3 SW Text Box

If you specify an option switch in this text box for conditional compilation, the program is compiled according to the specified switch setting.

## 3. 3 Debug Mode

When the Debug mode is selected, the menu bar is changed as shown in [Figure 3-8](#).

**Figure 3-8** Debug Mode Menu Bar



### 3. 3. 1 File Menu

The File menu includes six commands described below.

#### Load Hybrid C Program...

Loads the C program to be debugged.

➔ For more information, refer to [3. 2. 1. 1 "File Load Window" on page 3-7](#).

#### Load ATL Program...

Loads the ATL program to be debugged.

➔ For more information, refer to [3. 2. 1. 1 "File Load Window" on page 3-7](#).

#### Change Subr Program (down)

While the ATL program is being debugged, the selected sub-program is displayed.

While the C program is being debugged, the operation opposite to Reload Main Program (up) below is performed.

#### Reload Main Program (up)

While the ATL program is being debugged, the main-program is displayed.

While the C program is being debugged, the function that called for a break is displayed when it is suspended for a break.

#### Option...

Sets optional ASXDebugger functions.

➔ For more information, refer to [3. 5 "Option Window" on page 3-40](#).

#### Close

Quits ASXDebugger.

### 3. 3. 2 Execution Menu

The Execution menu includes nine commands described below.

#### Reset

Stops debugging the program.

#### Start

Starts debugging the program.

When a breakpoint is set in the program, ASXDebugger displays the breakpoint mark to the left of the program line where the breakpoint is located and stops debugging the program.



#### Step

Debugs only the line where the breakpoint caused the program to stop.

If the program reaches the line to call a function, the program goes to the function and stops.

#### Next

Debugs only the line where the breakpoint caused the program to stop.

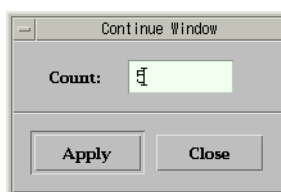
#### Continue

Restarts debugging the program.

#### skip to n'th Breakpoint...

When you select skip to n'th Breakpoint, the sub-window shown in [Figure 3-9](#) is displayed. This sub-window is use to specify the number of breakpoints to be ignored.

**Figure 3-9 Continue Window**



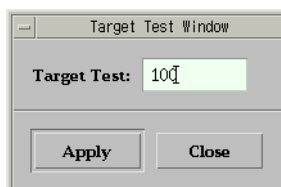
- Apply button  
Executes skip to n'th Breakpoint.
- Close button  
Closes the Continue Window.

#### Target Test...

When you select Target Test, the sub-window shown in [Figure 3-10](#) is displayed.

This sub-window is used to specify a test number you want to execute.

This function is equivalent to using Control Panel to specify a test number in the Target Test text box.

**Figure 3-10 Target Test Window**

- Apply button  
Sets the target test number.
- Close button  
Closes the Target Test Window.

**RON/ROF Monitor...**

Displays the line numbers of the RON or ROF change points in the executed program.

➔ For more information, refer to [3. 3. 2. 1 "RON/ROF Monitor" on page 3-13](#).

**Controller Mode**

Changes the controller operation mode.

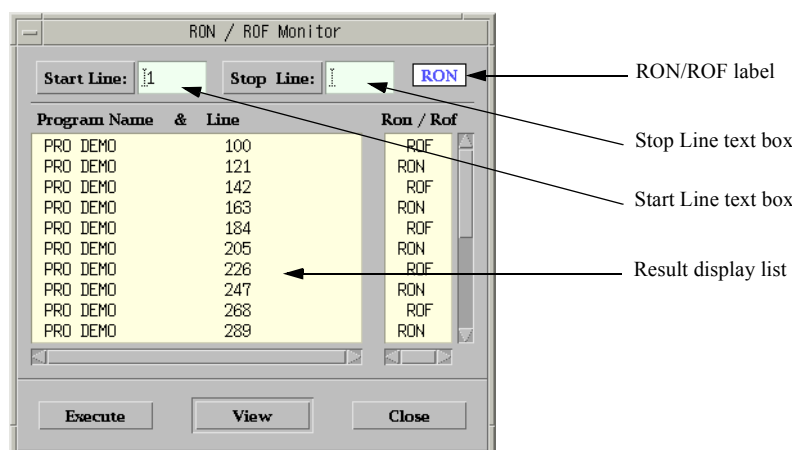
Selecting this command is equivalent to executing the /HYBRID command in the controller window.

If the hybrid controller operation environment is not prepared, this command is unavailable.

- Hybrid debug Mode  
Executes the /HYBRID/S command and displays the C program specified on the Control Panel main window.
- Hybrid Mode  
Executes the /HYBRID/E command and displays the ATL program from the memory to the main window.
- ATL Mode  
Executes the /HYBRID/D command and displays the ATL program from the memory to the main window.

**3. 3. 2. 1 RON/ROF Monitor**

Selecting RON/ROF Monitor... from the Execution menu displays [Figure 3-11](#). This sub-window displays a maximum of 200 line numbers of the RON or ROF change point (from the latest).

**Figure 3-11 RON/ROF Monitor****Start Line button**

Clicking this button specifies the start line. Highlight the relevant line in the main window and then click this button. The selected line number is stored in the Start Line text box.

**Start Line text box**

Specifies the start line. Only the ATL lines can be specified.

This text box stores the following line number when started.

- Stores 1 when the tester is in the ready state.
- When the program is being stopped at the breakpoint, the text box stores the relevant line number of the breakpoint.

**Stop Line button**

Clicking this button specifies the stop line. Highlight the relevant line in the main window and then click this button. The selected line number is stored in the Stop Line text box.

**Stop Line text box**

Specifies the stop line. Only the ATL lines can be specified.

The stop line can be omitted. When omitted, the lines up to the end of the program are counted.

**RON/ROF label**

Displays the RON or ROF state at the execution.

**Execute button**

Clicking this button starts monitoring the RON or ROF state. The monitoring starts from the following line:

When the tester is in the ready state, the monitoring starts from the line number specified in the Start Line text box.

When the program is being stopped at the breakpoint, the monitoring starts from the relevant line number of the breakpoint.

When started, all buttons except the Close button are disabled.



### Result display list

Displays the monitoring result of the RON or ROF state.

The list displays ATL program names, line numbers, and C function names.

Selecting the list displays the relevant line in the main window.

C functions are not displayed.

### View button

Clicking this button displays the monitoring result of the RON or ROF state in the result display list.

### Close button

Clicking this button closes the sub-window. If the program is being executed, the processing terminates.

#### Tip

- *If the program does not pass the start line, no result is displayed.*
- *If the program does not pass the stop line, the result is displayed when the program is complete.*
- *The buffer stores a maximum of 200 line numbers. If the number exceeds 200, the buffer numbers are overwritten, oldest first.*
- *If the program stops at a breakpoint, which is different from the breakpoint where the program should stop, re-execute the program by clicking the Continue button. If the program is re-executed by clicking the Execute button, the program re-starts, counting the line numbers from that breakpoint.*

### 3. 3. 3 BreakPoint Menu

The BreakPoint menu includes four commands described below.

#### Set BreakPoint

Sets a breakpoint to the specified line (to stop the program before executing the specified line).

When you set a breakpoint, the breakpoint mark shown below is displayed to the left of the specified line.



#### Delete BreakPoint

Deletes the breakpoint from the specified line.

When this command is selected, the breakpoint mark disappears from the left of the specified line.

#### Delete All BreakPoint

Deletes all breakpoints.

All breakpoint marks disappear.

#### Show BreakPoint

Displays line numbers where breakpoints are located.

#### Tip

*Before you set (or delete) breakpoints in a C program, start the C program once and stop it before the beginning of the test\_main() function.*

*The process for displaying (or deleting all) the breakpoints varies according to the program being debugged.*

- *When the C program is being debugged, only information from the C program is displayed (or all deleted).*
- *When the ATL program is being debugged, only information from the ATL program is displayed (or all deleted).*

*All breakpoints are deleted when the ASXDebugger terminates.*

### 3. 3. 4 Variable Menu

The Variable menu includes three commands described below.

#### Display...

Refers and changes the specified variables.

➔ For more information, refer to [3. 3. 4. 1 "Variable Window" on page 3-17](#).

#### Display all name...

Lists all variable names used in the program.

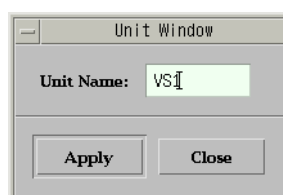
➔ For more information, refer to [3. 3. 4. 2 "Variable Table Window" on page 3-20](#).

#### Unit...

When you select Unit, the sub-window shown in [Figure 3-12](#) is displayed. Enter the unit name to be referred into this sub-window.

The result is displayed in the controller window.

**Figure 3-12 Unit Window**



- Apply button  
Executes referencing of the unit information.
- Close button  
Closes the Unit Window.

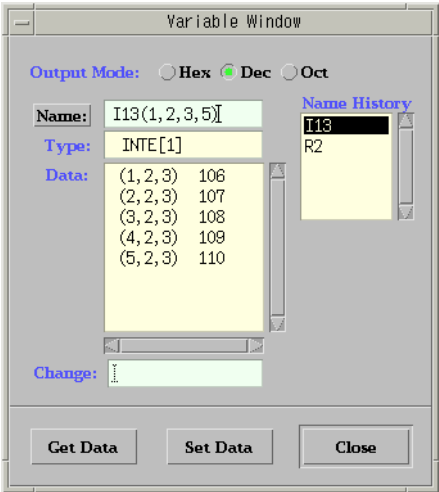
#### 3. 3. 4. 1 Variable Window

When you select Display... from the Variable menu, the sub-window shown in [Figure 3-13](#) is displayed. With this sub-window, you can refer and change variables.

However, you can refer and change variables only when the program is stopped at a breakpoint.

Clicking the Name button displays the Variable Table Window. You can refer to all variable names in the program using this window.

Figure 3-13 Variable Window



Output mode

Switches the display format of variables.

- Dec      Displays variables in decimal.
- Hex     Displays variables in hexadecimal. (A # supplied at the beginning of numeric data.)
- Oct     Displays variables in octal. (A "\$" is supplied at the beginning of numeric data.)

Tip

The format depends on the variable type.

Name button

Displays the list of variables in the other window (Variable Table Window).

➔ For more information, refer to [3. 3. 4. 2 "Variable Table Window" on page 3-20](#).

Name text box

Specifies the variable name to be referred. If you only specify a variable name, all elements of the variable are displayed.

For array variables, you can specify the position of the first and last elements and number of elements that will be displayed.

- One-dimensional array:      Vname (I1-I2, n)      (including string variables)
- Two-dimensional array:      Vname (I1-I2, J1-J2, n)
- Three-dimensional array:    Vname (I1-I2, J1-J2, K1-K2, n)

V: Variable names. I1, J1, K1: First element numbers. I2, J2, K2: Last element numbers. n: Number of elements.

➔ For more information, refer to [3. 8. 4 "Partial Reference to Array Variables" on page 3-50](#).

#### Type text box

Displays the variable type (entry disabled).

#### Data list

Displays the current data of the variable (entry disabled).

#### Change text box

Specifies the new data to be replaced with the current data.

When you enter data in hexadecimal, attach # to the beginning of the data.

When you enter data in octal, attach \$ to the beginning of the data.

#### Get Data button

Displays Data and Type of the variable specified in the Name box.

#### Set Data button

Replaces the current data (in the Data box) of the variable specified in the Name box with the new data specified in the Change box.

#### Name History list

A maximum of 20 variable names previously referred to are registered as a history.

If the target program to be debugged is changed, then the registered variable names are cleared.

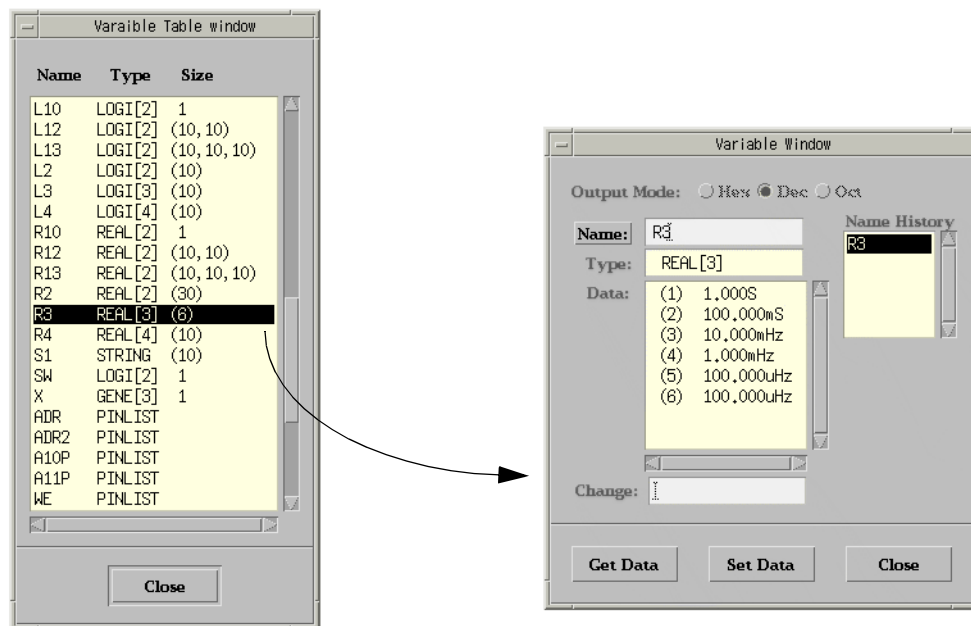
#### Close button

Closes the Variable Window.

### 3. 3. 4. 2 Variable Table Window

When you select Display all name... from the Variable menu, the sub-window shown in [Figure 3-14](#) is displayed. This sub-window displays the list of variables used in the current program. Clicking a variable name listed in the Variable Table Window opens the Variable Window. The Variable Window allows you to refer to the variable you specified.

**Figure 3-14 Variable Table Window**



#### Name list

The list of variables that are being used in the debugged program is displayed in the Name list.

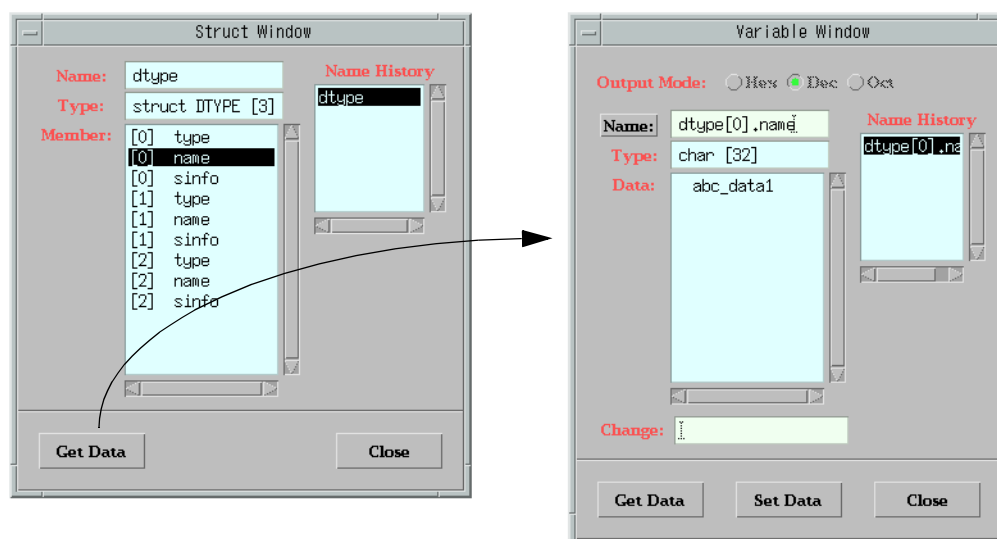
#### Close button

Closes the Variable Table Window.

### 3. 3. 4. 3 Struct Window

While the C program execution is suspended such as for the breakpoint, when you select the structured variables (hereafter referred to as struct variables) (the display is highlighted) and select Display... in the Variable menu, the sub-window (Struct Window) shown in Figure 3-15 is displayed. In the sub-window, you can refer to each struct variable element. When you select the element name from the Member list in the Struct Window and click the Get Data button, then the Variable Window where you can refer to the selected variable is displayed.

**Figure 3-15 Struct Window**



#### Name text box

The struct variable name is displayed in the Name text box.

#### Type text box

The struct variable type is displayed in the Type text box.

#### Member list

Each struct variable element is displayed in the Member list.

#### Name History

A maximum of 20 variable names previously referred to are registered as a history.

If the target program to be debugged is changed, then the registered variable names are cleared.

#### Get Data button

This button is used to display the selected element from the Member list in the Variable Window.

#### Close button

Closes the Struct Window.

### 3. 3. 5 TCM Menu

The TCM menu is used to display windows for reference and modification of TCM (Test Condition Monitor).

The TCM menu includes 11 commands described below.

CW...

Displays the Control Word monitor.

➔ For more information, refer to [4. 2 "Control Word Monitor" on page 4-7.](#)

Pin Clock...

Displays the Pin Clock monitor.

➔ For more information, refer to [4. 3 "Pin Clock Monitor" on page 4-9.](#)

Pin Level...

Displays the Pin Level monitor.

➔ For more information, refer to [4. 4 "Pin Level Monitor" on page 4-32.](#)

DC/VS...

Displays the DC/VS monitor.

➔ For more information, refer to [4. 5 "DC/VS Monitor" on page 4-39.](#)

Time Seq...

Displays the Time Sequence monitor.

➔ For more information, refer to [4. 6 "Time Sequence Monitor" on page 4-44.](#)

ALPG...

Displays the ALPG monitor.

➔ For more information, refer to [4. 7 "ALPG Monitor" on page 4-47.](#)

FM...

Displays the FM monitor.

➔ For more information, refer to [4. 8 "FM Monitor" on page 4-57.](#)

OBM...

Displays the OBM monitor.

➔ For more information, refer to [4. 9 "OBM Monitor" on page 4-69.](#)

Soft...

Displays the Soft monitor.

➔ For more information, refer to [4. 10 "Soft Monitor" on page 4-71.](#)

MCW...

Displays the MCW monitor.

➔ For more information, refer to [4. 11 "MCW Monitor" on page 4-74.](#)



### Test Head...

Displays the Test Head Status monitor.

➔ For more information, refer to [4. 12 "Test Head Status Monitor" on page 4-76](#).

### 3. 3. 6 Time Menu

When you select Time Measure from the Time menu, the Execution Time Measurement Tool (Time Scope) is activated.

➔ For more information, refer to [6. "Execution Time Measurement Tool\(Time Scope\)" on page 6-1.](#)

### 3. 3. 7 View Menu

The View menu includes five commands described below.

Insert...

Displays the insert file.

➔ For more information, refer to [3. 3. 7. 1 "Insert Window" on page 3-24.](#)

Pattern...

Displays the pattern file.

➔ For more information, refer to [3. 3. 7. 3 "Pattern Window" on page 3-28.](#)

Socket...

Displays the socket file.

➔ For more information, refer to [3. 3. 7. 4 "Socket Window" on page 3-31.](#)

Scram...

Displays the scramble file.

➔ For more information, refer to [3. 3. 7. 5 "Scram Window" on page 3-32.](#)

Header...

Displays the header file.

➔ For more information, refer to [3. 3. 7. 6 "Header Window" on page 3-33.](#)

#### 3. 3. 7. 1 Insert Window

When you select an insert file name (the display is highlighted) in the main window and then select the Insert command from the View menu, a sub-window as shown in [Figure 3-17](#) appears.

With this sub-window, you can refer to and modify the insert file and set breakpoints to the insert file.

To edit data, switch to the Edit mode (Edit button in the main window).

Edited data is automatically saved when it is compiled (by clicking the Compile button in the main window).

 Tip

*A maximum of five Insert Windows can be opened.*

Figure 3-16 Selecting the Insert File Name

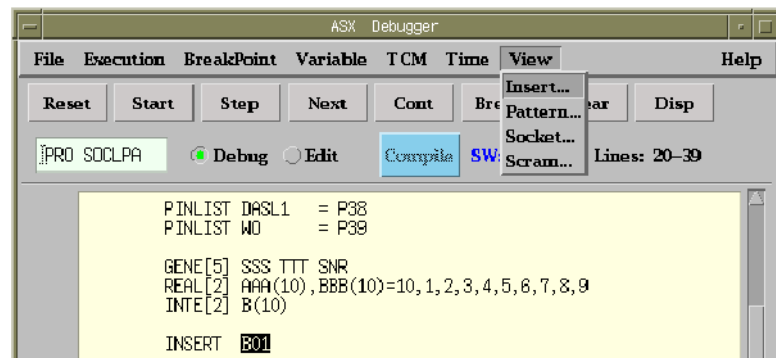
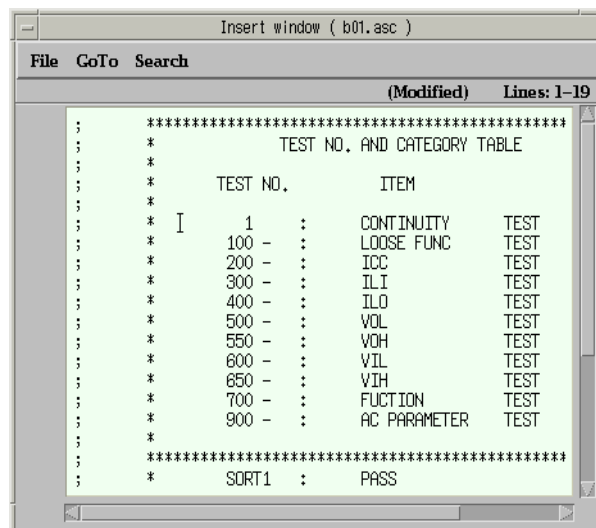


Figure 3-17 Insert Window



#### File Menu

- Close  
Closes the Insert Window.

#### Goto menu

- GoTo Line...  
Moves the cursor to the specified line and displays the line contents.  
➔ For more information, refer to [3. 4. 1. 1 "Select Line at Number Window" on page 3-36](#).

#### Search Menu

- Search String...

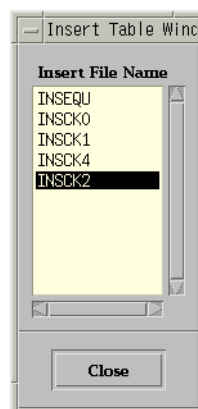
Searches for a specified character strings and replaces the found strings with other character strings.

➔ For more information, refer to [3. 4. 3. 1 "Search & Replace Window" on page 3-37](#).

### 3. 3. 7. 2 Insert Table Window

When the insert file name has been replaced with the EQUATE statement, and does not exist in the main program, the list of all insert file names used in the program is displayed in [Figure 3-18](#).

**Figure 3-18 Insert Table Window**



#### Close

Closes the window.

#### Insert File Name list

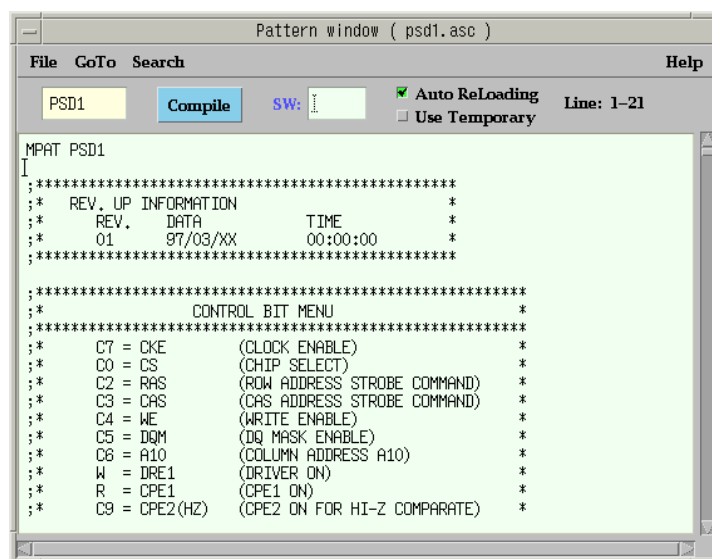
Lists all insert file names which are used in the program. If you click of the file names, that file is opened in the Insert Window.

### 3. 3. 7. 3 Pattern Window

When you select Pattern from the View menu, the sub-window shown in [Figure 3-19](#) is displayed. With this sub-window, pattern files are edited and compiled.

Like the Insert Window, if you have selected a pattern file name from the main window, it is displayed automatically.

**Figure 3-19 Pattern Window**



#### File Menu

The File menu includes five commands described below.

- New  
Clears the pattern text display section.
- Load...  
Loads the source pattern file to be edited.  
➔ For more information, refer to [3. 2. 1. 1 "File Load Window" on page 3-7.](#)
- Save  
Saves the currently displayed source pattern file.
- Save As...  
Saves the currently displayed source pattern file under a new file name (Creates a new file).  
➔ For more information, refer to [3. 2. 1. 2 "File Save Window" on page 3-8.](#)
- Close  
Closes the Pattern Window.

#### Goto menu

- GoTo Line...

Moves the cursor to the specified line and displays the line contents.

➔ For more information, refer to [3. 4. 1. 1 "Select Line at Number Window" on page 3-36](#).

### Search Menu

- Search String...

Searches for a specified character strings and replaces the found strings with other character strings.

For more information, refer to [3. 4. 3. 1 "Search & Replace Window" on page 3-37](#).

### Help Menu

- Pattern Statement...

Select a statement and then click this menu. The Online Manual is displayed and shows how to use the statement.

If Solaris 2.5.1 is used, the top page of the manual is displayed. If Solaris 7 or later is used, the page on which the selected statement is written is displayed.

➔ For more information, refer to [5. "Online Manual" on page 5-1](#).

### Object

Displays the name of the currently loaded pattern object file name.

### Compile button

Stores the program displayed in the pattern text display area in the source pattern file, and compiles it.

If a compilation error occurs, the error message and line number are displayed in the command display section.

➔ For more information on the command display part, refer to [3. 1 "Main Window" on page 3-2](#).

### Tip

*The compiler type depends on the test system as shown below.*

| Test System Model              | Cross Compiler |
|--------------------------------|----------------|
| T5500 series                   | PXTL51-00S     |
| T5377, T5376, T5375, and T5371 | PXTL51-36S     |

### SW text box

If you specify an option switch of the conditional compilation command using this text box, programs are compiled according to the specified switch condition.

### Auto ReLoading button (only Pattern Window)

When compilation has been successfully completed, if you have selected this button, the file is reloaded automatically.

Note that this function can only be used when ATLworks is used in Tester Online mode.

#### Use Temporary button (Pattern Window only)

If the Use Temporary button is selected, the edited pattern program is saved to a temporary file and then compiled when the Compile button is clicked. After it is compiled, a pattern object program is created and then saved to a temporary file. If the Auto ReLoading button is selected, the pattern object program is automatically reloaded.

Note that this function can only be used when ATLworks is used in Tester Online mode and the current test status is set to Pause.

#### Lines

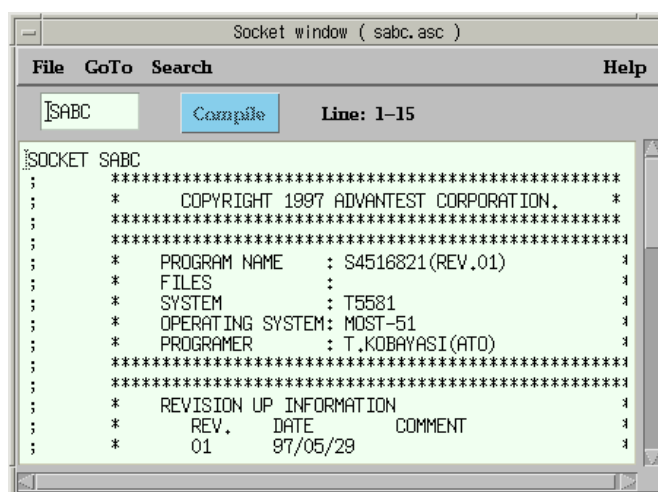
Displays the display range of the source program with line numbers.



### 3. 3. 7. 4 Socket Window

When you select Socket from the View menu, the sub-window shown in [Figure 3-20](#) is displayed. With this sub-window, socket files are edited and compiled.

**Figure 3-20 Socket Window**



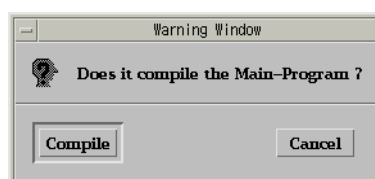
Menus are the same as those of the Pattern Window.

➔ For more information, refer to [3. 3. 7. 3 "Pattern Window"](#) on page 3-28.

However, the Auto ReLoading button is not provided.

When a socket file has been compiled normally, a confirmation message window asking you whether the main program should be recompiled is displayed (socket programs are not reflected unless the main program is recompiled).

**Figure 3-21 Recompile Confirmation Message Window**

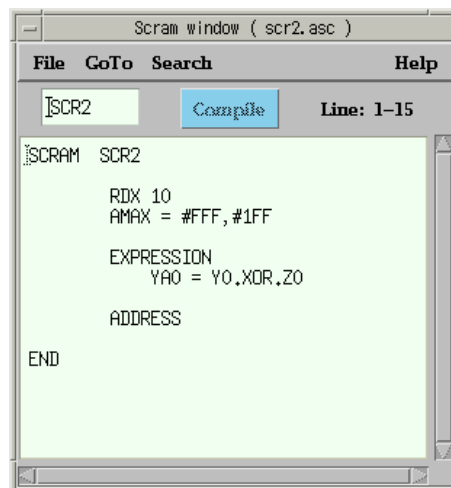


- Compile button  
Clicking the Compile button starts recompiling the main program.
- Cancel button  
Clicking the Cancel button cancels recompiling.

### 3. 3. 7. 5 Scram Window

When you select Scram from the View menu, the sub-window shown in [Figure 3-22](#) is displayed. With this sub-window, scramble files are edited and compiled.

**Figure 3-22 Scram Window**



Menus are the same as those of the Pattern Window.

➔ For more information, refer to [3. 3. 7. 3 "Pattern Window" on page 3-28](#).

However, the Auto ReLoading button is not provided.

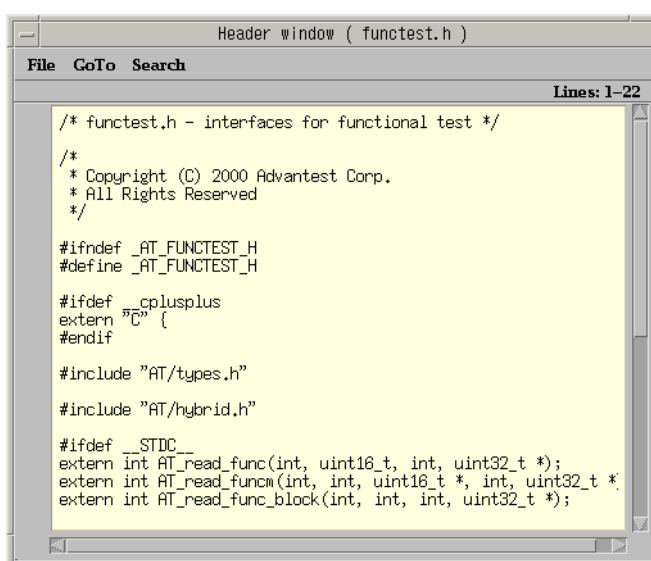
### 3. 3. 7. 6 Header Window

When the Header command is selected from the View menu, a sub-window, shown in [Figure 3-23](#), is displayed. The header file can be referred to and edited in the sub-window.

To edit data, switch to the Edit mode (click the Edit button in the main window).

Edited data is automatically saved when it is compiled (by clicking the Compile button in the main window).

**Figure 3-23 Header Window**



Menus are the same as those of the Pattern Window.

➔ For more information, refer to [3. 3. 7. 3 "Pattern Window"](#) on page 3-28.

### 3. 3. 8 Help Menu

The Help menu includes three commands described below.

#### ATL Statement...

Select an ATL statement and then click this menu. The Online Manual is displayed and shows how to use the statement.

#### Hybrid Function...

Select a Hybrid C function and then click this menu. The Online Manual is displayed and shows how to use the function.

#### Error Code...

Select an error code and then click this menu. The Online Manual is displayed and shows the error descriptions.

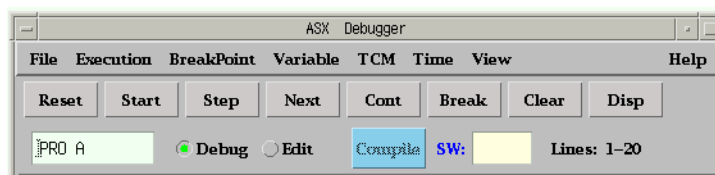
If Solaris 2.5.1 is used, the top page of the manual is displayed. If Solaris 7 or later is used, the selected item (statement, Hybrid C function, or error code) page is displayed.

➔ For more information, refer to [5. "Online Manual" on page 5-1](#).

### 3. 3. 9 Tool Bar

Buttons are provided for frequently used menu commands on the tool bar. Operation is the same as the corresponding menu commands. The tool bar can only be used in the Debug mode.

**Figure 3-24 Tool Bar**

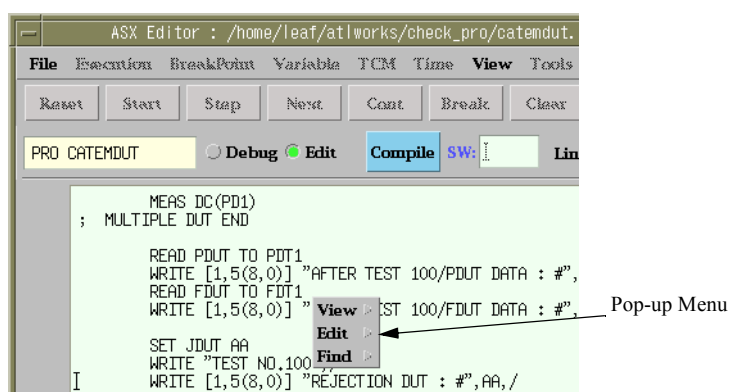


## 3. 4 Pop-Up Menu

When you press and hold the right mouse button on the text display area of the main window, the window shown in [Figure 3-25](#) is displayed.

To select items from the pop-up menu, press and hold the right mouse button, move the mouse pointer to the desired command, then release the button. If there is a sub-menu for the command, it opens automatically when you move the mouse pointer on the command. Move the mouse pointer to the desired command and release the button.

**Figure 3-25 Pop-up Menu**



### 3. 4. 1 View Menu

The View menu includes two commands described below.

#### Select Line at Number

Moves the insert cursor to the specified line, and displays it.

➔ For more information, refer to [3. 4. 1. 1 "Select Line at Number Window" on page 3-36](#).

#### What Line Number

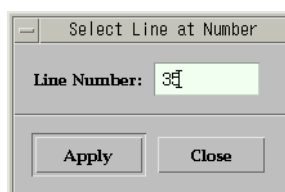
Displays the line number where the selected text begins.

➔ For more information, refer to [3. 4. 1. 2 "What Line Number Window" on page 3-36](#).

### 3. 4. 1. 1 Select Line at Number Window

When you select the Select Line at Number from the View menu, the sub-window shown in [Figure 3-26](#) is displayed. With this sub-window, the specified line is highlighted and then the insertion cursor is moved to the line.

**Figure 3-26 Select Line at Number Window**



Line Number

Specifies the line number to be searched for.

Apply button

Highlights the line specified by Line Number, and moves the insert cursor.

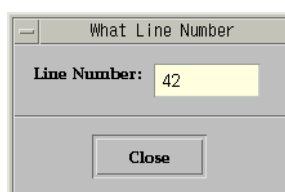
Close button

Closes the Select Line at Number Window.

### 3. 4. 1. 2 What Line Number Window

When you select What Line Number from the View menu, the sub-window shown in [Figure 3-27](#) is displayed. With this sub-window, the starting line number of the text selected in the text display section is displayed.

**Figure 3-27 What Line Number Window**



Line Number

Displays the starting line number of the text selected in the text display section.

Close button

Closes the What Line Number Window.

### 3. 4. 2 Edit Menu

The Edit menu includes four commands described below.

#### Undo

Cancels the edit operation last executed.

#### Copy

Copies the text selected in the text display section to the clipboard.

#### Paste

Pastes the contents of the clipboard to the insertion point in the text display section.

#### Cut

Deletes the text selected in the text display section and transfers it to the clipboard.

#### Tip

*The Edit menu is not available in the Debug mode.*

### 3. 4. 3 Find Menu

The Find menu includes two commands described below.

#### Find String

Searches for a specified character strings and replaces the found strings with other character strings.

➔ For more information, refer to [3. 4. 3. 1 "Search & Replace Window" on page 3-37](#).

#### Find Function

This function is used to specify the source program that you want to display, using the function name.

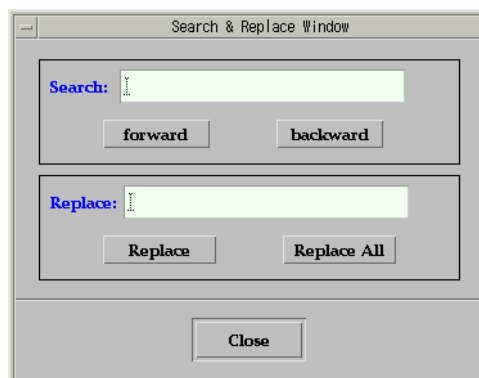
➔ For more information, refer to [3. 4. 3. 2 "Function View Window" on page 3-39](#).

#### Tip

*Find Function is not available in the Edit mode.  
It is available only when a C program is being debugged.*

### 3. 4. 3. 1 Search & Replace Window

When you select Find String from the Find menu, the sub-window shown in [Figure 3-28](#) is displayed. With this sub-window, the specified character string is searched for and highlighted. The searched character string can be replaced with another character string.

**Figure 3-28 Search & Replace Window****Tip**

*The Replace command is available only in the Edit mode.*

**Search text box**

Specifies the character string to be searched for.

**Forward button**

Searches for the character string specified in the Search text box in the forward direction (downward) and, if found, highlights it.

**Backward button**

Searches for the character string specified in the Search text box in the backward direction (upward) and, if found, highlights it.

**Replace text box**

Specifies a new character string to be replaced with.

**Replace button**

Replaces the character string found (highlighted) with the character string specified in the Replace text box.

**Replace All button**

Replaces all the character strings found (specified in the Search text box) with the character string specified in the Replace text box.

**Close button**

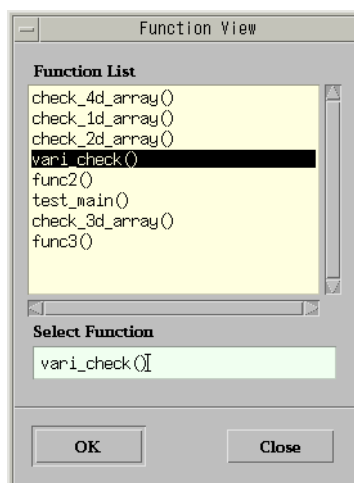
Closes the Search & Replace Window.



### 3. 4. 3. 2 Function View Window

When you select Find Function from the Find menu, the sub-window shown in [Figure 3-29](#) is displayed. The function specified in this sub-window is displayed in the main window.

**Figure 3-29 Function View Window**



#### Function List

The list of function names in the loaded program is displayed.  
However, static functions are not displayed.

#### Select Function text box

The function name to be displayed in the main window is specified from this Select Function text box.  
When you specify the function name in Function List, it is saved in this text box.  
With regard to static functions, directly enter function names.

#### OK button

This button is used to display the specified function in the main window and close the Function View window.

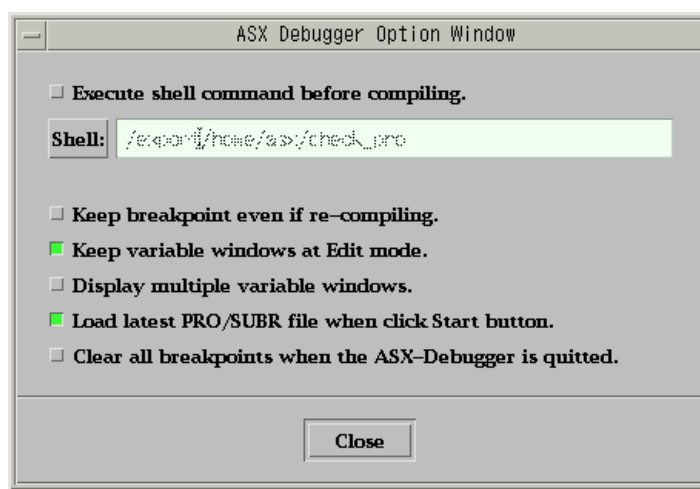
#### Close button

Closes the Function View Window.

## 3. 5 Option Window

When you select Option... from the File menu, the window shown in [Figure 3-30](#) is displayed. You can set each option function for the debugger from this window.

**Figure 3-30 Option Window**



**Execute shell command before compiling.**

When this button is ON, the shell program specified in the Shell text box is executed before the test plan program is compiled.

Clicking the Shell button displays the File Load Window, allowing you to specify a shell program using the mouse.

**Keep breakpoint even if re-compiling.**

When this button is ON, breakpoints are automatically reset to the same lines previously specified after the test plan program is compiled. Therefore, if some lines are shifted in the source program after it is compiled, you need to reset breakpoints afterward.

**Keep variable windows at Edit mode.**

When this button is ON, the Variable Window does not close, even if you change the debugger to the Debug mode.

**Display multiple variable windows.**

When this button is ON, multiple Variable Windows are displayed.

Up to five windows can be displayed.

**Load latest PRO/SUBR file when click start button.**

When this button is ON, the debugger checks if the current file is the most recent when you click the Start button. If the file has been updated, the most recent file is automatically loaded.

The target files are as follows:

ATL side: PRO, SUB, ASC  
C side: O

Clear all breakpoints when the ASXDebugger is quitted.

If this toggle button is selected, all breakpoints in the ATL program are deleted when the debugger is complete.

The breakpoints in the C program are deleted whether the toggle button is selected or not.

Close

Closes the Option Window.

### 3. 5. 1 Result from the Executed Shell

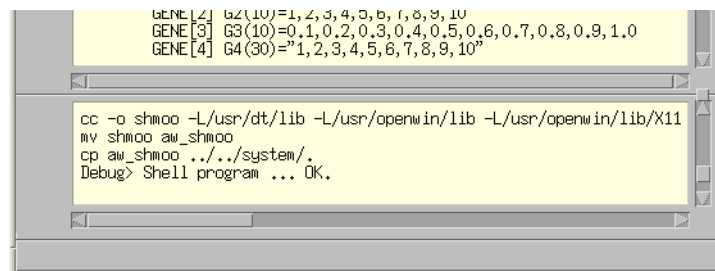
If the "Execute shell command before compiling." button is ON, the shell program (used to execute UNIX commands) specified in the Shell text box is executed before the test plan program is compiled.

The result from the executed shell program is displayed in the command display section.

When the result is Pass:

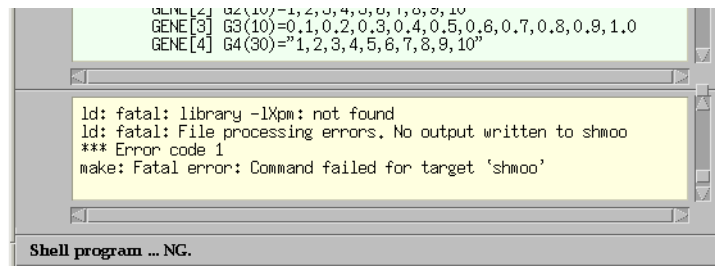
The message "Shell program ... OK." is displayed in the command display area, and the file being edited is compiled. The ASXDebugger goes into the Debug mode.

**Figure 3-31 Example of Pass**



When the result is Fail:

The message "Shell program ... NG." is displayed on the footer. The Edit mode of the ASXDebugger remains unchanged.

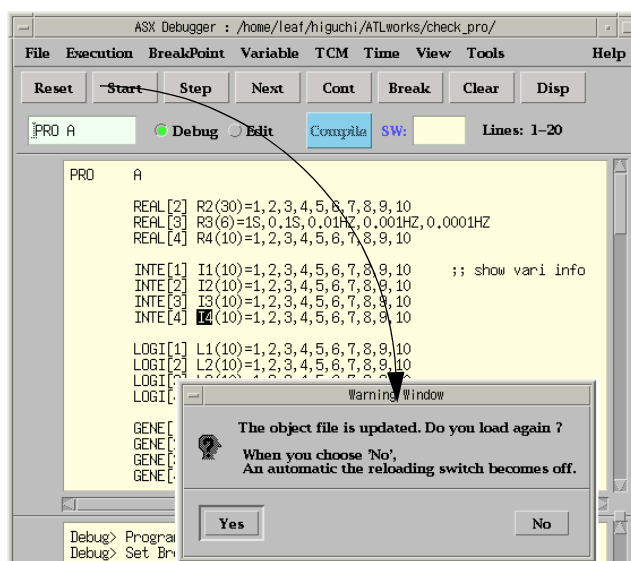
**Figure 3-32 Example of Fail****Tip**

*Set permission to shell program files.  
(% chmod +xshell program file name)*

### 3. 5. 2 Automatic Update of Files

If the "Load latest PRO/SUBR file when click start button." toggle button is ON, the debugger checks if the current file is the most recent one when you click the Start button. If the file has been updated, a dialog box appears asking if you want to load the most recent file.

**Figure 3-33 File Update Dialogue**



#### Yes button

- When the object file has been updated:  
Both source and object files are updated.
- When the source file only has been updated:  
To create an object file, ASXDebugger loads the most recent source file and compiles it in the Edit mode. ASXDebugger then goes into the Debug mode.  
Following that, if the "Keep breakpoint even if re-compiling." button is ON, breakpoints are reset.

#### No button

The confirmation dialog box is closed and the current displayed file is used.

The "Load latest PRO/SUBR file when click start button." button on the Option Window turns off.

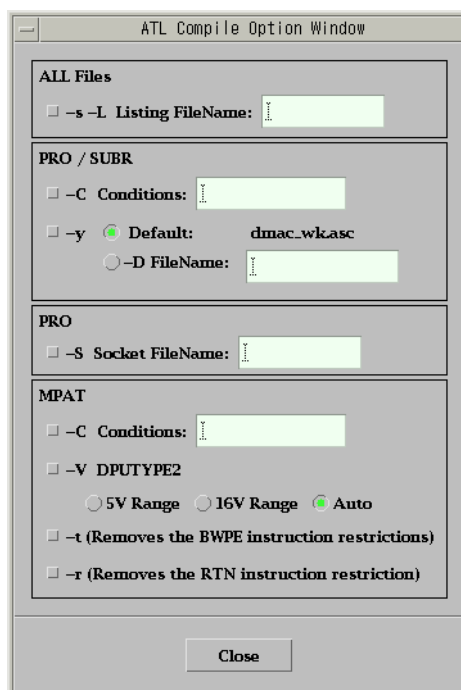
#### Tip

*The PRO and SUBR files, along with their source files (including the insert files), are updated. The pattern, socket, and scramble files are not updated. For C programs, only .O files are updated.*

## 3. 6 ATL Compile Option Window

If you select the ATL Compiler Option... command from the File menu, the ATL Compile Option Window shown below is displayed. You can set each option function for the compiler from this window.

**Figure 3-34 ATL Compile Option Window**



### ALL Files

Indicates that the compiler option is enabled for all attributes of a file when checked.

- -s -L Listing FileName

If you select the -s-L check box, the listing file containing the source program which uses the file name specified in the Listing FileName text box is displayed.

### PRO/SUBR

Indicates that the compiler option is enabled for the PRO/SUBR file when checked.

- - C Condition

If you select the -C check box, you can specify the compilation conditions.

Specify the necessary alphabet in the Condition text box as the compilation condition.

This setting is reflected on the switch of the main window SW text box. If you deselect the -C check box, the switch of SW text box is cleared.

- - y - D FileName

If you select the -y check box, a source file which is expanded from the DMACRO statement is created and then compiled. The default file name is set to dmac\_wk.asc.

If the -D check box is subsequently selected, the source file is created using the file name specified in the filename text box.

When compilation has been completed, the -y check box is deselected.

## PRO

Indicates that the compiler option is enabled for PRO file when checked.

- - S Socket FileName

If the -S check box is selected, you can specify the socket file to be referred to during the compilation.

If the specified socket file does not exist in the current directory, an error message is displayed and the text box data is cleared.

## MPAT

Indicates that the compiler option is enabled for the MPAT file when checked.

- - C Condition

If you select the -C check box, you can specify the compilation conditions.

Specify the necessary alphabet in the Condition text box as the compilation condition.

This setting is reflected on the switch of the pattern window SW text box. If you deselect the -C check box, the SW text box switch of the pattern window is cleared.

- - V DPUTYPE2

If you select the -V check box, you can specify the DPUTYPE2 range in the pattern program.

|           |                                         |
|-----------|-----------------------------------------|
| 5V Range  | DPUTYPE2 5V range is specified          |
| 16V Range | DPUTYPE2 16V range is specified         |
| Auto      | DPUTYPE2 range specification is omitted |

- -t (Removes the BWPE instruction restrictions)

If you select the -t check box, you can remove the following restrictions for the pattern program:

If you specify BDW, BCW or BMW for the interleaved mode, the BDW, BCW or BMW instruction cannot be specified using a separate specification (the instruction cannot be divided using a colon). If you specify BDW, BCW or BMW for the interleave mode, the data generation instruction, address generation instruction, X<RF instruction, and DSET instruction cannot be described in the same step.

- -r (Removes the RTN instruction restriction)

If you select the -r button, you can remove the following restrictions for the pattern program:

In the RTN instruction step, register operation instructions other than RF<RF+1 instruction cannot be specified.

➔ For more information on each function, refer to [ATL-51 Pattern Program Reference Manual\(No. : 8256292\)](#).

## 3. 7 C Compile Option Window

When you select Hybrid C Compile Option... in the File menu, the C Compile Option Window shown below is displayed. In this window, you can specify the makefile that is executed when the Compile button is clicked, or the makefile argument.

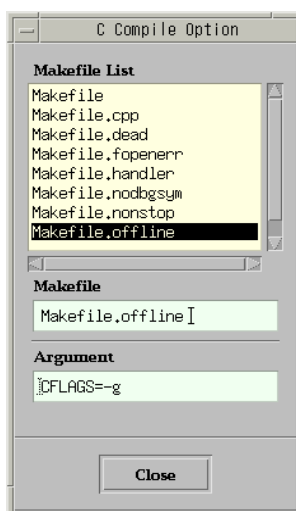
The target makefile to be executed is listed in the Makefile text box.

The target makefile argument is a character string displayed in the Argument text box.

When you specify Makefile text box→Makefile.offline, Argument text box→CFLAG=-g, the following command is executed:

```
% make -f Makefile.offline CFLAGS=-g
```

**Figure 3-35 C Compile Option Window**



### Makefile List

The makefiles in the current directory are displayed in the Makefile List.

### Makefile text box

The makefile to be executed is specified in the Makefile text box.

When you select a file in the Makefile List, it is saved in this text box.

You can specify another file name by entering the file name into this text box directly.

### Argument text box

The argument for the makefile is specified in this Target text box.

### Close

Closes the C Compile Option Window.



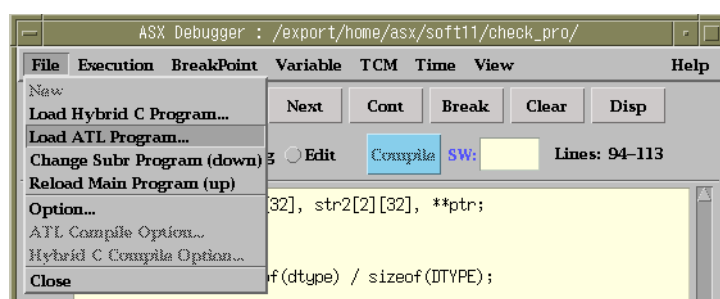
## 3. 8 Example of Operation

### 3. 8. 1 Loading a Program to Be Debugged

#### 1) Opening the File Load Window.

With the Edit mode, select the Load ATL Program... command from the File menu in the menu bar. The File Load Window opens.

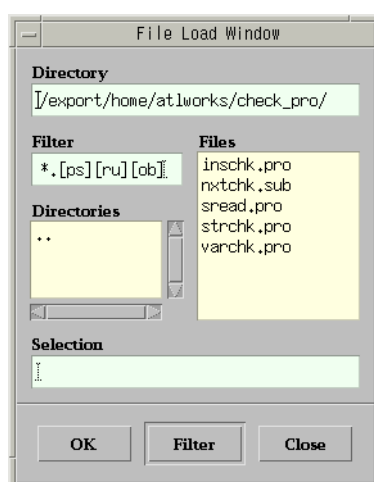
**Figure 3-36 File Menu Loading**



#### 2) Opening a source file.

Select the source file of the program to be debugged from the Directories list and Files list in the File Load Window and then click the OK button.

**Figure 3-37 File Load Window**



#### 3) Compiling a source file.

Click the Compile button in the menu bar to compile the source file. When compilation successfully completed, the object file name is displayed at Object box in the main window and the editing mode changes to Debug.

If compilation failed, a compilation error is displayed in the command display section of the main window.

#### Tip

*If an object file exists, load the file in the Debug mode.*

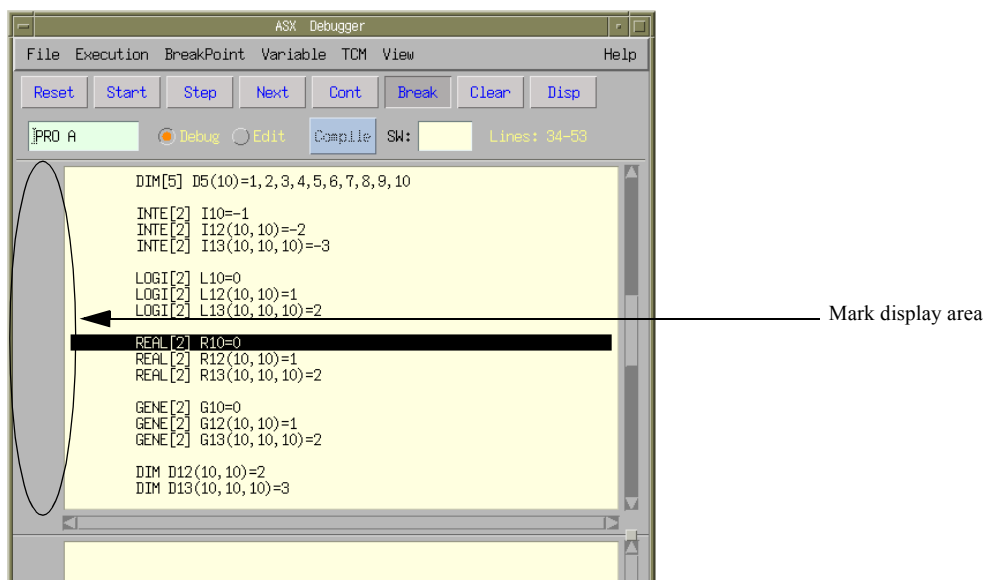
## 3. 8. 2 Setting and Canceling a Breakpoint

Setting a breakpoint

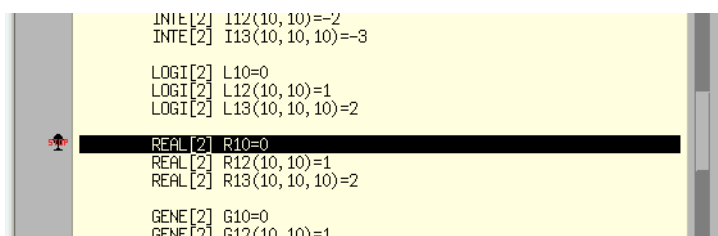
- 1) Select the line where you want to set a breakpoint.
- 2) Click the Break button in the tool bar to set a breakpoint.

You can also set a breakpoint by clicking the left mouse button on the mark display section.

**Figure 3-38 Setting a Breakpoint**



When you set a breakpoint, the breakpoint mark is displayed to the left of the specified line as shown in [Figure 3-39](#).

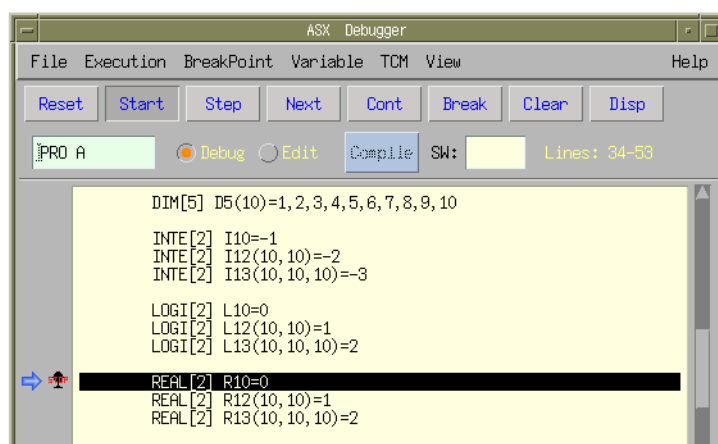
**Figure 3-39 After Setting a Breakpoint****Canceling a breakpoint**

- 1) Select the line at which you want to cancel the breakpoint.
- 2) Click the Clear button in the tool bar to cancel the breakpoint.

You can also cancel a breakpoint by clicking the left mouse button on the breakpoint mark.

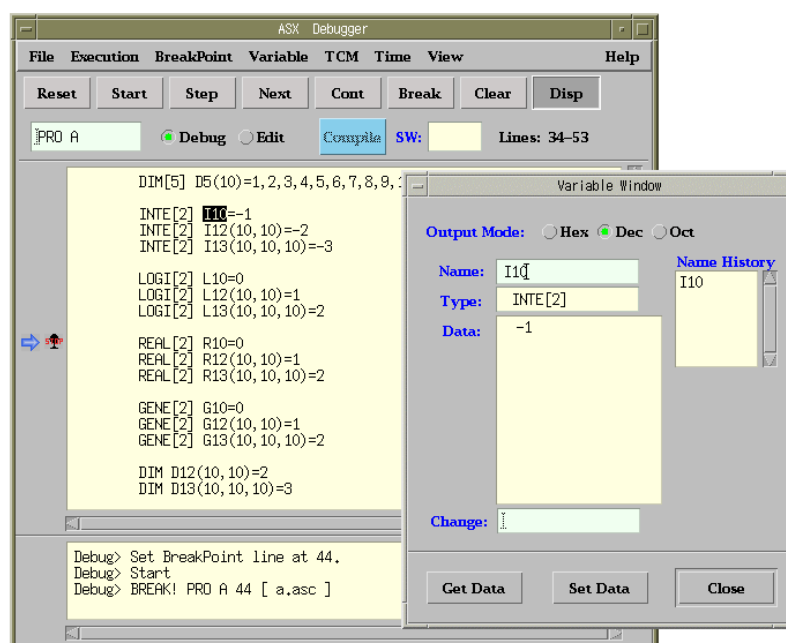
**3. 8. 3 Program Execution Referring to Variables**

- 1) Click the Start button in the tool bar to execute the program.
- 2) When the program is suspended at the breakpoint, an arrow is displayed as in [Figure 3-40](#).

**Figure 3-40 Executing the Program and Stopping at the Breakpoint**

- 3) Select a variable you want to refer and then click the Display button in the tool bar.

Figure 3-41 Referring to Variables



### 3. 8. 4 Partial Reference to Array Variables

To display some of the array variable elements, specify data as shown below in the Name text box.

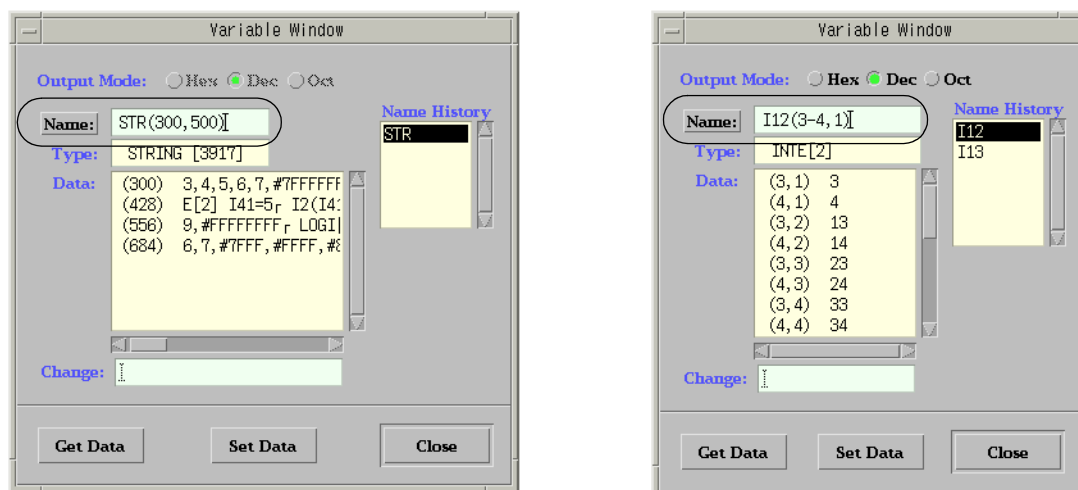
- Specifying the start element
    - For one-dimensional array V (I1, n) (Including string variables)
    - For two-dimensional array V (I1, J1, n)
    - For three-dimensional array V (I1, J1, K1, n)
  - Specifying the region of elements
    - For one-dimensional array V (I1-I2, n)
    - For two-dimensional array V (I1-I2, J1-J2, n)
    - For three-dimensional array V (I1-I2, J1-J2, K1-K2, n)
- V: Variable names. I1, J1, K1: First element numbers. I2, J2, K2: Last element numbers. n: Number of elements.  
 (Note that  $I1 \leq I2$ ,  $J1 \leq J2$ ,  $K1 \leq K2$ ,  $I1, I2, J1, J2, K1, K2, n > 0$ )

To display 500 elements starting from the 300th element of the string variable STR, enter STR (300, 500) into the Name text box.

For two-dimensional and three-dimensional variables, the third and fourth value specifies the number of elements, respectively. If the number of elements is not specified, all elements subsequent to the start element are displayed.

To display all elements of two-dimensional variable I12 whose first argument is 3 or 4, enter I12 (3-4, 1) into the Name text box (By connecting two numbers with -, the display section can be specified).

**Figure 3-42 Partial Specification of Array Variables**



**Tip**

If there is an error in partial specification, it is regarded as "no specification," and all elements are displayed.

(When numeric values or characters other than "-" and "," are used.)

### 3. 8. 5 Debugging a Hybrid C Program

The procedure for debugging a hybrid C program is described below:

- 1) Compile the C source program again with the -g option.
- 2) Execute /HYBRID/S on the test controller screen.
- 3) When the program starts, it is suspended at the beginning of the test\_main() function. The following message is displayed on the test controller screen:

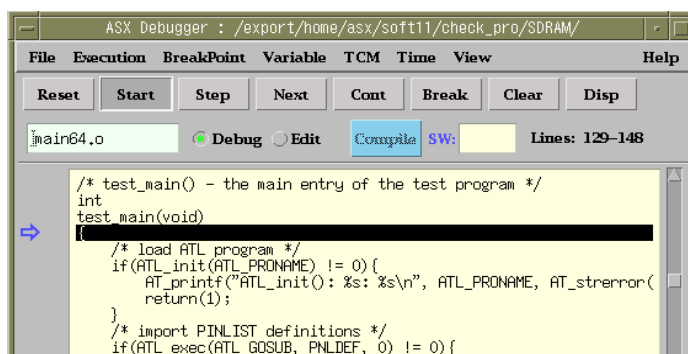
```
> H
Break at 0xdfffe58 : test_main()    task : tUsrMain
```

- 4) Click the Debugger button in the toolbox to activate the debugger.

When the debugger starts, an arrow is displayed that indicates the program is suspended at the beginning of the test\_main() function.

If the Continue button is clicked, then the program execution re-starts.

**Figure 3-43 Debugging a C Program**



- 5) Setting a breakpoint

Select a line then click the Break button to set the breakpoint such as debugging an ATL program.

➔ For more information, refer to [3. 8. 2 "Setting and Canceling a Breakpoint" on page 3-48](#).

 **Tip**

*Breakpoints in a C program can be set after stopping the program by test\_main().*

If the breakpoint setting function is not being displayed in the main window, you can set the breakpoint by loading a source program containing the target function.

You can also display the target function in Find View Window.

#### 6) Referring to variables

Select a variable then click the Disp button to refer to the variable (same as the procedure for debugging an ATL program).

➔ For more information, refer to [3. 8. 3 "Program ExecutionReferring to Variables" on page 3-49](#).

#### 7) Debugging an ATL program called from a C program

There are two ways, as follows:

- A way to set the breakpoints in the ATL program in advance.

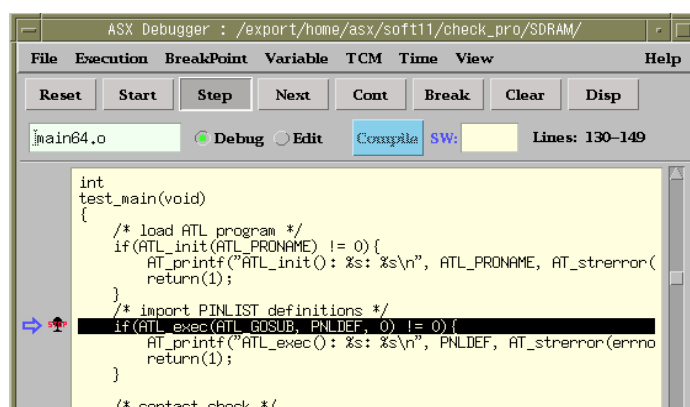
While the test controller is in a Ready state, select Load ATL Program... in the File menu and then load an ATL program in the main window to set the breakpoints.

When the program starts, it is suspended at the specified breakpoint in the ATL program.

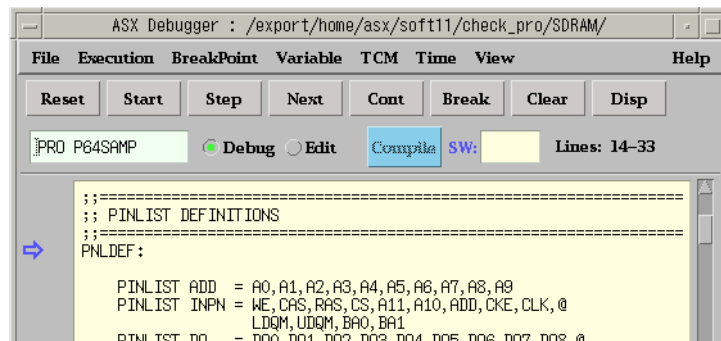
- A way to transfer from the C program to the ATL program by Step execution

While the program is being suspended at the beginning of the test\_main() function after the program starts, set the breakpoint to a line in the function (ATL\_exec) which calls the ATL program then click the Cont button. If the program is suspended at the ATL\_exec function, then click the Step button.

**Figure 3-44 Suspended in the C Program**



An ATL program to be called is displayed in the main window and an arrow is displayed at a line where the program is suspended. In this situation, The ATL program can be debugged.

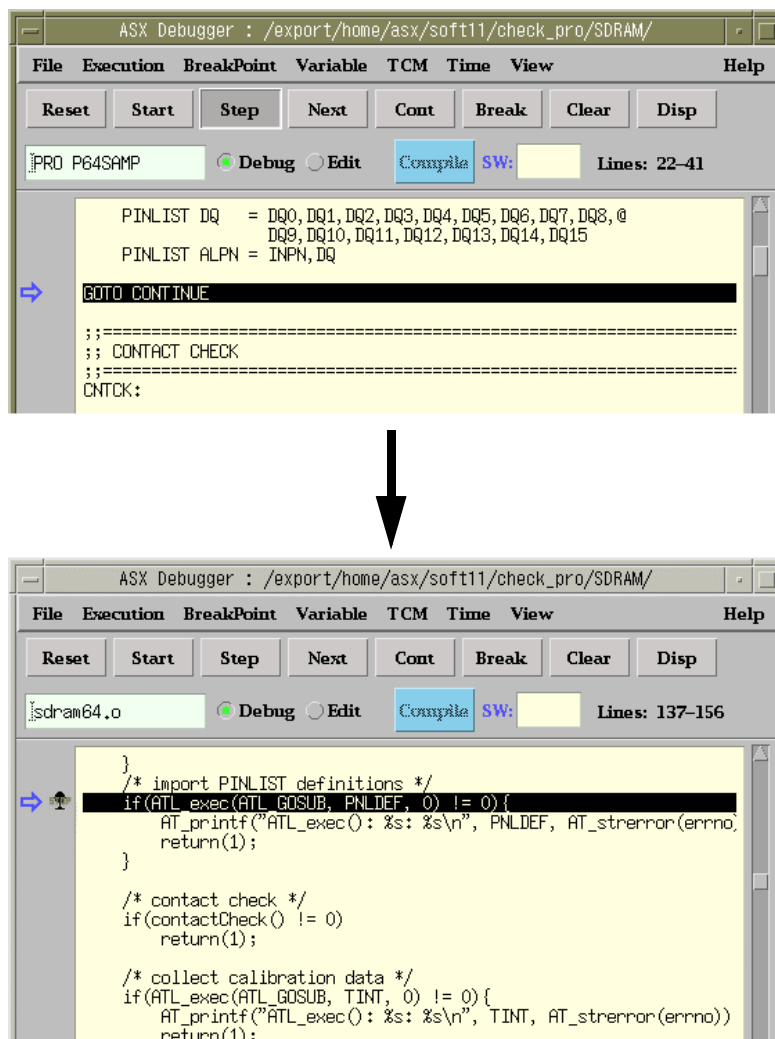
**Figure 3-45 Transfer to the ATL Program from the C Program**

To return to the C program, click the Step button at either the GOTO CONTINUE or RETURN C statements.

The program returns to ATL\_exec.



Figure 3-46 Return to the C Program from the ATL Program



#### Tip

When a breakpoint is set and Next is executed at the ATL\_exec line in the ATL program, ATL\_exec halts in the ATL program before proceeding to the next line in the C program.

In this situation, executing Step at either the GOTO CONTINUE or RETURN statements does not return to the ATL\_exec line. Next is executed and the program halts at the next line of ATL\_exec.

Executing Continue instead of Step also halts the program at the next line of ATL\_exec.

## 3. 9 Error Message List

{Keyword} not found.  
Cannot find the keyword.

File open error.  
An error occurred during file open operation.

File read error.  
An error occurred during file reading.

File write error.  
An error occurred during file writing.

File name error.  
The attribute of the specified file is inappropriate.

Memory allocate error.  
A memory allocation error occurred.

Break-point is multiple defined.  
Breakpoint has already been set at the specified line.

Break-point cannot be set on line.  
Cannot set breakpoint at the specified line.

Break-point buffer overflow.  
The number of breakpoint settings exceeds the limit (128).

Please select a {Keyword}.  
Select a keyword.

Tester offline! cannot change mode.  
In the off-line mode, cannot select the debug mode.

During Testing! cannot change mode.

Cannot change the mode during testing.

During Testing! cannot load file.  
Cannot load a file during testing.

DP0 file cannot be executed.  
Cannot execute the DP0 file.

Directory Name get error.  
Could not get the directory name.

Change directory error.  
Could not move the directory.

Variable data read error.  
Could not refer variable data.

Insert Window Overflow. (Max5)  
The number of Insert Windows exceeds the limit (5).

File get error.  
Could not load the file.

Peer error.  
A data communication error occurred.

Continue count error.  
The retry number is incorrect.

Illegal line number.  
The line number is inappropriate.

Illegal variable.  
The variable name is inappropriate.

Illegal change data.  
The modified data is inappropriate.

The search string not input.

The character string to be searched for has not been entered.

The selection string is not match.

The character string to be replaced with is missing.

Output source filename error. (DMACRO expanded)

The name of source file to be output is inappropriate.

(A same file name is used for both the file that are being used in the DMACRO statement and the source file to be compiled.)

Debug file not found.

The debug file is not found.

Cancel RON/ROF Monitor.

Stops monitoring RON or ROF.

During C Debugging! cannot execute.

While the C program is debugged, the program cannot be executed.

During ATL Debugging! cannot execute.

While the ATL program is debugged, the program cannot be executed.

No debugging symbols found.

The loaded C object has not been compiled with -g.

Object module contains undefined symbols.

The loaded C object contains undefined symbols that are not linked.

No source file named ???.

The loaded C object does not contain the target source file.

Program is not running.

The program has not been executed.

Stopping at the test\_main().

The program has already been suspended at the beginning of the test\_main().

Makefile not found.

There is no Makefile in the current directory.

## 4. Test Condition Monitor

Test Condition Monitor (TCM) is used to refer and modify the tester condition. You can save the current tester settings to a text file.

### Restrictions

- Each TCM monitor cannot operate correctly if the type of tester is changed using /CONFIG while ATLworks is operating on TESIM. If you change the tester type, the warning message dialog box shown below is displayed. Terminate TCM by clicking the Quit TCM button, and restart it.

**Figure 4-1 Warning Message**



- While Pin Clock Monitor is being refreshed (that is, a process during which data is loaded from the tester and displayed on the monitor), the window shown below is displayed. Note that if you perform other operations while refreshing, the system may not operate correctly. Please wait to perform other operations until this window disappears.

**Figure 4-2 Refresh Message**



- The window shown below is displayed while the tester settings are saved to a text file. Note that if you perform other operations while data is being saved, the system may not operate correctly. Please wait to perform other operations until this window disappears.

Figure 4-3 Save Message



- TCM cannot be used with the T5336 or T5311.
- A cross compiler is required to change setting conditions of the tester. The compiler type depends on the test system as shown below.

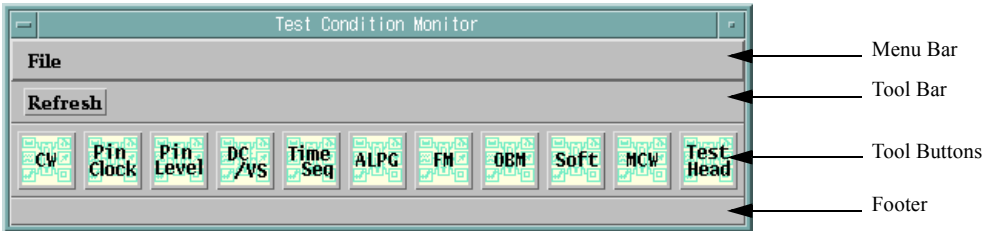
| Test System Model              | Cross Compiler |
|--------------------------------|----------------|
| T5500 series                   | PXTL51-00S     |
| T5377, T5376, T5375, and T5371 | PXTL51-36S     |

- If the Online Manual (which can be started from the Help menu on each monitor) is used in Solaris 2.5.1, the top page of the manual is displayed. If Solaris 7 or later is used, the page on which the selected statement is written is displayed.
- ➔ For more information, refer to [5. "Online Manual" on page 5-1](#).

4. 1 TCM Main Window

The TCM main window is displayed when you click the Test Condition Monitor button in ToolBox of ATLworks.

Figure 4-4 TCM Main Window



- Menu Bar
- The Menu Bar has the File menu. The File menu includes the following menu commands:
- Save...  
Saves setting conditions of the tester in a text file.
- ➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).

- Close

Terminates the TCM main window and all the monitors which were activated using the tool buttons.

#### Tool Bar

The tool bar includes the Refresh button. When you click this button, all the monitors which were activated using the tool buttons are redrawn. (With Refresh, the current tester setup condition is loaded and then the display contents of each monitor are updated.)

#### Tool Buttons

Clicking a button starts the corresponding monitor.

Shaded buttons are disabled for use.

#### Footer

If a communication error occurs between the tester and monitor, an error message is displayed in the footer with a beep sound.

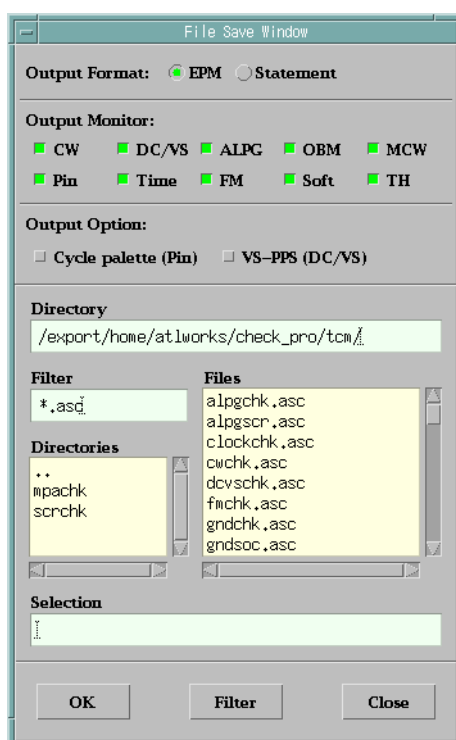


### 4. 1. 1 File Save Window

When you select Save... from the File menu of the TCM main window or on each monitor, the File Save Window shown below appears.

When you specify a file name and click the OK button, test conditions of the tester are saved to a text file.

**Figure 4-5 File Save Window**



#### Tip

- Save data while the program is stopped at a breakpoint, etc.  
If data is saved while the program is running, settings for each monitor may differ from the saved data.
- The current test condition of the tester is saved as is. Therefore, the system operation may differ depending on the execution order of statements.

#### Output Format

Selects a file format.

- EPM  
Saves data in the same format as the pin monitor (EPM).
- Statement  
Saves data as statements that can be compiled using XATL.

In this case, insert "PRO<file name><socket name>" at the beginning of the file, and "END" at the end of the file.

### Output Monitor

Selects monitors from which you want to save data.

Data for the monitor whose toggle button is set to ON is saved.

Note that if Statement is specified for Output Format, you cannot select the Soft and the TH monitor.

### Output Option

- Cycle Palette (Pin)

If you select the Cycle palette (Pin) toggle button, all cycle palette data is saved. Note that this can be selected only when all of the following conditions are satisfied:

- EPM is selected for Output Format.
- Pin is selected for Output Monitor.
- The T5593, T5592, T5377, T5376, T5375, or T5371 test system is used.

- VS-PPS (DC/VS)

If you select the VS-PPS (DC/VS) toggle button, each PPS setting used for the VS is saved.

Note that this can be selected only when all of the following conditions are satisfied:

- EPM is selected for Output Format.
- DC/VS is selected for Output Monitor.

### Selection

Specifies the file name where test conditions of the tester are saved.

Specify the file name with 36 characters or less, including the file extension .asc.

If you do not include .asc in the file name, it is automatically added.

### OK button

Saves test conditions of the tester to the file specified using the Selection text box.

If you specify a previously saved file, the dialog box shown below appears.

**Figure 4-6 Confirmation Message**



- Overwrite button

Overwrites the specified file.

- Append button

Appends data to the specified file.

Note that if statement is specified for Output Format, "PRO<file name><socket name>" and "END" are not added.

- **Cancel button**  
Cancels saving data.

**Filter button**

Extracts files from the selected directory according to the filter setting and displays them in the Files list.

**Close button**

Closes the File Save Window.

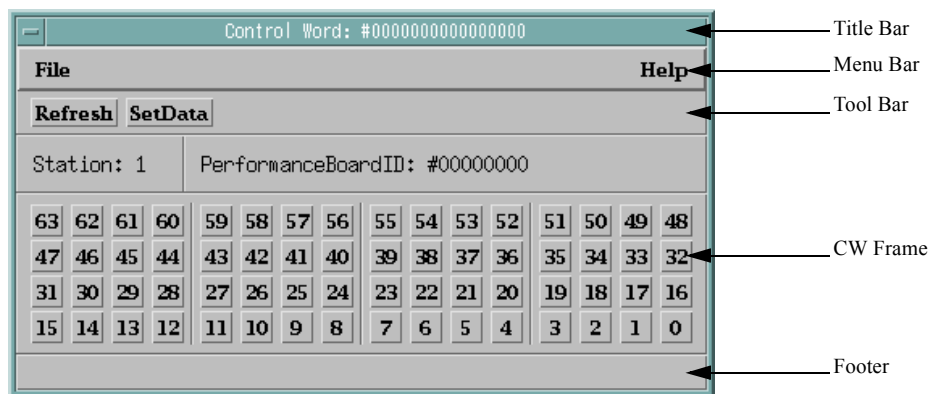
## 4. 2 Control Word Monitor

The Control Word monitor (CW) is used to refer and change the tester Control Word setup condition.

### 4. 2. 1 Main Window

When you click the CW button of the monitor button in the TCM main window, [Figure 4-7](#) is displayed.

**Figure 4-7 Control Word Monitor (Screen of T5581)**



#### Tip

32 Buttons from 32 to 63 cannot be used for the T5592, T5591, T5581P, or T5571P.

#### Title Bar

In the title bar, the button condition for the current CW frame is displayed with a hexadecimal number.

#### Menu Bar

The menu bar consists of File and Help menus.

##### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the Control Word monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the Control Word monitor.

##### Help Menu

The help menu is used to display the Online Manual for the Control Word monitor.

The Help menu is a pull-down menu consisting of the following commands:

- Control Word...

### Tool Bar

The tool bar includes the following buttons:

- Refresh button

When you click the Refresh button, the current tester setup condition is loaded and the contents of the Control Word Monitor display are updated.

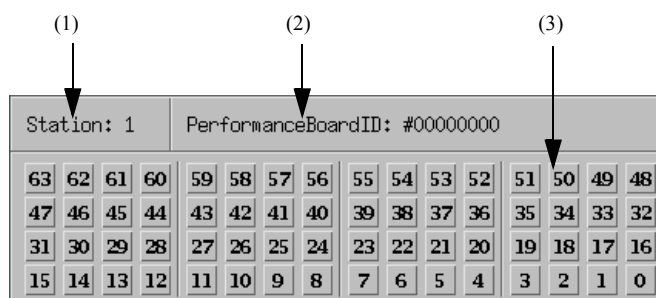
- SetData button

The modifications made with the CW frame are set to the tester.

### CW Frame

The CW frame is shown in [Figure 4-8](#).

**Figure 4-8 CW Frame**



1. Displays the station number (decimal notation) currently being used.  
For two station parallel testing, "1, 2" is displayed.
2. Displays the ID number (hexadecimal notation) of the currently mounted performance board.  
For two station parallel testing, "#00000000, #00000000" is displayed.  
If you uses the T5593, T5586, T5585, T5377, T5376, or T5375, "PerformanceBoardID1" and "PerformanceBoardID2" are displayed.
3. Buttons from 0 to 63 corresponding to each bit of Control Word of the tester are prepared.  
Each button has two statuses: bit 1 and bit 0.  
When the bit is 1, the corresponding button is pressed.  
When the bit is 0, the corresponding button is released.

### Footer

When you click the SetData button, if the modifications made with the CW frame cannot be set to the tester, an error message is displayed in the footer with a beep sound.

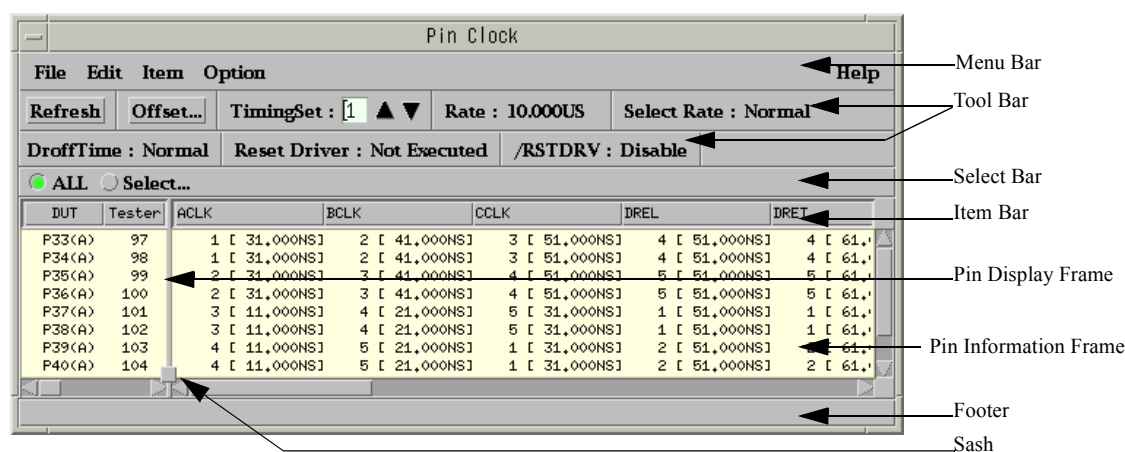
4. 3 Pin Clock Monitor

With the Pin Clock monitor, you can display and set the information on the pin clock set on the tester.

4. 3. 1 Main Window

When you click the Pin Clock button in the TCM main window, [Figure 4-9](#) is displayed.

Figure 4-9 Pin Clock Monitor (Activated)



**Menu Bar**  
The menu bar consists of File, Edit, Item, Option and Help menus.

- File Menu**  
The File menu includes the following menu commands:
- Save...  
Saves data from the Pin Clock monitor and Pin Level monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
  - Close  
Terminates.

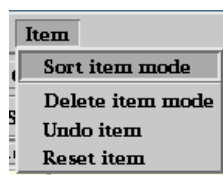
**Edit Menu**  
When you select the Edit menu in the menu bar, a pull-down menu opens showing four commands as in [Figure 4-10](#).

**Figure 4-10 Edit Menu**

- Select Pin...  
When you select the Select Pin... command, the Select Pin window is displayed.  
→ For more information, refer to [4. 3. 4 "Select Pin Window" on page 4-25](#).
- Select Offset...  
When you select the Select Offset... command, the Select Offset window is displayed.  
→ For more information, refer to [4. 3. 6 "Select Offset Window" on page 4-29](#).
- Setting Pin Condition ...  
When you select the Setting Pin Condition... command, the Setting Pin Condition window is displayed.  
→ For more information, refer to [4. 3. 8 "Setting Pin Condition Window" on page 4-31](#).
- Setting Pin Clock...  
When you select the Setting Pin Clock... command, the Setting Pin Clock window is displayed.  
→ For more information, refer to [4. 3. 5 "Setting Pin Clock Window" on page 4-28](#).

**Item Menu**

When you select the Item menu in the menu bar, a pull-down menu opens showing four commands as in [Figure 4-11](#).

**Figure 4-11 Item Menu**

- Sort item mode  
When you select the Sort item mode, you can sort items clicked in the item bar.  
→ For more information, refer to [4. 3. 2 "Sort Item Mode" on page 4-22](#).
- Delete item mode  
When you select the Delete item mode, you can delete items clicked in the item bar.  
→ For more information, refer to [4. 3. 3 "Delete Item Mode" on page 4-24](#).
- Undo item

The Undo item command can redisplay the items you deleted from the list with Delete in reverse order of deleting. (If all items are displayed, the Undo item command is disabled.)

- Reset item

The Reset item command can redisplay all the items you deleted into the item bar. (If all items are displayed, the Reset item command is disabled.)

You can select either Sort item mode or Delete item mode. When the current mode is selected again, the mode is turned off and the selections in the item bar are canceled. Neither mode is selected by default.

### Option menu

The Option menu includes the following commands:

- Cycle Palette Data ...

Displays all cycle palette data for the pins selected in the pin display frame or pin information frame.

➔ For more information, refer to [4. 3. 7 "Cycle Palette Data Window" on page 4-30](#).

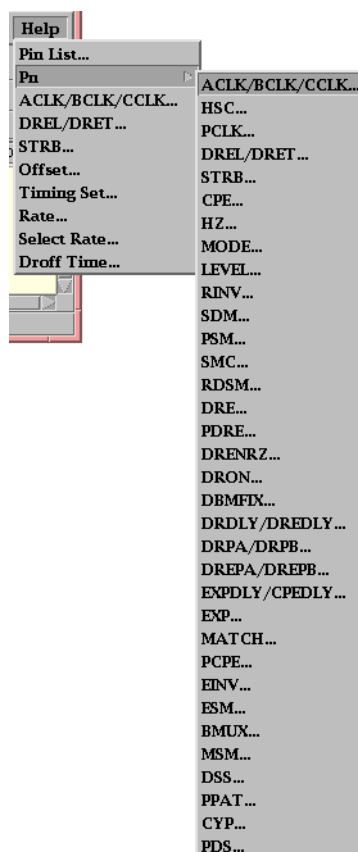
Note that this command can only be used for the T5593, T5592, T5377, T5376, T5375, and T5371.

### Help Menu

When you select the Help menu in the menu bar, a pull-down menu opens showing ten commands as in [Figure 4-12](#). Items from (a.) to (ag.) related to pin connecting conditions are displayed in the Pn menu. When you select each item, the related Online Manual is displayed.



Figure 4-12 Help Menu



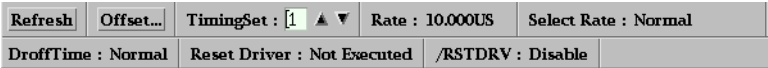
1. Pin List...
2. Pn...
  - a. ACLK/BCLK/CCLK...
  - b. HSC...
  - c. PCLK...
  - d. DREL/DRET...
  - e. STRB...
  - f. CPE...
  - g. HZ...
  - h. MODE...
  - i. LEVEL...
  - j. RINV...
  - k. SDM...
  - l. PSM...
  - m. SMC...

- n. RDSM...
- o. DRE...
- p. PDRE...
- q. DRENZR...
- r. DRON...
- s. DBMFX...
- t. DRDLY/DREDLY...
- u. DRPA/DRPB...
- v. DREPA/DREPB...
- w. EXPDLY/CPEDLY...
- x. EXP...
- y. MATCH...
- z. PCPE...
- aa.EINV...
- ab.ESM...
- ac.BMUX...
- ad.DSS...
- ae.PPAT...
- af.CYP...
- ag.PDS...
- 3. ACLK/BCLK/CCLK...
- 4. DREL/DRET...
- 5. STRB...
- 6. Offset...
- 7. Timing Set...
- 8. Rate...
- 9. Select Rate...
- 10.Droff Time...

Tool Bar

The tool bar of the PinClock monitor is shown in [Figure 4-13](#).

**Figure 4-13 PinClock Tool Bar**



**Refresh**

When you click the Refresh button, the current tester setup condition is loaded and the contents of the Pin Clock Monitor display are updated.

**Offset...**

The Select Offset window is displayed.

➔ For more information, refer to [4. 3. 6 "Select Offset Window" on page 4-29](#).

**TimingSet**

Displays the timing set number. Enter the Timing Set value (1 to 16) directly or change the value by clicking the up or down arrow on the right side.

**RATE**

Displays the test rate setting.

**Select Rate (Displayed for other than the T5593, T5592, T5377, T5376, T5375, and T5371.)**

Displays the mode which is specified by the SELECT RATE statement (Normal, Fix or High Resolution).

**Droff Time**

Displays the setting of the minimum off time for the driver (Normal or HS).

**Reset Driver**

Displays either "Executed" or "Not Executed," which indicates whether or not the RESET DRIVER statement (used to initialize driver levels) has been executed.

**/RSTDRV**

Displays either "Enable" or "Disable," which indicates whether or not driver output levels are automatically initialized.

**Select Bar**

The select bar of PinClock monitor is shown in [Figure 4-14](#).

**Figure 4-14 PinClock Select Bar**

- ALL  
Displays all pins currently being used in the pin display frame.
- Select...  
The Select Pin window is displayed.

➔ For more information, refer to [4. 3. 4 "Select Pin Window" on page 4-25](#).

**Item Bar**

The item bar of the PinClock monitor is shown in [Figure 4-15](#).

**Figure 4-15 PinClock Item Bar**

| DUT | Tester | ACLK | BCLK | CCLK | DREL | DRET |
|-----|--------|------|------|------|------|------|
|-----|--------|------|------|------|------|------|

The item bar is displayed at the top of the pin display frame and pin information frame, listing the item names.

### Pin Display Frame

The pin display frame is shown in [Figure 4-16](#).

**Figure 4-16 Pin Display Frame**

| DUT    | Tester | PinList     | Signal |
|--------|--------|-------------|--------|
| P33(A) | 97     | IOPIN_GROUP | PI033  |
| P34(A) | 98     | IOPIN_GROUP | PI034  |
| P35(A) | 99     | IOPIN_GROUP | PI035  |
| P36(A) | 100    | IOPIN_GROUP | PI036  |
| P37(A) | 101    | IOPIN_GROUP | PI037  |
| P38(A) | 102    | IOPIN_GROUP | PI038  |
| P39(A) | 103    | IOPIN_GROUP | PI039  |
| P40(A) | 104    | IOPIN_GROUP | PI040  |

If a socket file is specified in the test program, you can assign DUT pin numbers and DUT pin names to tester pins. DUT pin numbers and DUT pin names are displayed in the DUT and Signal columns in the pin display frame, respectively. When the socket file is not specified, nothing is displayed in the Signal column. DUT pin numbers are followed by a bank number, which will be (A) if setting conditions are the same.

- DUT  
Displays DUT pin numbers. They can be sorted in ascending order.
- Tester  
Displays tester pin numbers. They can be sorted in ascending order.
- PinList  
Multiple names can be assigned to a certain channel, so the first pin list name you defined in the program is displayed in the PinList column. Pin list names can be sorted in alphabetical order.
- Signal  
Displays DUT pin names (pin signal names). They can be sorted in alphabetical order and then in ascending numeric order.

### Pin Information Frame

The pin information frames of the PinClock monitor are shown in [Figure 4-17](#), [Figure 4-18](#), and [Figure 4-19](#) (Figures of the T5591).

'-' in the frame means nothing is set.

**Figure 4-17 Pin Information Frame (Part 1/3)**

| ACLK          | BCLK          | CCLK          | PCLK | DREL          | DRET          | STRB1         | CPE HZ | STRB2 |
|---------------|---------------|---------------|------|---------------|---------------|---------------|--------|-------|
| 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | -    | 4 [ 71,000NS] | 4 [ 61,000NS] | 5 [ 51,000NS] | 1 -    | -     |
| 5 [ 11,000NS] | 1 [ 21,000NS] | 2 [ 31,000NS] | -    | 3 [ 51,000NS] | 3 [ 61,000NS] | - [ ]         | 1 -    | 4     |
| -             | -             | -             | -    | -             | -             | -             | 1 -    | -     |

**Figure 4-18 Pin Information Frame (Part 2/3)**

| STRB2 | CPE | HZ       | STRB3 | CPE | HZ | STRB4 | CPE | HZ | MODE | LEVEL | RINV  | PSH   | SHC   | RDSH | DRE  | DREN2 |
|-------|-----|----------|-------|-----|----|-------|-----|----|------|-------|-------|-------|-------|------|------|-------|
| -     | [   | 1        | -     | -   | [  | 1     | -   | -  | [    | 1     | -     | NRZBC | LINIT | ON   | -    | DRE1  |
| 4     | [   | 51,000NS | 2     | -   | -  | [     | 1   | -  | -    | NRZBC | LINIT | ON    | -     | -    | DRE1 | -     |
| -     | -   | 1        | -     | -   | -  | 1     | -   | -  | -    | -     | -     | -     | -     | -    | -    | -     |

**Figure 4-19 Pin Information Frame (Part 3/3)**

| DRON | DRDLY | DREDLY | EXP1 | EXP3 | EXPDLY | CPEDLY | PCPE1 | PCPE3 | EINV1 | EINV3 | ARID1 | ARID3 | BHUX | DSS | PPAT | PDS         |
|------|-------|--------|------|------|--------|--------|-------|-------|-------|-------|-------|-------|------|-----|------|-------------|
| -    | 0     | 0      | EXPB | EXPB | 0      | 0      | -     | -     | -     | -     | -     | -     | -    | 1   | -    | <FL,-,-,FL> |
| -    | 0     | 0      | EXPB | EXPB | 0      | 0      | -     | -     | -     | -     | -     | -     | -    | 1   | -    | <FL,-,-,FL> |
| -    | -     | -      | EXPB | EXPB | 0      | 0      | -     | -     | -     | -     | -     | -     | -    | 4   | -    | <FL,-,-,FL> |

**ACLK** (Displayed for other than the T5593 and T5592)

Displays the phase number of ACLK and its timing edge.

**BCLK**

Displays the phase number of BCLK and its timing edge.

**CCLK**

Displays the phase number of CCLK and its timing edge.

**HSC** (Displayed for the T5593)

Displays the high speed clock driver pin condition. (M/S/-).

**PCLK** (Displayed for the T5591)

The PCLK specification is displayed. (ON/INV/-)

**DREL** (DREL1 for the T5593 and T5592)

Displays the DREL number (DREL1 to DREL128) and its timing edge.

**DREL3** (Displayed for the T5593 and T5592)

Displays the DREL number (DREL129 to DREL256) and its timing edge.

**DRET** (DRET1 for the T5593 and T5592)

Displays the DRET number (DRET1 to DRET128) and its timing edge.

**DRET3** (Displayed for the T5593 and T5592)

Displays the DRET number (DRET129 to DRET256) and its timing edge.

**STRB1**

For the T5500 series:

Displays the strobe numbers (odd strobe numbers for STRB1 to STRB127) and their timing edge settings. For WSTRB, displays 'W' before the strobe numbers for STRB1 and STRB2.

- CPE

Displays the selected value of the control pattern (1 to 4/-).

- HZ

Displays the Hi-Z comparison or real time Hi-Z comparison setting (ON/A/B/-).

For the T5377, T5376, T5375, and T5371:

Displays the strobe numbers and the timing edge settings for STRB1 or WSTRB1. For WSTRB, displays 'W' before the strobe numbers.

When MWS is specified, displays 'M' before the strobe numbers.

- CPE  
Displays the selected value of the control pattern (1 to 4/A to D/-).
- HZ2  
Displays the Hi-Z comparison setting (ON/-).

#### STRB2 (Displayed for other than the T5592)

##### For the T5500 series:

Displays the strobe numbers (even strobe numbers for STRB2 to STRB128) and their timing edge settings. For WSTRB, displays 'W' before the strobe numbers for STRB1 and STRB2.

- CPE  
Displays the selected value of the control pattern (1 to 4/-).
- HZ  
Displays the Hi-Z comparison or real time Hi-Z comparison setting (ON/A/B/-).

##### For the T5377, T5376, T5375, and T5371:

Displays the strobe numbers and the timing edge settings for STRB2 or WSTRB2. For WSTRB, displays 'W' before the strobe numbers.

When MWS is specified, displays 'M' before the strobe numbers.

- CPE  
Displays the selected value of the control pattern (1 to 4/A to D/-).
- HZ2  
Displays the Hi-Z comparison setting (ON/-).

#### STRB3 (Displayed for other than the T5581, T5581H, T5581P, and T5571P)

##### For the T5500 series:

Displays the strobe numbers (odd strobe numbers for STRB129 to STRB255) and their timing edge settings. For WSTRB, displays 'W' before the strobe numbers for STRB3 and STRB4.

- CPE  
Displays the selected value of the control pattern (1 to 4/-).
- HZ  
Displays the Hi-Z comparison or real time Hi-Z comparison setting (ON/A/B/-).

##### For the T5377, T5376, T5375, and T5371:

Displays the strobe numbers and the timing edge settings for STRB3 or WSTRB3. For WSTRB, displays 'W' before the strobe numbers.

When MWS is specified, displays 'M' before the strobe numbers.

- CPE  
Displays the selected value of the control pattern (1 to 4/A to D/-).
- HZ2  
Displays the Hi-Z comparison setting (ON/-).

STRB4 (Displayed for other than the T5592, T5581, T5581H, T5581P, and T5571P)

For the T5500 series:

Displays the strobe numbers (even strobe numbers for STRB130 to STRB256) and their timing edge settings. For WSTRB, displays 'W' before the strobe numbers for STRB3 and STRB4.

- CPE

Displays the selected value of the control pattern (1 to 4/-).

- HZ

Displays the Hi-Z comparison or real time Hi-Z comparison setting (ON/A/B/-).

For the T5377, T5376, T5375, and T5371:

Displays the strobe numbers and the timing edge settings for STRB4 or WSTRB4. For WSTRB, displays 'W' before the strobe numbers.

When MWS is specified, displays 'M' before the strobe numbers.

- CPE

Displays the selected value of the control pattern (1 to 4/A to D/-).

- HZ2

Displays the Hi-Z comparison setting (ON/-).

MODE

Displays the waveform mode.

LEVEL

Displays the driver output initial level which is specified by a pin statement (HINIT/LINIT).

A value displayed in parentheses indicates the hardware driver output level (0 or 1).

0 indicates the low level and 1 indicates the high level.

For the T5591, the driver output initial level specified by a pin statement only is displayed.

RINV

The real time invert specification is displayed. (ON or -)

SDM (Displayed for other than the T5593, T5592 and T5591.)

The data swap mode specification is displayed. (ON, F, P, or -)

PSM

The pattern swap mode specification is displayed. (ON or -)

SMC

The swap mode control specification is displayed. (ON or -)

RDSM

The real time data swap mode specification is displayed. (ON or -)

DRE (DRE1 for the T5593 and T5592)

The driver enable patterns for DRECLK1 to DRECLK128 are displayed. (DRE1, DRE2, PDRE, C12, C13, C14, C15, or -)

**DRE3 (Displayed for the T5593 and T5592)**

The driver enable patterns for DRECLK129 to DRECLK256 are displayed. (DRE1, DRE2, PDRE, C12, C13, C14, C15, or -)

**PDRE (Displayed for the T5593 and T5592)**

The PDRE specification is displayed. (ON or -)

**DRENZR**

The waveform mode of the driver enable clock is displayed. (ON, D or -)

ON is displayed when DRENZR is specified. D is displayed when DREDNRZ is specified.

**DRON**

ON is displayed when the initial condition of the IOC pin is driver. (ON or -)

**DBMFI (Displayed for the T5593)**

Displays the mode used to inhibit changing data to ALPG data using the signal from the ALPG (INHDBM signal) (ON/-).

**DRDLY**

Displays the number of delay cycles of the driver pattern (0 to 31/-).

**DRPA (Displayed for the T5593)**

Displays the driver pattern of a dummy cycle, which is to be sent to port A (FIXH/FIXL/-).

**DRPB (Displayed for the T5593)**

Displays the driver pattern of a dummy cycle, which is to be sent to port B (FIXH/FIXL/-).

**DREDLY**

Displays the number of delay cycles of the DRE pattern (0 to 31/-).

**DREPA (Displayed for the T5593)**

Displays the DRE pattern of a dummy cycle for DREA (FIXH/FIXL/-).

**DREPB (Displayed for the T5593)**

Displays the DRE pattern of a dummy cycle for DREB (FIXH/FIXL/-).

**EXP (EXP1 is displayed for the T5593, T5592, T5591, T5586, and T5585.)**

The port output specification for the expected pattern of STRB1 to STRB128 is displayed (EXPB, EXPA, EXPL, EXPH, EXPAB/-).

**EXP3 (Displayed for the T5593, T5592, T5591, T5586, and T5585.)**

The port output specification for the expected pattern of STRB129 to STRB256 is displayed (EXPB, EXPA, EXPL, EXPH/-).

**EXPDLY**

Displays the number of delay cycles of the expected pattern (0 to 31/-).

**CPEDLY**

Displays the number of delay cycles of the CPE pattern (0 to 31/-).

**MATCH (Displayed for other than the T5593, T5592 and T5591.)**

The match mode specification is displayed. (ON or -)



PCPE (PCPE1 is displayed for the T5593, T5592, T5591, T5586, and T5585. Nothing is displayed for the T5377, T5376, T5375, or T5371.)

The PCPE condition of STRB1 to STRB128 is displayed (PCPE/CCPE/ACPE/-).

PCPE3 (Displayed for the T5593, T5592, T5591, T5586, and T5585.)

The PCPE condition of STRB129 to STRB256 is displayed (PCPE/CCPE/ACPE/-).

EINV (EINV1 is displayed for the T5593, T5592, T5591, T5586, and T5585.)

The inversion mode of the expected value data is displayed for STRB1 to STRB128. (ON/-)

EINV3 (Displayed for the T5593, T5592, T5591, T5586, and T5585.)

The inversion mode of the expected value data is displayed for STRB129 to STRB256. (ON/-)

ARID (ARID1 is displayed for the T5592, T5591, T5586, and T5585. Nothing is displayed for the T5593.)

The enable mode of ARIRAM of STRB1 to STRB128 is displayed (ON/-).

ARID3 (Displayed for the T5592, T5591, T5586, and T5585.)

The enable mode of ARIRAM of STRB129 to STRB256 is displayed (ON/-).

ESM (Displayed for other than the T5593, T5592, T5591, T5377, T5376, T5375, and T5371.)

The swap mode condition of the expected value data is displayed. (ON or -)

BMUX (Displayed for other than the T5592, T5377, T5376, T5375, and T5371.)

The conditions of the main and sub- banks are displayed. (MUX, MAIN, SUB, or -)

MSM (Displayed for the T5593.)

Displays the PG MAIN/SUB mode setting (ON/-).

DSS (Displayed for other than the T5377, T5376, T5375, and T5371.)

The data selection mode is displayed (1 to 4/-).

PPAT (Displayed for other than the T5377, T5376, T5375, and T5371.)

The DBM specification for the pattern to be used for driver waveform generation is displayed. (ON or -)

CYP (Displayed for other than the T5593, T5592, T5377, T5376, T5375, and T5371.)

The cycle palette number for the PDS data, which is displayed in the pin information frame, is displayed (1/A1/B1/-).

For the T5593, "A1" is displayed for CYP A and "B1" is displayed for CYP B.

"1" is displayed for test systems other than the T5593.

If data is not displayed for PDS (DC pins), "-" is displayed.

## PDS

The data selector setting used to assign ALPG signals to pins is displayed.

### Tip

*If the T5593, T5592, T5377, T5376, T5375 or T5371 is used, the data selector setting for cycle palette 1 is displayed.*

If you want to refer the data selector settings for cycle palette 2 and subsequent cycle palettes, use Cycle Palette Data Window.

➔ For more information, refer to 4. 3. 7 "Cycle Palette Data Window" on page 4-30.

Footer

If the format or value of the pin information frame is out of range, an error message is displayed in the footer with a beep sound.

Sash

The sash is used to adjust the width of the pin display frame. It can be dragged and moved with the mouse.

Figure 4-20 shows the position of the sash when the PinClock monitor is activated. In Figure 4-21 and Figure 4-22, the sash is moved.

Figure 4-20 Position of the Sash When the Monitor is Activated

| DUT    | Tester | ACLK          | BCLK          | CCLK          | DREL          | DRET          |
|--------|--------|---------------|---------------|---------------|---------------|---------------|
| P33(A) | 97     | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 51,000NS] | 4 [ 61,000NS] |
| P34(A) | 98     | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 51,000NS] | 4 [ 61,000NS] |
| P35(A) | 99     | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 51,000NS] | 5 [ 61,000NS] |
| P36(A) | 100    | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 51,000NS] | 5 [ 61,000NS] |
| P37(A) | 101    | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |
| P38(A) | 102    | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |
| P39(A) | 103    | 4 [ 11,000NS] | 5 [ 21,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] |
| P40(A) | 104    | 4 [ 11,000NS] | 5 [ 21,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] |

Figure 4-21 The Sash is Moved to the Right Edge

| DUT    | Tester | PinList     | Signal | ACLK          |
|--------|--------|-------------|--------|---------------|
| P33(A) | 97     | IOPIN_GROUP | PI033  | 1 [ 31,000NS] |
| P34(A) | 98     | IOPIN_GROUP | PI034  | 1 [ 31,000NS] |
| P35(A) | 99     | IOPIN_GROUP | PI035  | 2 [ 31,000NS] |
| P36(A) | 100    | IOPIN_GROUP | PI036  | 2 [ 31,000NS] |
| P37(A) | 101    | IOPIN_GROUP | PI037  | 3 [ 11,000NS] |
| P38(A) | 102    | IOPIN_GROUP | PI038  | 3 [ 11,000NS] |
| P39(A) | 103    | IOPIN_GROUP | PI039  | 4 [ 11,000NS] |
| P40(A) | 104    | IOPIN_GROUP | PI040  | 4 [ 11,000NS] |

Figure 4-22 The Sash is Moved to the Left Edge

| DUT    | ACLK          | BCLK          | CCLK          | DREL          | DRET          | S |
|--------|---------------|---------------|---------------|---------------|---------------|---|
| P33(A) | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 51,000NS] | 4 [ 61,000NS] |   |
| P34(A) | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 51,000NS] | 4 [ 61,000NS] |   |
| P35(A) | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 51,000NS] | 5 [ 61,000NS] |   |
| P36(A) | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 51,000NS] | 5 [ 61,000NS] |   |
| P37(A) | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |   |
| P38(A) | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |   |
| P39(A) | 4 [ 11,000NS] | 5 [ 21,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] |   |
| P40(A) | 4 [ 11,000NS] | 5 [ 21,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] |   |

### 4. 3. 2 Sort Item Mode

When Sort item mode is selected from the Item menu, sort mode is enabled.

Clicking an item button on the item bar highlights the corresponding column and data is sorted. For example, numbers are sorted in ascending order and character strings are sorted in alphabetical order.

The list can be sorted using more than one item. Select one item button and then select another one. The list is sorted first on the initial item and then on the latter item (see the example below).

The selection of the sort item can be canceled by clicking the item button highlighted.

When the Sort item mode is turned off, all the selected item buttons are no longer highlighted. However, the order of pin numbers or pin information items does not change after the sort item selection is canceled.

#### When sorting Multiple Items

The following explains the procedure for sorting tester pins whose RINV item and SMC item are ON.

**Figure 4-23 Initial Screen**

| DUT    | Tester | MODE  | LEVEL    | RINV | SDM | PSM | SMC | RDSM | DRE  | DRENZ | DRON | DRDLY | DREDLY | EXP  |
|--------|--------|-------|----------|------|-----|-----|-----|------|------|-------|------|-------|--------|------|
| P33(A) | 97     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P34(A) | 98     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P35(A) | 99     | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P36(A) | 100    | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P37(A) | 101    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P38(A) | 102    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P39(A) | 103    | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P40(A) | 104    | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |

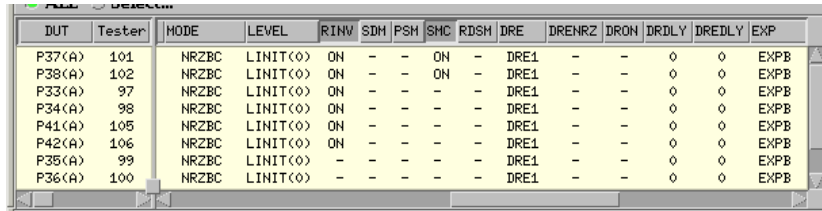
1) Select the RINV button.

**Figure 4-24 RINV Selection Screen**

| DUT    | Tester | MODE  | LEVEL    | RINV | SDM | PSM | SMC | RDSM | DRE  | DRENZ | DRON | DRDLY | DREDLY | EXP  |
|--------|--------|-------|----------|------|-----|-----|-----|------|------|-------|------|-------|--------|------|
| P33(A) | 97     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P34(A) | 98     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P37(A) | 101    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P38(A) | 102    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P41(A) | 105    | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P42(A) | 106    | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P35(A) | 99     | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P36(A) | 100    | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |

2) Select the SMC button.

Figure 4-25 SMC Selection Screen



The screenshot shows a software window titled "ALL DETECT" containing a table with test parameters. The table has 14 columns: DUT, Tester, MODE, LEVEL, RINV, SDH, PSH, SMC, RDSH, DRE, DRENZ, DRON, DRDLY, DREDLY, and EXP. There are 8 rows of data, all with MODE set to NRZBC and LEVEL set to LINIT(0). The SMC column contains values ON, -, -, -, -, -, -, and -. The DRE column contains DRE1 for all rows. The EXP column contains EXPB for all rows.

| DUT    | Tester | MODE  | LEVEL    | RINV | SDH | PSH | SMC | RDSH | DRE  | DRENZ | DRON | DRDLY | DREDLY | EXP  |
|--------|--------|-------|----------|------|-----|-----|-----|------|------|-------|------|-------|--------|------|
| P37(A) | 101    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P38(A) | 102    | NRZBC | LINIT(0) | ON   | -   | -   | ON  | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P33(A) | 97     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P34(A) | 98     | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P41(A) | 105    | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P42(A) | 106    | NRZBC | LINIT(0) | ON   | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P35(A) | 99     | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |
| P36(A) | 100    | NRZBC | LINIT(0) | -    | -   | -   | -   | -    | DRE1 | -     | -    | 0     | 0      | EXPB |

### 4. 3. 3 Delete Item Mode

When Delete item mode is selected from the Item menu, delete mode is enabled.

When you click the item button on the item bar, information on the item is deleted.

#### Tip

*The last item list cannot be deleted.*

*The DUT list cannot be deleted.*

To redisplay the deleted item column, select Undo from the Item menu.

To redisplay all deleted item columns, select Reset from the Item menu.

#### When deleting Items

The following example shows the procedure for deleting the BCLK item.

**Figure 4-26 Initial Screen**

| DUT    | Tester | ACLK          | BCLK          | CCLK          | DREL          | DRET          |
|--------|--------|---------------|---------------|---------------|---------------|---------------|
| P41(A) | 105    | 5 [ 11,000NS] | 1 [ 21,000NS] | 2 [ 31,000NS] | 3 [ 51,000NS] | 3 [ 61,000NS] |
| P42(A) | 106    | 5 [ 11,000NS] | 1 [ 21,000NS] | 2 [ 31,000NS] | 3 [ 51,000NS] | 3 [ 61,000NS] |
| P33(A) | 97     | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 61,000NS] | 4 [ 61,000NS] |
| P34(A) | 98     | 1 [ 31,000NS] | 2 [ 41,000NS] | 3 [ 51,000NS] | 4 [ 61,000NS] | 4 [ 61,000NS] |
| P35(A) | 99     | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 61,000NS] | 5 [ 61,000NS] |
| P36(A) | 100    | 2 [ 31,000NS] | 3 [ 41,000NS] | 4 [ 51,000NS] | 5 [ 61,000NS] | 5 [ 61,000NS] |
| P37(A) | 101    | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |
| P38(A) | 102    | 3 [ 11,000NS] | 4 [ 21,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] |

1) Select the BCLK button.

**Figure 4-27 BCLK Item Deletion Screen**

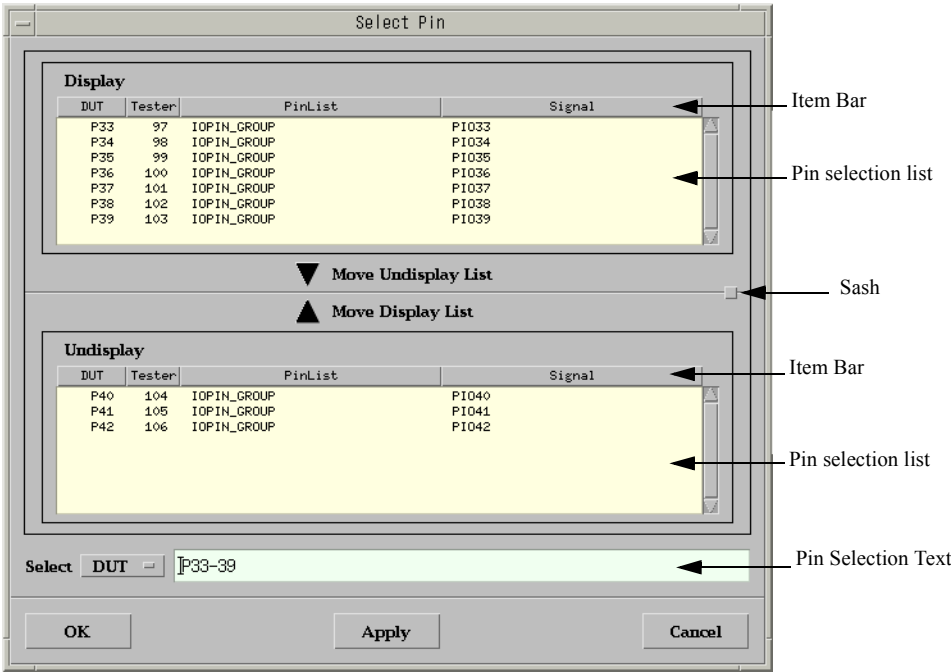
| DUT    | Tester | ACLK          | CCLK          | DREL          | DRET          | STRB1         |
|--------|--------|---------------|---------------|---------------|---------------|---------------|
| P33(A) | 97     | 1 [ 31,000NS] | 3 [ 51,000NS] | 4 [ 61,000NS] | 4 [ 61,000NS] | 5 [ 61,000NS] |
| P34(A) | 98     | 1 [ 31,000NS] | 3 [ 51,000NS] | 4 [ 61,000NS] | 4 [ 61,000NS] | 5 [ 61,000NS] |
| P35(A) | 99     | 2 [ 31,000NS] | 4 [ 51,000NS] | 5 [ 61,000NS] | 5 [ 61,000NS] | 1 [ 61,000NS] |
| P36(A) | 100    | 2 [ 31,000NS] | 4 [ 51,000NS] | 5 [ 61,000NS] | 5 [ 61,000NS] | 1 [ 61,000NS] |
| P37(A) | 101    | 3 [ 11,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] | - [ 61,000NS] |
| P38(A) | 102    | 3 [ 11,000NS] | 5 [ 31,000NS] | 1 [ 51,000NS] | 1 [ 61,000NS] | - [ 61,000NS] |
| P39(A) | 103    | 4 [ 11,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] | 3 [ 61,000NS] |
| P40(A) | 104    | 4 [ 11,000NS] | 1 [ 31,000NS] | 2 [ 51,000NS] | 2 [ 61,000NS] | 3 [ 61,000NS] |

The BCLK column is deleted then the CCLK column is shifted to the left.

4. 3. 4 Select Pin Window

When you click Select Pin... in the Edit menu or the Select... button in the select bar, [Figure 4-28](#) is displayed. In this window, conditions on the number of pins displayed in the pin display frame are set.

Figure 4-28 Select Pin Window



Item Bar

When you click each button, the list is sorted based on the selected item column. The list is sorted in the same way as the list in the pin display frame.

➔ For more information, refer to [4. 3. 2 "Sort Item Mode" on page 4-22](#).

Sash

By dragging the sash up or down, the number of displayed lines in the [Display] and [Undisplay] lists can be changed.

Pin selection list

The Display list and the Undisplay list are displayed in the Pin Select window. The Display list contains the information currently displayed in the pin display frame. The Undisplay list contains the information not displayed in the pin display frame. If you drag the mouse in the list, the region between the start and stop positions of the drag operation is highlighted. Clicking the right or left arrow button adjacent to the list moves the highlighted information into the Display or Undisplay list depending on the direction of the arrow. The selection text varies with the contents of the pin selection list.

- Control key

When you hold down the Control key and click the mouse button, you can select more than one pin.

- Shift key

When you select a pin, hold down the Shift key and select another pin, you can select many pins listed between your first and second selections.

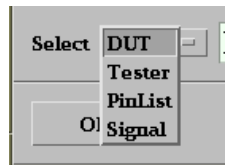
### Pin Selection Text

The menu button and text box are provided. Click the option menu button to select a pin format and type pin numbers in the text box. Pressing the Return key displays the related pin information in the Display list.

#### 1. Option Menu Button

As shown in [Figure 4-29](#), there are four pin formats: DUT, Tester, PinList, and Signal. When the socket file is not specified in the program, Signal is disabled. When the pin list is not specified, PinList is disabled. When the status has changed to Disabled from Enabled, DUT is automatically specified.

**Figure 4-29 Option Menu Button**



#### 2. Definition of Pin Specification

- Single specification  
Number specification
- Consecutive specification  
Delimit the start number and stop number with a hyphen ('-'). Ascending and descending order can be specified.
- Multiple specification  
Delimit multiple single or consecutive specifications by commas (',').

#### 3. Pin Format

- DUT Format  
Specify DUT pin numbers next to P. (A single number, consecutive numbers and multiple numbers can be specified.)  
(Example) P1-3, P10-4, 11
- Tester Format  
Specify tester pin numbers. A single number, consecutive numbers and multiple numbers can be specified.  
(Example) 1-4, 30-11, 44
- PinList Format  
Specify pin list names. More than one name can be specified.

(Example) INPIN, OUTPIN, IOPIN

- Signal Format

Specify DUT pin names. A single number, consecutive numbers and multiple numbers can be specified.

(Example) IN1-6, OUT4-8, DC50

4. Text Box

Specify pin numbers in the pin format selected from the option menu.

When you change the contents of the text box, the pin selection list is also changed at the same time.

5. Text Box Error Handling

When you specify the pin in the text box and press the enter key, the extra spaces are ignored.

If extra hyphens or commas are entered or an incorrect format is used, the error message dialog box is displayed. If you click the OK button in the dialog box, the setting in the text box goes back to the previous condition and the pin selection list is not updated.

OK, Apply, and Cancel buttons

- OK

Updates the pin display frame to the current setting and closes the Select Pin window.

- Apply

Updates the pin display frame to the current setting. The Pin Select window does not close.

- Cancel

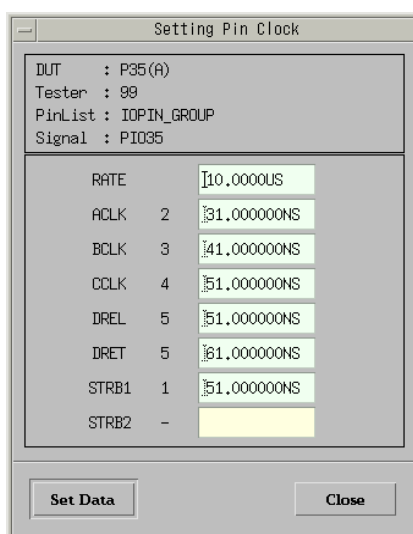
Restores the list to the condition after the last OK or Apply was clicked and closes the Select Pin window.



### 4. 3. 5 Setting Pin Clock Window

When you click the Setting Pin Clock... in the Edit menu, or double click the line to which you want to perform setting in the pin display frame or pin information frame, [Figure 4-30](#) is displayed.

**Figure 4-30 Setting Pin Clock Window**

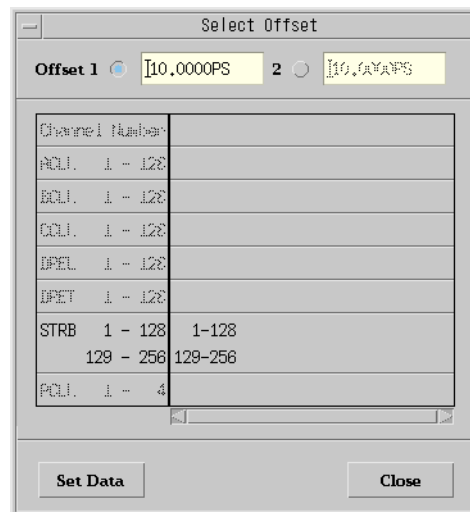


- Set Data button  
Determines the timing setting for the clock currently selected.
- Close button  
Closes the Setting Pin Clock window without setting data.

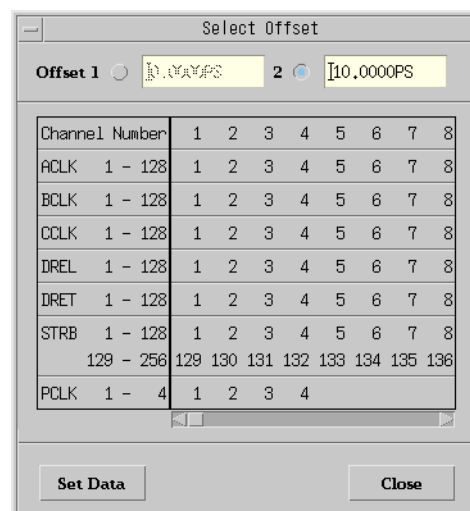
### 4. 3. 6 Select Offset Window

When you click Select Offset... in the Edit menu or the Offset... button on the tool bar, [Figure 4-31](#) and [Figure 4-32](#) are displayed. The timing settings for Offset 1 or Offset 2 and the selected phase numbers are displayed in this window.

**Figure 4-31 Select Offset Window (When Offset1 is selected)**



**Figure 4-32 Select Offset Window (When Offset2 is selected)**



- Set Data button  
Sets the timings for the selected Offset. The offset value set is added to the timings set in the Select Offset Window.
- Close button  
Closes the Select Offset window without setting data.

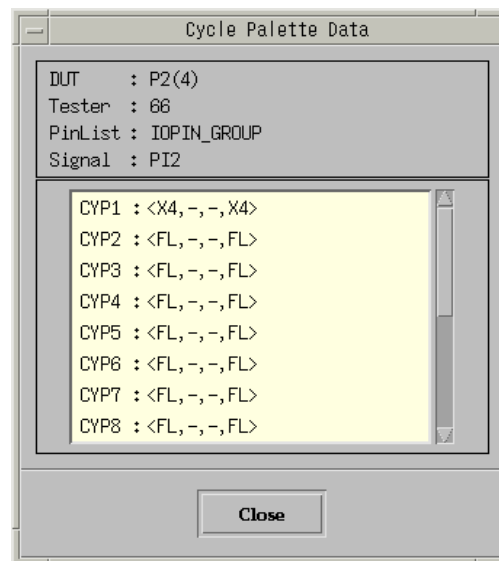
### 4. 3. 7 Cycle Palette Data Window

When you select Cycle Palette Data... from Option menu, [Figure 4-33](#) is displayed.

The Cycle Palette Data window displays cycle palette data for the current selected pin.

To change the target pin, select the desired pin in the pin display frame or pin information frame.

**Figure 4-33 Cycle Palette Data Window**



- Close button  
Closes the Cycle Palette Data window.

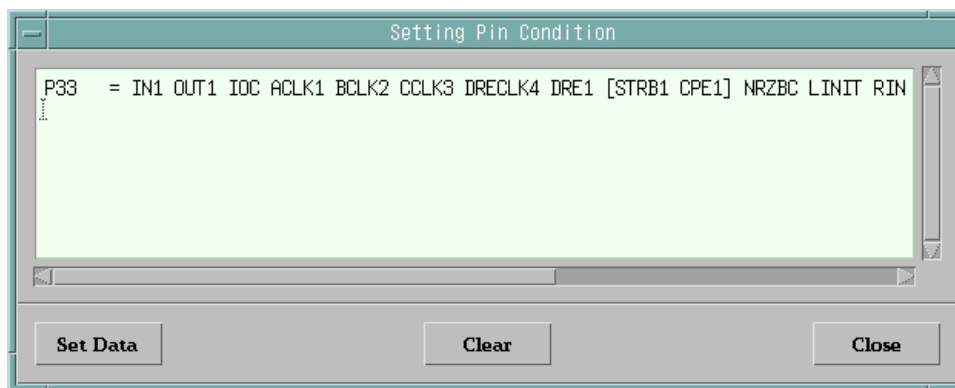
 **Tip**

*The Cycle Palette Data window can be used for the T5593, T5592, T5377, T5376, T5375, and T5371 only.*

### 4. 3. 8 Setting Pin Condition Window

When you select Setting Pin Condition... from the Edit menu, [Figure 4-34](#) is displayed. The currently selected pin connecting conditions are displayed in the text box in this window. You can change the settings by changing the connecting conditions and clicking the SetData button. When you display additional connecting conditions of other pins in the text box, select pins in the pin display frame or pin information frame.

**Figure 4-34 Setting Pin Condition Window**



- Set Data button  
This button is used to set the pin connection conditions displayed in the text box.
- Clear button  
This button is used to clear the test box.
- Close button  
This button is used to close the Setting Pin Condition window without updating the settings.

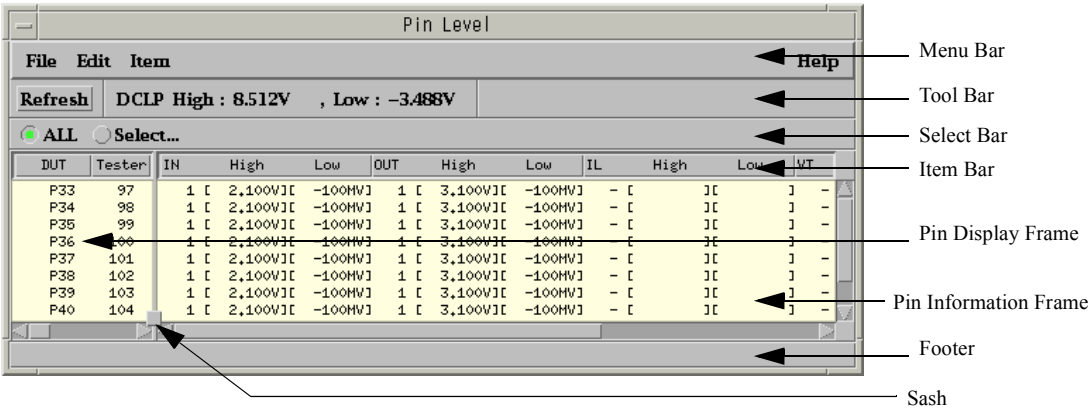
## 4. 4 Pin Level Monitor

The Pin Level monitor is used to display and set the information on the pin level set to the tester.

### 4. 4. 1 Main Window

When you click the Pin Level button in the TCM main window, [Figure 4-35](#) is displayed.

Figure 4-35 Pin Level Monitor



#### Menu Bar

The menu bar consists of File, Edit, Item and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the Pin Clock monitor and Pin Level monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the Pin Level monitor.

#### Edit Menu

When you select the Edit menu in the menu bar, a pull-down menu opens showing three commands as in [Figure 4-36](#).

Figure 4-36 Edit Menu



- Select Pin...

When you select the Select Pin... command, the Select Pin window is displayed.

➔ For more information, refer to [4. 3. 4 "Select Pin Window" on page 4-25](#).

- Setting Pin Condition ...

When you select the Setting Pin Condition... command, the Setting Pin Condition window is displayed.

➔ For more information, refer to [4. 3. 8 "Setting Pin Condition Window" on page 4-31](#).

- Setting Pin Level...

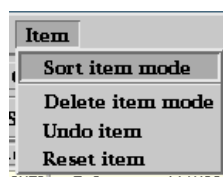
When you select the Setting Pin Level... command, the Setting Pin Level window is displayed.

➔ For more information, refer to [4. 4. 2 "Setting Pin Level Window" on page 4-37](#).

#### Item Menu

When you select the Item menu in the menu bar, a pull-down menu opens showing four commands as in [Figure 4-37](#).

**Figure 4-37 Item Menu**



- Sort item mode

When you select the Sort item mode, you can sort items clicked in the item bar.

➔ For more information, refer to [4. 3. 2 "Sort Item Mode" on page 4-22](#).

- Delete item mode

When you select the Delete item mode, you can delete items clicked in the item bar.

➔ For more information, refer to [4. 3. 3 "Delete Item Mode" on page 4-24](#).

- Undo item

The Undo item command can redisplay the items you deleted from the list with Delete in reverse order of deleting. (If all items are displayed, the Undo item command is disabled.)

- Reset item

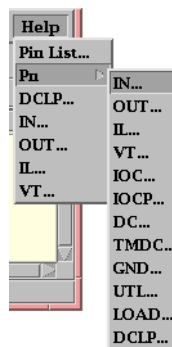
The Reset item command can redisplay all the items you deleted with Delete into the item bar. (If all items are displayed, the Reset item command is disabled.)

You can select either Sort item mode or Delete item mode. When the current mode is selected again, the mode is turned off and the selections in the item bar are canceled. Neither mode is selected by default.

### Help Menu

When you select the Help menu in the menu bar, a pull-down menu opens showing seven commands as in [Figure 4-38](#). Items from (a.) to (l.) related to pin connecting conditions are displayed under (2) Pn. When you select each item, the related Online Manual is displayed.

**Figure 4-38 Help Menu**

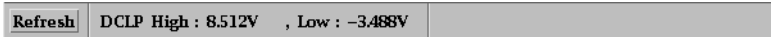


1. Pin List...
2. Pn...
  - a. IN...
  - b. OUT...
  - c. IL...
  - d. VT...
  - e. IOC...
  - f. IOCP...
  - g. DC...
  - h. TMDC...
  - i. GND...
  - j. UTL...
  - k. LOAD...
  - l. DCLP...
3. DCLP...
4. IN...
5. OUT...
6. IL...
7. VT...

### Tool Bar

The tool bar of the PinLevel monitor is shown in [Figure 4-39](#).

Figure 4-39 PinLevel Tool Bar



- Refresh  
When you click the Refresh button, the current tester setup condition is loaded and the contents of the Pin Level Monitor display are updated.
- DCLP High (Displayed for other than the T5593)  
This button is used to display the high-level voltage value of the dynamic clamp.
- DCLP Low (Displayed for other than the T5593)  
This button is used to display the low-level voltage value of the dynamic clamp.

Select Bar

The select bar of PinLevel monitor is shown in [Figure 4-40](#).

Figure 4-40 PinLevel Select Bar



- ALL  
Displays all pins currently being used in the pin display frame.
- Select...  
The Select Pin window is displayed.  
➔ For more information, refer to [4. 3. 4 "Select Pin Window" on page 4-25](#).

Item Bar

The item bar of PinLevel monitor is shown in [Figure 4-41](#).

Figure 4-41 PinLevel Item Bar



The item bar is displayed at the top of the pin display frame and pin information frame, listing the item names.

Pin Display Frame

The pin display frame is shown in [Figure 4-42](#).

Figure 4-42 Pin Display Frame

| DUT    | Tester | PinList     | Signal |
|--------|--------|-------------|--------|
| P33(A) | 97     | IOPIN_GROUP | PI033  |
| P34(A) | 98     | IOPIN_GROUP | PI034  |
| P35(A) | 99     | IOPIN_GROUP | PI035  |
| P36(A) | 100    | IOPIN_GROUP | PI036  |
| P37(A) | 101    | IOPIN_GROUP | PI037  |
| P38(A) | 102    | IOPIN_GROUP | PI038  |
| P39(A) | 103    | IOPIN_GROUP | PI039  |
| P40(A) | 104    | IOPIN_GROUP | PI040  |



If a socket file is specified in the test program, you can assign DUT pin numbers and DUT pin names to tester pins. DUT pin numbers and DUT pin names are displayed in the DUT and Signal columns in the pin display frame, respectively. When the socket file is not specified, nothing is displayed in the Signal column.

- DUT  
Displays DUT pin numbers. They can be sorted in ascending order.
- Tester  
Displays tester pin numbers. They can be sorted in ascending order.
- PinList  
Multiple names can be assigned to a certain channel, so the first pin list name you defined in the program is displayed in the PinList column. Pin list names can be sorted in alphabetical order.
- Signal  
Displays DUT pin names (pin signal names). They can be sorted in alphabetical order and then in ascending numeric order.

#### Pin Information Frame

The pin information frame of the PinLevel monitor is shown in [Figure 4-43](#) (Figure for the T5591).

'-' in the frame means nothing is set.

**Figure 4-43 Pin Information Frame**

| IN | High   | Low    | OUT | High   | Low    | VT | CP     | DR     | IOC | DC | TERM | LOAD | DCLP |
|----|--------|--------|-----|--------|--------|----|--------|--------|-----|----|------|------|------|
| 1  | 2,100V | -100mV | 1   | 3,100V | -100mV | 1  | 1,100V | 1,100V | ON  | ON | ON   | ON   | ON   |

- IN  
Displays IN unit numbers connected and their levels (High/Low) (1 to 16/-).  
For INH, displays 'H' before the IN numbers.
- OUT  
Displays OUT unit numbers connected and their levels (High/Low) (1 to 16/-).
- IL (Displayed for other than the T5593, T5592 and T5591.)  
Displays programmable load numbers used and their levels (High/Low) (1 to 16/-).  
For ILALL, displays 'A' before the IL numbers.
- VT  
Displays terminator level numbers used and their levels (High/Low) (1 to 16/-).  
For the T5593, T5592 and T5591, the comparator voltage (CP) and driver voltage (DR) are displayed.
- IOC  
Displays the I/O common pin setting (ON/-).
- IOCP (Displayed for the T5593)  
Displays the I/O comparator setting (ON/-).

- DC  
Displays the connecting condition of the DC measurement unit. (ON or -)
- TMDC (Displayed for the T5593.)  
Displays the setting to connect DC to a terminator pin for the driver (the pin to which VT and DC for the driver are connected) (ON/-).
- GND (Displayed for other than the T5593, T5592 and T5591.)  
Displays the connecting condition of the GND relay. (ON or -)
- UTL (Displayed for the T5581, T5581H, T5581P, and T5571P.)  
Displays the connecting condition of the UTL relay. (ON or -)
- TERM  
Displays the connecting condition of the terminator. (ON, C, D, or -)
- LOAD (Displayed for other than the T5593 and T5592)  
Displays the connecting condition of the LOAD relay. (ON or -)
- DCLP (Displayed for other than the T5593)  
Displays the condition of the dynamic clamp. (ON or -)

#### Footer

If the format or value of the pin information frame is illegal, an error message is displayed in the footer with a beep sound.

#### Sash

The sash is used to adjust the width of the pin display frame. It can be dragged and moved with the mouse.

➔ For more information, refer to ["Sash" on page 4-21](#).

## 4. 4. 2 Setting Pin Level Window

When you click the Setting Pin Level... in the Edit menu, or double-click the line to which you want to perform setting in the pin display frame or pin information frame, [Figure 4-44](#) is displayed.

**Figure 4-44 Setting Pin Level Window (Screen of the T5591)**

|         |   | High   | Low     |
|---------|---|--------|---------|
| DUT     | : | P34    |         |
| Tester  | : | 98     |         |
| PinList | : |        |         |
| Signal  | : | PI034  |         |
| DCLP    |   | 5.000V | -2.000V |
| IN      | 1 | 2.000V | -200MV  |
| OUT     | 1 | 3.000V | -300MV  |
| VT      | 1 | 1.000V | 1.000V  |

CP DR

Set Data Close

- Set Data button  
Sets the data level for the channel currently selected.
- Close button  
Closes the Setting Pin Level window without setting data.

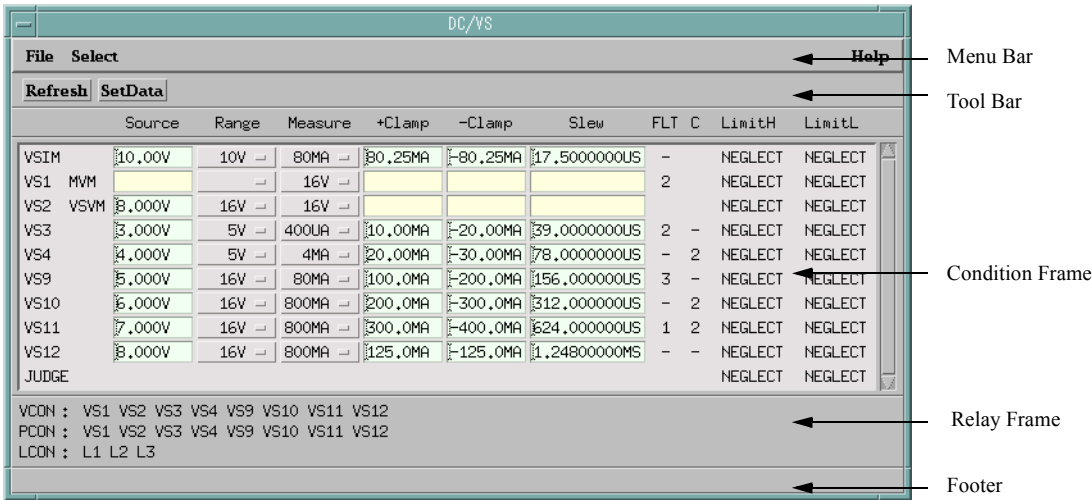
## 4. 5 DC/VS Monitor

The DC/VS monitor is used to refer and change the DC and VS setup conditions.

### 4. 5. 1 Main Window

When you click the DC/VS button in the TCM main window, [Figure 4-45](#) is displayed.

**Figure 4-45 DC/VS Monitor (Screen of the T5591)**



#### Menu Bar

The menu bar consists of File, Select and Help menus.

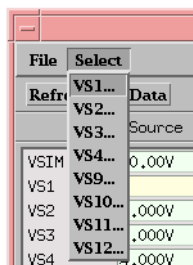
#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the DC/VS monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the DC/VS monitor.

#### Select menu command

The VS to which PPS is connected is displayed in the pull-down menu.

**Figure 4-46 DC/VS Select Menu**

When you select each VSm... button, the VS\_ PPS window is displayed with the selected VS number m.

### Help Menu

The Online Manual regarding the DC/VS statement is displayed.

The Help menu is a pull-down menu consisting of the following commands:

- ISVM...
- VSIM...
- HVSIM...
- VS...
- LIMIT...
- VCON...
- PCON...
- LCON...

### Tool Bar

The following describes the function of each button in the tool bar of the DC/VS monitor.

- Refresh  
When you click or the Refresh button, the current tester setup condition is loaded and the contents of the DC/VS Monitor display are updated.
- SetData  
When you click this button after changing each parameter, the settings changed are reflected in the tester.

### Condition Frame

Figure 4-47 DC/VS Condition Frame

|          | Source | Range | Measure | +Clamp  | -Clamp   | Slew          | FLT | C | LimitH  | LimitL  |
|----------|--------|-------|---------|---------|----------|---------------|-----|---|---------|---------|
| VSIM     | 10.00V | 10V   | 80MA    | 80.25MA | -80.25MA | 17.5000000US  | -   |   | NEGLECT | NEGLECT |
| VS1 MVM  |        |       | 16V     |         |          |               | 2   |   | NEGLECT | NEGLECT |
| VS2 VSVM | 8.000V | 16V   | 16V     |         |          |               |     |   | NEGLECT | NEGLECT |
| VS3      | 3.000V | 5V    | 400UA   | 10.00MA | -20.00MA | 39.0000000US  | 2   | - | NEGLECT | NEGLECT |
| VS4      | 4.000V | 5V    | 4MA     | 20.00MA | -30.00MA | 78.0000000US  | -   | 2 | NEGLECT | NEGLECT |
| VS9      | 5.000V | 16V   | 80MA    | 100.0MA | -200.0MA | 156.0000000US | 3   | - | NEGLECT | NEGLECT |
| VS10     | 6.000V | 16V   | 800MA   | 200.0MA | -300.0MA | 312.0000000US | -   | 2 | NEGLECT | NEGLECT |
| VS11     | 7.000V | 16V   | 800MA   | 300.0MA | -400.0MA | 624.0000000US | 1   | 2 | NEGLECT | NEGLECT |
| VS12     | 8.000V | 16V   | 800MA   | 125.0MA | -125.0MA | 1.24800000MS  | -   | - | NEGLECT | NEGLECT |
| JUDGE    |        |       |         |         |          |               |     |   | NEGLECT | NEGLECT |

The items which are referred and modified with the condition frame are described below.

- ISVM, VSIM, HVSIM

Condition setup for the DC measurement unit.

ISVM, VSIM, or HVSIM is selected.

- Source

Specifies the applied current (is) for ISVM or the applied voltage (vs) for VSIM and HVSIM.

- Range

Specifies the applied current range (R(ir)) for ISVM or the applied voltage range (R(vr)) for VSIM and HVSIM.

- Measure

Specifies the measurement range (M(vr) or M(ir)).

- +Clamp, -Clamp

Specifies the current (voltage) clamp ((vp, vm) or (ip, im)).

- Slew

Specifies the slew rate (T(t)) (VSIM only).

- FLT

Displays the connection of the low-pass filter (FLT) (entry disabled).

- Limit H, Limit L

Displays the maximum and minimum values for DC Pass and Fail judgment at the time of JUDGE execution (entry disabled).

- VS<sub>n</sub>

Sets the voltage value to be applied to the power supply pin of the DUT and the current limit value.

Condition setup for the power current measurement function.

Only the currently enabled VS is displayed.

- VSVM

Displayed when specified for VSVM (entry disabled).

- MVM (Displayed for other than the T5581 and T5581H)

Displayed when specified for MVM (entry disabled).

- Source  
Specifies the applied voltage value (vs) or OPEN.
- Range  
Specifies the applied voltage range (R(vr)).
- Measure  
Specifies the measurement range (M(ir)).
- +Clamp, -Clamp  
Specifies the current clamp (ip, im).
- Slew  
Specifies the slew rate (T(t)).
- FLT  
Displays the connection of the low-pass filter (FLT) (entry disabled).
- C  
Displays the guaranteed current capacity (C(k)) for the range from 2μA to 4 mA (entry disabled).
- Limit H, Limit L  
Displays the maximum and minimum values for the Pass/Fail judgment at the time of JUDGE execution (entry disabled).
- JUDGE
  - Limit H, Limit L  
Displays the maximum and minimum values for the Pass/Fail judgment at the time of JUDGE execution (entry disabled).

## Relay Frame

**Figure 4-48 DC/VS Relay Connection Frame**

```
VCON : VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
PCON : VS1 VS2 VS3 VS4 VS9 VS10 VS11 VS12
LCON : L1 L2 L3
```

The items which are referred with the relay frame are described below.

- VCON  
Displays the open condition of the power relay for devices (entry disabled).
- PCON  
Displays the bypass capacitor connection relay on the performance board (entry disabled).
- LCON  
Displays the load relay drive signal on the performance board (entry disabled).

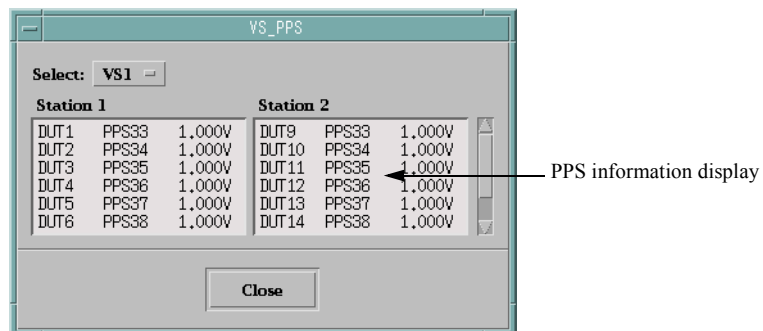
## Footer

If a tester communication error occurs, an error message is displayed.

## 4. 5. 2 VS\_PPS Window

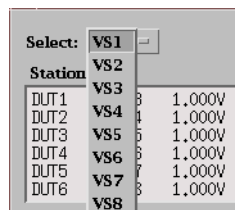
The VS\_PPS window is used to refer the PPS setup condition for each DUT of VS.

**Figure 4-49 VS\_PPS Window**



- ◆ **Select Option Menu**  
The VS to which PPS is connected can be selected.

**Figure 4-50 Select Option Menu**



- ◆ **PPS Information Display**  
Individual voltage value setting for PPS assigned to the selected VS is displayed for each DUT of the station currently being executed.  
For the parallel drive VS, the minimum PPS number is displayed.
- ◆ **Close Button**  
Closes the VS\_PPS window.



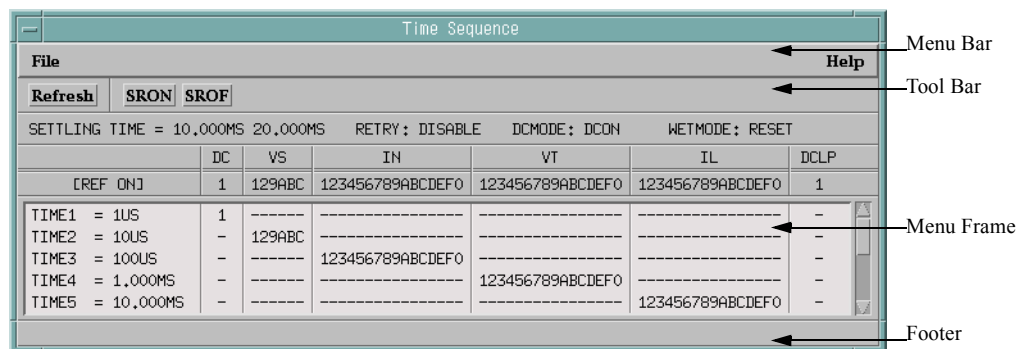
## 4. 6 Time Sequence Monitor

The Time Sequence monitor is used to refer the RON or ROF sequence and wait time settings specified by the TIME statement.

### 4. 6. 1 Main Window

When you click the TimeSeq button in the TCM main window, [Figure 4-51](#) is displayed.

**Figure 4-51 Time Sequence Monitor (Screen of T5581)**



#### Menu Bar

The menu bar consists of File and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Save data from the Time sequence monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the Time Sequence monitor.

#### Help Menu

The Online Manual regarding the Time Sequence is displayed.

The Help menu is a pull-down menu consisting of the following commands:

- SRON/SROF...
- SETTLING TIME...
- RETRY...
- DCMODE...
- WETMODE...
- TIME...

#### Tool Bar

The function of each button in the tool bar of the Time Sequence monitor is described below.

- Refresh  
When you click the Refresh button, the current tester setup condition is loaded and the contents of the Time Sequence Monitor display are updated.
- SRON  
Performs reference ON in accordance with the specification of the TIME statement.
- SROF  
Performs reference OFF in accordance with the specification of the TIME statement.

### Main Frame

**Figure 4-52 Time Sequence Main Frame**

|                                   |    |                |                  |                  |                  |                |  |
|-----------------------------------|----|----------------|------------------|------------------|------------------|----------------|--|
| SETTLING TIME = 10.000MS 20.000MS |    | RETRY: DISABLE |                  | DCMODE: DCON     |                  | WETMODE: RESET |  |
|                                   | DC | VS             | IN               | VT               | IL               | DCLP           |  |
| [REF ON]                          | 1  | 129ABC         | 123456789ABCDEF0 | 123456789ABCDEF0 | 123456789ABCDEF0 | 1              |  |
| TIME1 = 1US                       | 1  | -----          | -----            | -----            | -----            | -              |  |
| TIME2 = 10US                      | -  | 129ABC         | -----            | -----            | -----            | -              |  |
| TIME3 = 100US                     | -  | -----          | 123456789ABCDEF0 | -----            | -----            | -              |  |
| TIME4 = 1.000MS                   | -  | -----          | -----            | 123456789ABCDEF0 | -----            | -              |  |
| TIME5 = 10.000MS                  | -  | -----          | -----            | -----            | 123456789ABCDEF0 | -              |  |

- SETTLING TIME  
Displays the settling time at the DC test.
- RETRY  
Displays the AUTO range measurement mode.
- DCMODE  
Displays the control mode of reference ON and OFF for the DC measurement unit.
- WETMODE  
Displays the wet mode and DC wet mode.
- REF ON  
Displays the unit for which reference ON is to be performed.
- TIME1 to 11  
Displays the sequence control for reference ON/OFF.
  - DC
  - VS1 to 12  
The number of VSs depends on the tester type as follows:  
For the T5581 and T5581H:  
VS1, 2, 9 to 12  
For the T5593, T5592, T5591, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, and T5371:  
VS1 to 12
  - IN1 to 16

- VT1 to 16
- IL1 to 16
- DCLP

**Footer**

If a tester communication error occurs, an error message is displayed here.

## 4. 7 ALPG Monitor

The ALPG monitor is used to refer and change the setup condition of the ALPG.

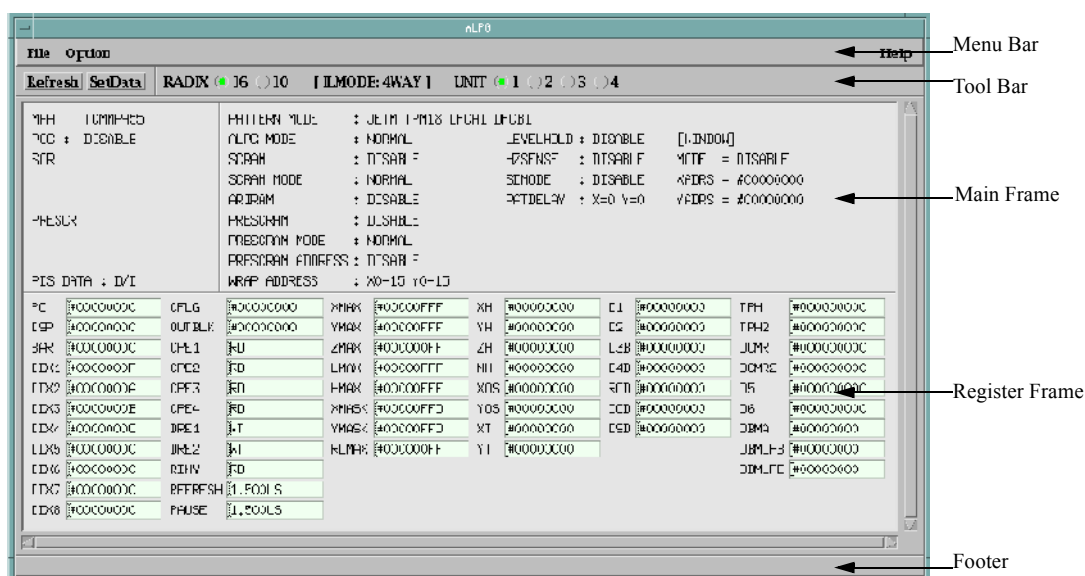


*If no pattern program file is transferred to the PG, the ALPG setup condition cannot be changed.*

### 4. 7. 1 Main Window

When you click the ALPG button in the TCM main window, [Figure 4-53](#) is displayed.

**Figure 4-53 ALPG Monitor (Screen of the T5592)**



#### Menu Bar

The menu bar consists of File, Option and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the ALPG monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the ALPG monitor.

#### Option menu

The Option menu includes the following commands:

- Send Pattern & Scram File...

Sends pattern files and scramble files, and sets the address descrambler operation mode.

➔ For more information, refer to [4. 7. 2 "Send Pattern & Scram File Window" on page 4-55](#).

### Help Menu

Displays the Online Manual regarding the ALPG.

The Help menu is a pull-down menu consisting of the following commands:

- REG MPAT...

### Tool Bar

The function of each button in the tool bar of the ALPG monitor is described below.

- Refresh

When you click the Refresh button, the current tester setup condition is loaded and the contents of the ALPG Monitor display are updated.

- SetData

When you click this button after changing each parameter, the settings changed are reflected in the tester.



Tip

*This button cannot be used if no pattern program file is transferred to the PG.*

- RADIX

Changes the radix for each value in the main frame and register frame.

When 16 is selected, values are displayed in hexadecimal.

When 10 is selected, values are displayed in decimal.

- ILMODE

Displays the interleave mode using nWAY.

For the T5571P, T5377, T5376, T5375, and T5371:

Not displayed.

For the T5593, T5586, T5585, T5581, T5581H, and T5581P:

Displays 1WAY or 2WAY.

For the T5592 and T5591:

Displays 1WAY, 2WAY, or 4WAY.

- UNIT

Selects the unit number of the ALPG to be displayed.

For the T5571P, T5377, T5376, T5375, and T5371:

Cannot select.

For the T5586, T5585, T5581, T5581H, and T5581P:

Selects 1 or 2.

For the T5591:

1 Selects a number from 1 to 4 (However, when the interleave mode is 1WAY, numbers from 3 and 4 are not available).

For the T5592:

Selects a number from 1 to 4. (However, numbers from 2 to 4 are not available when the interleave mode is 1WAY, and 2 and 4 are not available when the interleave mode is 2WAY.)

For the T5593:

Selects 1 or 2.

Selects 1 to 4 for the T5593 with the SUB PG option installed. (3 and 4 cannot be used in the 1WAY interleave mode.)

## Main Frame

**Figure 4-54 ALPG Main Frame**

|          |           |                  |                          |           |                             |
|----------|-----------|------------------|--------------------------|-----------|-----------------------------|
| MPA      | TCMPA55   | PATTERN MODE     | : JETM TPM18 DFCB1 DFCB1 |           |                             |
| PCC      | : DISABLE | ALPG MODE        | : NORMAL                 | LEVELHOLD | : DISABLE [WINDOW]          |
| SCR      |           | SCRAM            | : DISABLE                | HZSENSE   | : DISABLE MODE = DISABLE    |
|          |           | SCRAM MODE       | : NORMAL                 | SDMODE    | : DISABLE XADRS = #00000000 |
|          |           | ARIRAM           | : DISABLE                | PATDELAY  | : X=0 Y=0 YADRS = #00000000 |
| PRESCR   |           | PRESCRAM         | : DISABLE                |           |                             |
|          |           | PRESCRAM MODE    | : NORMAL                 |           |                             |
|          |           | PRESCRAM ADDRESS | : DISABLE                |           |                             |
| PDS DATA | : D/D     | WRAP ADDRESS     | : X0-15 Y0-15            |           |                             |

The items which are referred with the main frame are described below.

- MPA

Displays the pattern program names that are transferred to the PG.

- PCC (Displayed for other than the T5593, T5377, T5376, T5375, or T5371.)

Displays the setting mode of the PCC.

- SCR

Displays the scramble program files that are used for the PG.

For the T5377, when SEPARATE is specified to SCRAM MODE, scramble files can be sent to DUTs and the FM. [DUT] is displayed at the end of DUT file names and [FM] is displayed at the end of FM file names.

- PDS DATA

Displays the test data to be output to the PDS.

- PRESCR

Displays the scramble program files which are transferred to the pre-scrambler.

- PATTERN MODE

Displays the operation mode of the PG.

- ALPG MODE

Displays the operation mode of the ALPG.

- SCRAM  
Displays ENABLE/DISABLE of the scrambler.
- SCRAM MODE (not displayed for the T5376, T5375, or T5371.)  
Displays the DIVIDE mode of the scrambler.
- ARIRAM  
Displays the operation mode of the ARIRAM.
- PRESCRAM  
Displays ENABLE/DISABLE of the pre-scrambler.
- PRESCRAM MODE  
Displays the operation mode of the pre-scrambler.
- PRESCRAM ADDRESS  
Displays the mode setting which determines which addresses are sent from the prescrambler or ALPG to the DUTs, FM, or FP function.
- WRAP ADDRESS  
Displays the valid bits of the address to be output from the pre-scrambler.
- LEVELHOLD  
Displays the initialization mode of the driver output level.
- HZSENSE (Displayed other than for the T5593, T5377, T5376 or T5375.)  
Displays the HZ mode.
- SDMODE  
Displays the multi-strobe mode.
- PATDELAY (not displayed for the T5371)  
Displays the delay cycle number for X and Y addresses, and TP data.  
The TP data delay cycle number is displayed only for the T5593, T5377, T5376 and T5375.
- WINDOW MODE (Displayed other than for the T5593)  
Displays the window mode.
- WINDOW XADRS (Displayed other than for the T5593)  
Displays the window X address.
- WINDOW YADRS (Displayed other than for the T5593)  
Displays the window Y address.

Register frame

**Figure 4-55 ALPG Register Frame (Screen of the T5592)**

|      |           |         |           |       |           |     |           |     |           |        |           |
|------|-----------|---------|-----------|-------|-----------|-----|-----------|-----|-----------|--------|-----------|
| PC   | #00000000 | CFLG    | #00000000 | XMAX  | #000000FF | XH  | #00000000 | D1  | #00000000 | TPH    | #00000000 |
| ISP  | #00000000 | OUTBLK  | #00000000 | YMAX  | #000000FF | YH  | #00000000 | D2  | #00000000 | TPH2   | #00000000 |
| BAR  | #00000000 | CPE1    | RD        | ZMAX  | #000000FF | ZH  | #00000000 | D3B | #00000000 | DCMR   | #00000000 |
| IDX1 | #0000000F | CPE2    | RD        | LMAX  | #000000FF | NH  | #00000000 | D4B | #00000000 | DCMR2  | #00000000 |
| IDX2 | #0000000A | CPE3    | RD        | HMAX  | #000000FF | XDS | #00000000 | RCD | #00000000 | D5     | #00000000 |
| IDX3 | #0000000B | CPE4    | RD        | XMASK | #000000FD | YDS | #00000000 | CCD | #00000000 | D6     | #00000000 |
| IDX4 | #0000000C | DRE1    | HT        | YMASK | #000000FD | XT  | #00000000 | DSI | #00000000 | DBMA   | #00000000 |
| IDX5 | #00000000 | DRE2    | HT        | RLMAX | #000000FF | YT  | #00000000 |     |           | DBMLPB | #00000000 |
| IDX6 | #00000000 | RINV    | RD        |       |           |     |           |     |           | DBMLPE | #00000000 |
| IDX7 | #00000000 | REFRESH | 1.500US   |       |           |     |           |     |           |        |           |
| IDX8 | #00000000 | PAUSE   | 1.500US   |       |           |     |           |     |           |        |           |

The items which are referred and changed with the register frame are described below.  
The setting cannot be changed if no pattern program file is transferred to the PG.

- PC  
Specifies the start address of the ALPG.
- ISP  
Specifies the address to be jumped when an interrupt occurs during the auto-refresh test.
- BAR (Displayed other than for the T5593)  
Specifies the address to be jumped for the FLGLI1 flag sense instruction.
- IDX1 to IDX8  
Specifies the number of loops for the loop instruction (JNIn).
- IDX9 to IDX16 (Displayed for the T5593, T5377, T5376 and T5375.)  
Specifies data used for the LDIn instruction that changes the IDX1 to IDX8 register values.
- CFLG  
Specifies the branching condition for the JNCn (1 to 8) instruction.
- OUTBLK  
Specifies the start address at the time of OUT operation.
- CPE1 to CPE4  
Specifies the control signal for logical comparison cycle designation.
- DRE1, DRE2  
Specifies the control signal for driver enable for I/O pins.
- RINV  
Specifies the signal for real-time invert control.
- REFRESH  
Specifies the time at the time of the refresh test.
- PAUSE  
Specifies the time at the time of the pause test.
- ND (Displayed for the T5593, T5377, T5376 and T5375.)  
Specifies data used for the D register address operation.



- BD (Displayed for the T5593, T5377, T5376 and T5375.)  
Specifies data used for the B register address operation.
- BH (Displayed for the T5593, T5377, T5376 and T5375.)  
Specifies a B register initial value.
- FPSEL (Displayed for the T5593, T5377, T5376 and T5375.)  
Specifies a function name used to specify an operation mode of the address function FPX.
- RHZA (Displayed for the T5593)  
Specifies the MUT signal used to specify a cycle where the real time Hi-Z comparison is made using STRB specified in the RHZAn (1, 2, 129, or 130) in the pin statement.
- RHZB (Displayed for the T5593)  
Specifies the MUT signal used to specify a cycle where the real time Hi-Z comparison is made using STRB specified in the RHZBn (1, 2, 129, or 130) in the pin statement.
- ASCNT (Displayed for the T5593)  
Specifies the MUT signal used to specify a cycle where STRB and clock values can be incremented or decremented using the auto-scan clock function.
- ASINI (Displayed for the T5593)  
Specifies the MUT signal used to specify a cycle where the initial values are set for the STRB and clock values using the auto-scan clock function.
- XMAX, YMAX, ZMAX (ZMAX is displayed for other than the T5371)  
Specifies the valid bits for each of the X, Y, and Z addresses to be applied to the DUT.
- LMAX, HMAX  
Specifies the valid bits when using X, Y, and Z registers in linked form.
- XMASK, YMASK  
Specifies the data used for address functions FP7 (Mask Parity), FP8 (Row Bar), FP9 (Column Bar), and FP10 (Row/Column Bar).
- RLMAX  
Specifies the refresh counters REFRESHA n and REFRESHB n.
- TRAS (Displayed for the T5377, T5376, T5375, and T5371.)  
Specifies the time when the TRAS test is executed.
- MR1 (Displayed for the T5377, T5376, T5375, and T5371.)  
Specifies a Mode Register value used to control the PG and PG timer operation modes.
- ND2 to ND7 (Displayed for the T5593 and T5377)  
Specifies data used for the N register address operation.
- XH, YH, ZH, NH  
Specifies the initial value of XB, YB, Z, and the N register.
- XOS, YOS  
Specifies the offset value for the X and Y addresses.
- XT, YT

Specifies the address data to be output in the cycle to which the XT or YT instruction is set.

- ZD (Displayed for the T5593, T5377, T5376, T5375, and T5371.)  
Specifies the data used for the Z address operation.
- BD2 to BD7 (Displayed for the T5593 and T5377)  
Specifies data used for the B register address operation.
- D1, D2, D3B, D4B  
Specifies the data used for operation of each of the XC, XS, XK, YC, YS, and YK registers.
- RCD, CCD  
Specifies the data used for FP8 (Row Bar), FP9 (Column Bar), and FP10 (Row or Column Bar).
- DSD  
Specifies the value used as shift data for (Diagonal) and FP4 (Inverted Diagonal).
- D1B to D1H (Displayed for the T5593 and T5377)  
Specifies data used for XB, XC, XS, XK, YB, YC, YS, and YK register operations.
- D2B to D2D (Displayed for the T5593 and T5377)  
Specifies data used for XB, XC, XS, XK, YB, YC, YS, and YK register operations.
- TPH  
Specifies the data used to initialize the TP(TP1) register.
- TPH2  
Specifies the data used to initialize the TP2 registers.
- DCMR, DCMR2  
Specifies the mask bits for determining whether or not inversion by /D or /D2 is to be made.
- D5  
Specifies the data used for TP1 operation.
- D6  
Specifies the data used for TP2 operation.
- DBMA (Displayed for other than the T5377, T5376, T5375, and T5371.)  
Specifies the data used to initialize the address pointer of the DBM.
- DBMLPB (Displayed for the T5593 and T5592)  
Specifies the loop start address when DBM patterns are repeatedly generated in DBM loop mode.
- DBMLPE (Displayed for the T5593 and T5592)  
Specifies the loop end address when DBM patterns are repeatedly generated in DBM loop mode.
- DBMBA (Displayed for the T5593)  
Displays the offset address added in the tester controller when sending DBM patterns in the DBM memory.  
This setting cannot be changed.

---

 Limitation

---

A DBM unit corresponding to the SUB PG option for the T5593 is not included. Accordingly, the DBM registers (DBMA, DBMLPB, DBMLPE, and DBMBA) cannot be used for the SUB PG.

Therefore, the DBM registers cannot be used when UNIT radio button 3 or 4 is selected for the T5593.

---

- TPH1B to TPH1D (Displayed in the T5377)  
TPH (TPH1 and TPH1A to TPH1D) Specifies an initial register value.
- TPH2B to TPH2D (Displayed in the T5377)  
TPH2 (TPH2A to TPH2D) Specifies an initial register value.

**Footer**

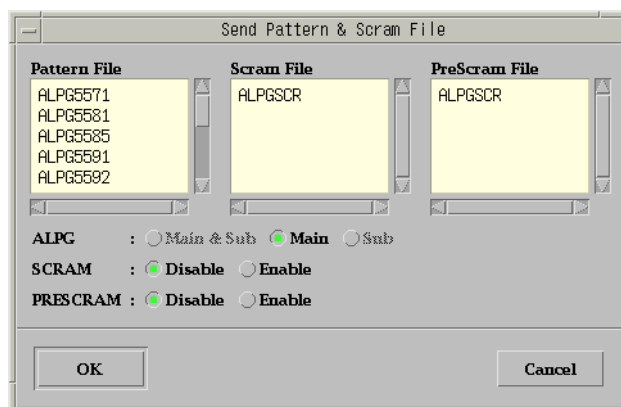
If a tester communication error occurs, an error message is displayed here.

## 4. 7. 2 Send Pattern & Scram File Window

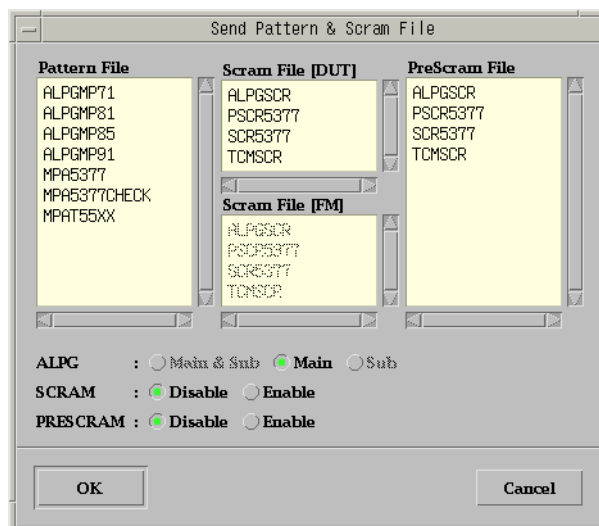
When you select Send Pattern & Scram File Window... from the Option menu, the following window is displayed.

This window is used to send pattern files and scramble files, and to set the address descrambler operation mode.

**Figure 4-56 Send Pattern & Scram File Window (Except for the T5377)**



**Figure 4-57 Send Pattern & Scram File Window (T5377)**



### Pattern File list

Used to select pattern files sent to the pattern generator.

If nothing is selected in the list, a pattern file is not sent.

### Scam File list

Used to select scramble files sent to the address descrambler.

If nothing is selected in the list, a scramble file is not sent.

For the T5377, when SEPARATE is specified to SCRAM MODE, scramble files can be sent to DUTs and the FM. The Scram File list has an upper and a lower section. The upper section displays the DUT files and the lower section displays the FM files. If no DUT files are selected, the FM list is unavailable.

#### PreScram File list

Used to select scramble files sent to the pre-scrambler.

If nothing is selected in the list, a scramble file is not sent.

#### ALPG radio button

Selects the target ALPG to which the scramble file is sent and the operation mode is set.

- Main & Sub  
Selects both main and sub ALPGs.
- Main  
Selects the main ALPG.
- Sub  
Selects the sub ALPG.

#### SCRAM radio button

Selects the address descrambler operation mode.

- Disable  
The address descrambler is disabled.
- Enable  
The address descrambler is enabled.

#### PRESCRAM radio button

Selects the operation mode of the address descrambler using pre-scrambler.

- Disable  
The pre-scrambler is disabled.
- Enable  
The pre-scrambler is enabled.

#### OK button

Sends pattern files and scramble files. Sets the address descrambler operation mode, and closes the Send Pattern & Scram File Window.

#### Cancel button

Closes the Send Pattern & Scram File Window without changing the current settings.

## 4. 8 FM Monitor

The FM monitor is used to refer the setup condition of the failure analysis memory.

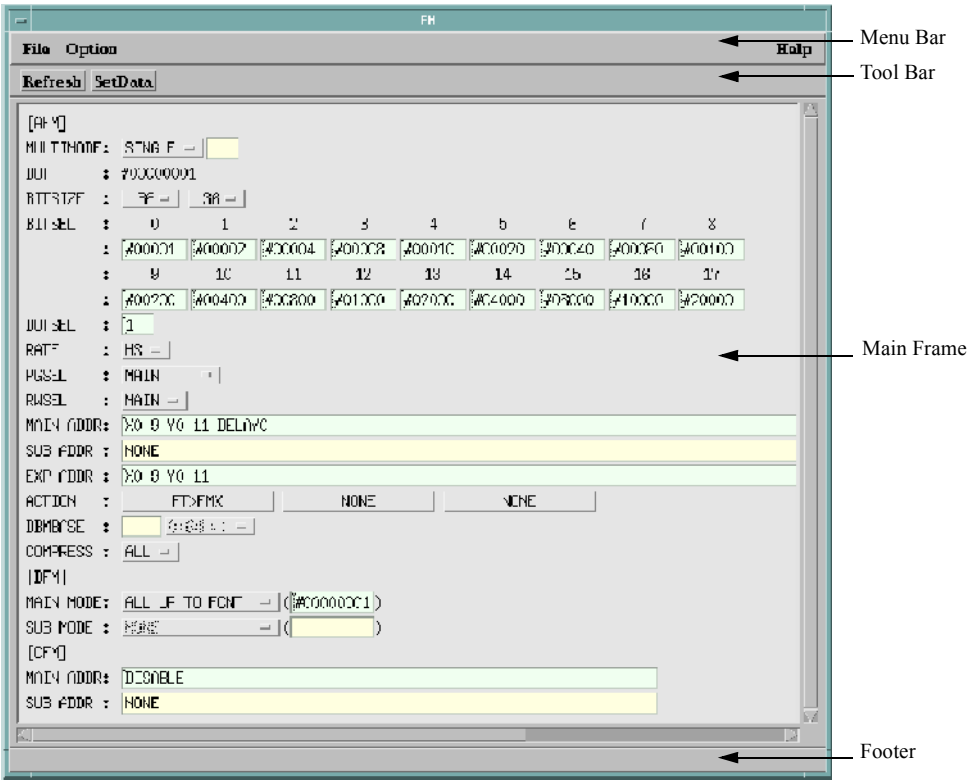
 Tip

If the AFM board is not installed in the tester being used, the FM monitor cannot be used to change the current settings.

### 4. 8. 1 Main Window

When you click the FM button in the TCM main window, the FM monitor shown in [Figure 4-58](#) is displayed.

Figure 4-58 FM Monitor (Screen of the T5581)



- Menu Bar
- The menu bar consists of File, Option and Help menus.
- File Menu
- The File menu includes the following menu commands:
- Save...

Saves data from the FM monitor to a text file.

➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).

- Close

Exits the FM monitor.

### Option menu

The Option menu includes the following commands:

- RESET FM...

Resets the AFM, CFM, MASK, or EXP memory.

➔ For more information, refer to [4. 8. 2 "RESET FM Window" on page 4-62](#).

- AFM MOVE...

Sends the failure data from the AFM to the pattern memory.

➔ For more information, refer to [4. 8. 3 "AFM MOVE Window" on page 4-63](#).

Note that this command cannot be used with the T5593, T5592 or T5591.

- AFM REMOVE...

Obtains logical OR for the failure memory data and the MASK mode memory data, and then stores the results in the MASK mode memory.

➔ For more information, refer to [4. 8. 4 "AFM REMOVE Window" on page 4-64](#).

- CFM MOVE...

Sends the fail block data from the CFM to the BBM.

➔ For more information, refer to [4. 8. 5 "CFM MOVE Window" on page 4-65](#).

This menu can be used only for the T5377, T5376, T5375 and T5371.

- CFM REMOVE...

Performs a logical sum of the fail block data between the CFM and the BBM, and sends the result to the BBM.

➔ For more information, refer to [4. 8. 6 "CFM REMOVE Window" on page 4-66](#).

This menu can be used only for the T5377, T5376, T5375 and T5371.

### Help Menu

Displays the Online Manual related to the failure analysis memory.

The Help menu is a pull-down menu consisting of the following commands:

- AFM

The AFM command includes the following commands.

MULTIMODE...

BITSIZE...

BITSEL...

DUTSEL...

RATE...

PGSEL...

RWSEL...

ADDRESS...

ACTION...

DBMBASE...

COMPRESS...

- DFM

The DFM command includes the following command.

MODE...

- CFM

The CFM command includes the following command.

ADDRESS...

- RESET AFM...

- RESET CFM...

- AFM MOVE...

- AFM REMOVE...

- CFM MOVE...

- CFM REMOVE...

### Tool Bar

The function of each button in the tool bar of the FM monitor is described below.

- Refresh

When you click the Refresh button, the current tester setup condition is loaded and the contents of the FM Monitor display are updated.

- SetData

When you click this button after changing each parameter, the settings changed are reflected in the tester.



Tip

*If the AFM board is not installed in the tester being used, the FM monitor cannot be used to change the current settings.*

---

### Main Frame



Figure 4-59 FM Main Frame

[FM]  
 MULTIMODE: SINGLE  
 DUT : #0000001  
 BITSIZE : 00 08  
 DITSEL : 0 1 2 3 4 5 6 7 8  
           : #00001 #00002 #00004 #00008 #00010 #00020 #00040 #00080 #00100  
           : 9 10 11 12 13 14 15 16 17  
           : #00200 #00400 #00800 #01000 #02000 #04000 #08000 #10000 #20000  
 DUTSEL : 1  
 RATE : HS  
 PGSEL : MAIN  
 RWSEL : MAIN  
 MAIN ADDR: X0-9 Y0-11 DELAY  
 SJB ADDR : NONE  
 EXP ADDR: X0-9 Y0-11  
 ACTION : T>MX NONE NONE  
 LUMBASE : 00000000  
 LUMPRESS : ALL  
 [U+M]  
 MAIN MODE: ALL JP TO FCNT (#0000001)  
 SJB ADDR: NONE  
 [L+FM]  
 MAIN ADDR: DTSARIF  
 SJB ADDR : NONE

The items that can be displayed and edited using the FM main frame are described below.

#### AFM

- MULTIMODE  
Specifies the mode regarding the number of DUTs of the AFM.
- DUT  
Displays the DUT number specified by CONFIGn of MULTIMODE.  
This setting cannot be changed.
- BITSIZE  
Specifies the number of bits of the DUT and the number of bits to be captured to the AFM.
- BITSEL  
Specifies which device pin is assigned to which AFM bit.
- DUTSEL  
Specifies the target DUT number.  
Note that this data can only be edited when SINGLE is specified for MULTIMODE.
- RATE  
Specifies the test rate for the AFM.
- PGSEL  
Specifies the PG to be connected with the AFM.
- RWSEL  
Specifies the AFM area where data is read from or written to.

- MAIN ADDR, SUB ADDR, EXP ADDR  
Assigns the X and Y address bits from the PG to the AFM address.
- ACTION  
Specifies the action mode for each AFM memory board.  
➔ For changing specification, refer to [4. 8. 7 "Action Setting Window" on page 4-67](#).
- DBMBASE  
Specifies the block number for clearing DBM, writing, reading, data transfer, and data generation in the text box. Also specifies the size of one block in the Option menu.  
This item can be specified only for the T5377, T5376, T5375 and T5371.
- COMPRESS  
Specifies the current bit compression mode for the AFM and CFM.

#### DFM

- MAIN MODE  
Specifies the store data and operation mode of the DFM on the MAIN side.
- SUB MODE  
Specifies the store data and operation mode of the DFM on the SUB side.

#### CFM

- MAIN ADDR, SUB ADDR  
Assigns the desired address bits for X and Y addresses specified using the AFM ADDRESS statement to the CFM address.

#### Footer

If a tester communication error occurs, an error message is displayed here.

## 4. 8. 2 RESET FM Window

When you select the RESET FM... command from the Option menu, the window shown below appears.

This window is used to reset the AFM, CFM, BBM, MASK, or EXP memory.

**Figure 4-60 RESET FM Window**



### AFM toggle button

Specifies the memory set for Action.

### CFM toggle button

Specifies the CFM.

### CFM & BBM toggle button

Specifies the CFM and the BBM.

### MASK toggle button

Specifies the MASK memory.

### EXP toggle button

Specifies the EXP memory.

### OK button

Resets the data in the memory area selected using each option button, and then closes the RESET FM window.

### Tip

*If the AFM board is not installed in the tester being used, this operation cannot be executed.*

### Cancel button

Cancels the operation and closes the RESET FM window.

### 4. 8. 3 AFM MOVE Window

When you select the AFM MOVE... command from the Option menu, the window shown below appears.

This window is used to send failure data from the AFM to the pattern memory.

**Figure 4-61 AFM MOVE Window**



#### Action list

Specifies an action mode. Failure data which is set for the specified action mode is copied.

#### DUT Number text box

Specifies the target device number.

#### OK button

Sends failure data, which is set for the action mode selected in the Action list, to the pattern memory, and then closes the AFM MOVE window.



Tip

*If the AFM board is not installed in the tester being used, this operation cannot be executed.*

#### Cancel button

Cancels the operation and closes the AFM MOVE window.



Tip

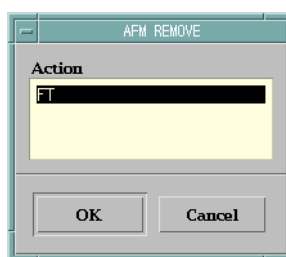
*The AFM MOVE window cannot be used with the T5593, T5592 or T5591.*

## 4. 8. 4 AFM REMOVE Window

When you select the AFM REMOVE... command from the Option menu, the window shown below appears.

This window is used to obtain logical OR between data of the failure data memory and the MASK mode memory, and stores the result in the MASK mode memory.

**Figure 4-62 AFM REMOVE Window**



### Action list

Specifies an action mode. Failure data which is set for the specified action mode is copied.

### OK button

Sends failure data, which is set for the action mode selected in the Action list, to the memory specified for the MASK mode, and then closes the AFM REMOVE window.

### Tip

*If the AFM board is not installed in the tester being used, this operation cannot be executed.*

### Cancel button

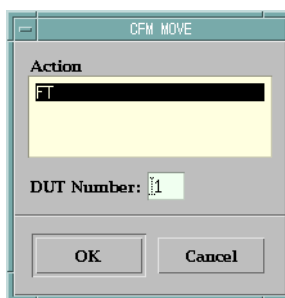
Cancels the operation and closes the AFM REMOVE window.

## 4. 8. 5 CFM MOVE Window

If you select the CFM MOVE... command from the Option menu, the CFM MOVE window shown below is displayed.

This window is used to send the fail block data from the CFM to the BBM.

**Figure 4-63 CFM MOVE Window**



### Action list

Selects an action mode for the fail block data.

### DUT Number text box

Specifies the target device number.

### OK button

Used to send the fail block data in the action mode which is specified by the Action list, and close the CFM MOVE window.

#### Tip

*If the AFM board is not installed in the tester being used, this operation cannot be executed.*

### Cancel button

Used to close the CFM MOVE window without sending the fail block data.

#### Tip

*This window can be used only for the T5377, T5376, T5375 and T5371.*

## 4. 8. 6 CFM REMOVE Window

If you select the CFM REMOVE... command from the Option menu, the CFM REMOVE window shown below is displayed.

This window is used to perform a logical sum of the fail block data between the CFM and the BBM, and sends the result to the BBM.

**Figure 4-64 CFM REMOVE Window**



### Action list

Selects an action mode for the fail block data.

### DUT Number text box

Specifies the target device number.

### OK button

Used to send the fail block data in the action mode which is specified by the Action list, and close the CFM REMOVE window.

#### Tip

*If the AFM board is not installed in the tester being used, this operation cannot be executed.*

### Cancel button

Used to close the CFM REMOVE window without sending the fail block data.

#### Tip

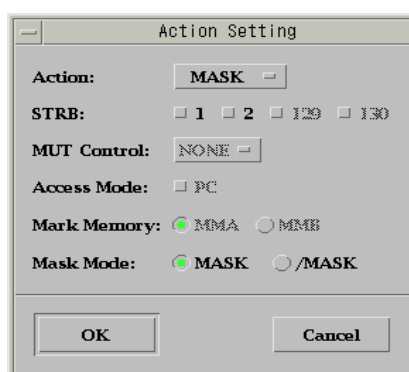
*This window can be used only for the T5377, T5376, T5375 and T5371.*

## 4. 8. 7 Action Setting Window

When you click the push button in the ACTION section of the FM main frame, the window shown below appears.

This window is used to specify the action mode of each AFM memory board.

**Figure 4-65 Action Setting Window**



### Action option menu

Specifies an action mode.

When you select NONE, no control bit is specified.

### STRB toggle button

Specifies a strobe number.

When nothing is selected, a strobe number is not specified.

### MUT Control option menu

Specifies a control bit.

When you select NONE, no control bit is specified.

### Access Mode toggle button

Specifies an access mode. When the button is on, the PC (Program Counter) of the MPAT operates the AFM memory board, when the button is off, the AP (Address Pointer) of the DBM operates the AFM memory board.

### Mark Memory radio button

Selects a signal name used for the AFM and CFM data clear operations.

### Mask Mode radio button

Selects the mode when the data stored in the AFM or BBM is generated as a mask pattern.

/MASK is selected when the data is inverted and generated as a mask pattern.

### OK button

Displays the specified action mode at ACTION in the FM main frame, and then closes the Action Setting window.



### Cancel button

Cancels the action mode setting and closes the Action Setting window.

---

#### Tip

*The action mode specified using the Action Setting window is only applied to the FM main frame display. To load the setting into the tester, click the SetData button on the tool bar.*

---

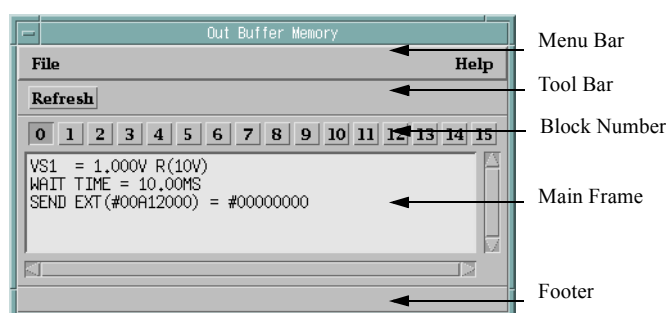
## 4. 9 OBM Monitor

The OBM monitor is used to refer the settings of the burst test condition.

### 4. 9. 1 Main Window

When you click the OBM button in the TCM main window, [Figure 4-66](#) is displayed.

**Figure 4-66 OBM Monitor**



#### Menu Bar

The menu bar consists of File and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the OBM monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Exits the OBM monitor.

#### Help Menu

Displays the Online Manual regarding the OBM.

The Help menu is a pull-down menu consisting of the following commands:

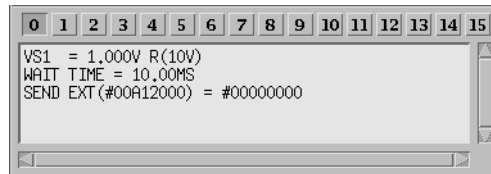
- BURST BLOCK...

#### Tool Bar

The function of each button in the tool bar of the OBM monitor is described below.

- Refresh  
When you click the Refresh button, the current tester setup condition is loaded and the contents of the OBM Monitor display are updated.

#### Main Frame

**Figure 4-67 OBM Main Frame**

Displays the burst block data of the selected block number (0 to 15).

**Footer**

If a tester communication error occurs, an error message is displayed here.

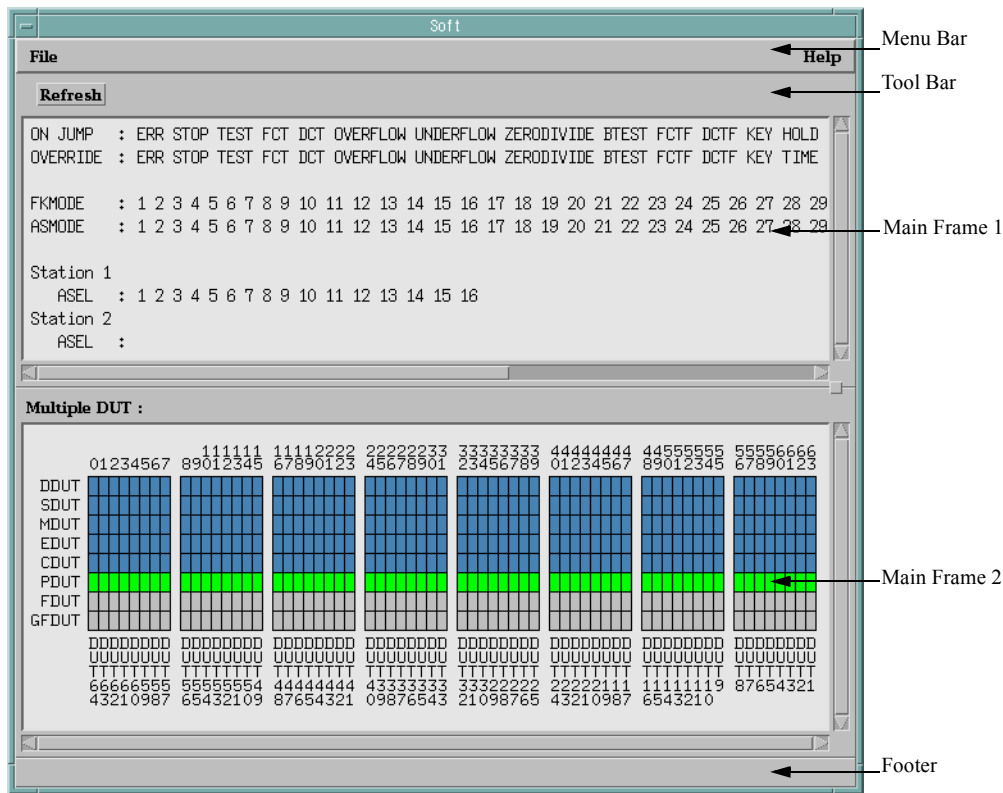
## 4.10 Soft Monitor

The Soft monitor is used to refer the software status.

### 4.10. 1 Main Window

When you click the Soft button in the TCM main window, [Figure 4-68](#) is displayed.

**Figure 4-68 Soft Monitor**



#### Menu Bar

The menu bar consists of File and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the Soft monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4.](#)
- Close  
Exits the Soft monitor.

### Help Menu

Displays the Online Manual regarding the software status.

The Help menu is a pull-down menu consisting of the following commands:

- Read mDUT...

### Tool Bar

The function of each button in the tool bar of the Soft monitor is described below.

- Refresh

When you click the Refresh button, the current tester setup condition is loaded and the contents of the Soft Monitor display are updated.

### Main Frame

#### Main Frame 1

- ON JUMP

Displays the event names used in the ON statement.

Displays \* at the beginning of the event names for which a program branch caused by the ON statement is valid.

- OVERRIDE

Displays the event names for which program branch caused by the ON statement is inhibited.

- FKMODE

Displays the setup condition for the function key mode.

- ASMODE

Displays the setup condition for the action selector mode.

- ASEL

Displays the setup condition for the action selector switch.

#### Main Frame 2

- DDUT

Displays DUTs specified in the program as the objects of simultaneous multiple device tests.

The DUTs defined in the socket program as the objects of simultaneous tests are displayed as they are.

- SDUT

Displays DUTs which were actually tested.

That is, these are the DUTs set on the handler or prober. DUTs which were jammed or which were not ON WAFER are excluded.

- MDUT

Displays object DUTs of the current tests.

DUTs on which a failure stop occurred or DUTs excluded from the test objects by the SET REJECTION statement are excluded.

- EDUT  
Displays object DUTs of the current tests.  
DUTs on which failure stop occurred or DUTs excluded from the test objects by the SET REJECTION or SET EXCLUSION statement are excluded.
- CDUT  
Displays object DUTs of the current tests.  
Usually this is the same as EDUT. However, during execution from the MULTIPLE DUT PARALLEL, SINGLE, DOUBLE, or VSPARALLEL statement to the MULTIPLE DUT END statement, only the bits corresponding to the object DUTs of these statements are displayed in order of the DUT number.
- PDUT  
Displays DUTs that passed the previous test.
- FDUT  
Displays DUTs that failed the previous test.
- GFDUT  
Displays DUTs that failed tests up to present.

**Footer**

If a tester communication error occurs, an error message is displayed.

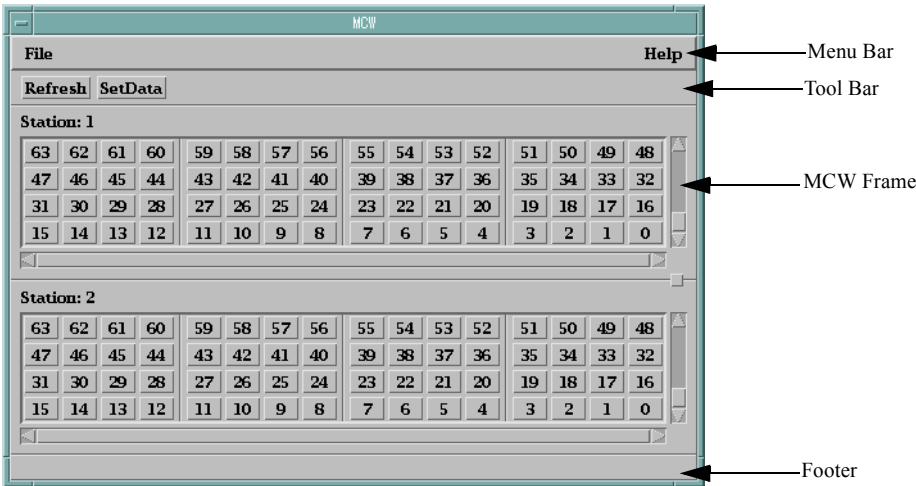
## 4.11 MCW Monitor

MCW Monitor is used to refer to or modify the setting conditions of the tester MCW.

### 4.11. 1 Main Window

When you click the MCW button in the TCM main window, [Figure 4-69](#) is displayed.

Figure 4-69 MCW Monitor



#### Tip

For the T5591, T5581P, and T5571P, button 0 to 31 can be used.  
For the T5592, T5586, 5585, T5581, T5581H, and T5371, button 0 to 63 can be used.  
For the T5593, button 0 to 127 can be used.  
For the T5377, T5376, and T5375, button 0 to 255 can be used.

#### Menu Bar

The menu bar consists of File and Help menus.

#### File Menu

The File menu includes the following menu commands:

- Save...  
Saves data from the MCW monitor to a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).
- Close  
Closes MCW Monitor.

### Help Menu

Displays the Online Manual regarding the MCW.

The Help menu is a pull-down menu consisting of the following commands:

- MCW...

### Tool Bar

The tool bar includes the following buttons:

- Refresh

When you click the Refresh button, the current tester setup condition is loaded and the contents of the MCW Monitor display are updated.

- SetData

Sets the contents modified in the MCW frame to the tester.

### MCW Frame

The MCW frame provides buttons 0 to 255 corresponding to tester MCW bits used for each test station.

Each button has two statuses: bit 1 and bit 0.

When the bit is 1, the corresponding button is pressed.

When the bit is 0, the corresponding button is released.

### Footer

If a tester communication error occurs, an error message is displayed here.



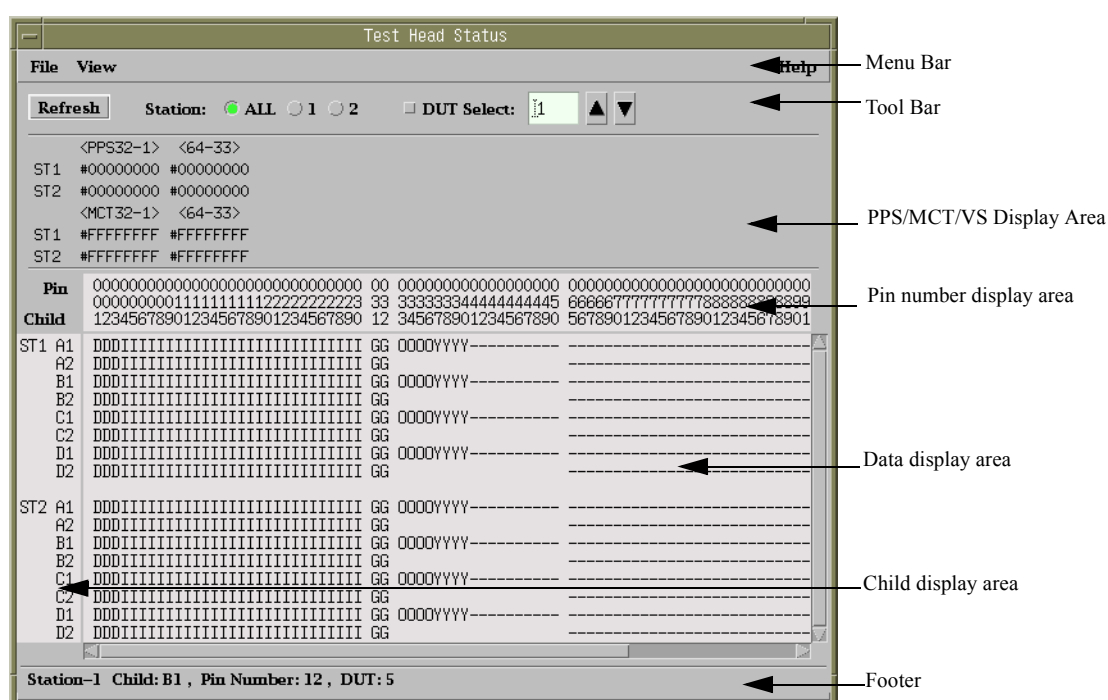
#### 4.12 Test Head Status Monitor

The Test Head Status monitor is used to display the current relay conditions on the test head.

### 4.12. 1 Main Window

When you click the Test Head button in the TCM main window, [Figure 4-70](#) is displayed.

**Figure 4-70 Test Head Status Monitor**



## Menu Bar

This is a group of menu commands that can be used in the Test Head Status screen. The menu bar consists of File, View and Help menus.

## File Menu

The File menu includes the following menu commands:

- Save ...  
Saves the displayed information in a text file.  
➔ For more information, refer to [4. 1. 1 "File Save Window" on page 4-4](#).  
The output format is the same as the one used for /THTYPE.
- Close  
Quit the Test Head Status monitor.

### View Menu

The View menu is a pull-down menu consisting of the following commands.

- Select Relay Status ...  
Highlights a specified symbol in the data display area.  
➔ For more information, refer to [4. 12. 2 "Relay Status Window" on page 4-80](#).
- VS-PPS Table ...  
Displays PPS numbers assigned to VSs for each DUT.  
This command can be used only when the DUT Select check box on the tool bar is selected.  
➔ For more information, refer to [4. 12. 3 "VS-PPS Table Window" on page 4-82](#).
- PPS/MCT Number ...  
Displays the PPS and MCT numbers in decimal.  
➔ For more information, refer to [4. 12. 4 "PPS/MCT Number Window" on page 4-83](#).

### Help Menu

The Online Manual related to the Test Head Status screen is displayed.

The Help menu is a pull-down menu consisting of the following commands:

- THTYPE ...

### Tool Bar

Tool bar consists of the following functions.

#### Refresh

Obtains the current tester setup conditions and updates the contents of the Test Head Status Monitor display.

#### Station

Specifies a display target Station.

Only Station 1 is displayed for the T5591.

- ALL  
Displays relay conditions set for both Stations.
- 1  
Displays relay conditions set for Station 1.
- 2  
Displays relay conditions set for Station 2.

#### DUT Select

Highlights the pin relay conditions set for a selected DUT in the data display area.

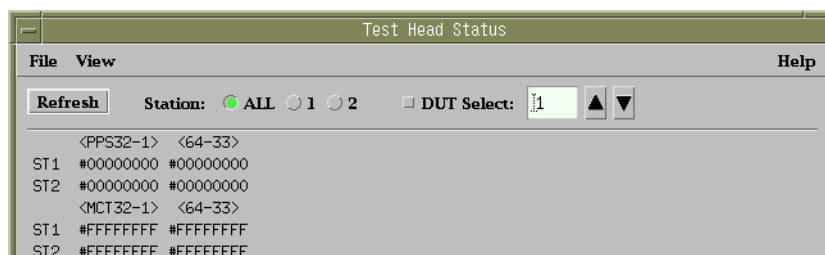
- ➔ For more information, refer to [4. 12. 5 "Highlighting Pin Relay Conditions for Each DUT" on page 4-84](#).
- DUT Select check box

When the DUT Select check box is selected, the DUT Select function is enabled. When cleared, the function is disabled.

- Text Box  
Displays a selected DUT number.
- Upper arrow button  
Increments the displayed DUT number by 1 in the text box.
- Lower arrow button  
Decrements the displayed DUT number by 1 in the text box.

#### PPS/MCT/VS Display Area

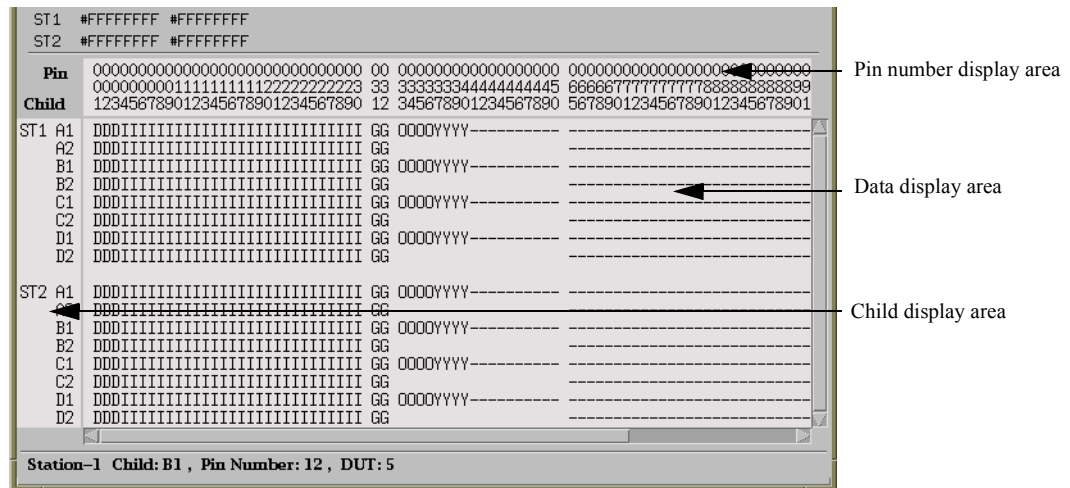
**Figure 4-71 PPS/MCT/VS Display Area**



- PPS256-1  
Displays PPS mask relay conditions for the number of PPSs which differs for testers.  
The relay conditions are displayed in hexadecimal and the PPSs which are set to 1 are electrically masked.
- MCT128-1  
Displays the MCT conditions on the performance board in hexadecimal for the number of MCTs which differs for testers.  
MCT is not displayed for the T5593. PPS320-257 is displayed.
- VS12-9  
Displays the VS9 to VS12 mask relay conditions in hexadecimal.  
This works for the T5581 and T5581H only.

#### Data display area

Figure 4-72 Data Display Area



#### Pin number display area

Displays the pin numbers which can be used with the tester.

#### Child number display area

Displays the Child numbers which can be used with the tester.

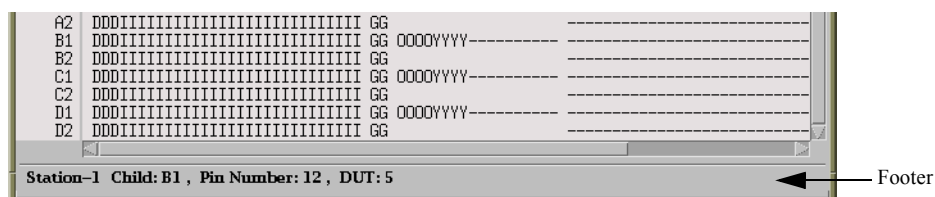
#### Data display area

Displays the current relay conditions on the test head.

The data display area displays the following relay conditions.

- DC relay
- CAL relay
- POUT relay
- IOC relay
- DR relay
- CMP relay
- PPS mask (and PCON relay)
- MCT signal
- Limit Fail
- GND

#### Footer



The footer area displays the following messages.

#### Error message

When an error (for example, a communication error of a tester) occurs, an error message is displayed here.

#### DUT number

The number of the DUT at the location of the mouse cursor is displayed here.

This works only when a used socket program defines parallel tests.

When a driver is used for two or more DUTs in parallel, the smallest DUT number is displayed.

#### ◆ For 16-device Parallel Tests Using Two Test Stations

- When placing a mouse cursor in the Station1 area.  
Station-1 Child: A1 , Pin Number: 2 , DUT: 2
- When placing a mouse cursor in the Station2 area.  
Station-2 Child: A1 , Pin Number: 2 , DUT: 2 (Total DUT: 10)

## 4.12. 2 Relay Status Window

When Select Relay Status... is selected from the View menu, the following window will be displayed.

This window shows the meanings of symbols displayed in the data display.

When clicking a toggle button for a symbol, the selected symbol is highlighted in the data display area.



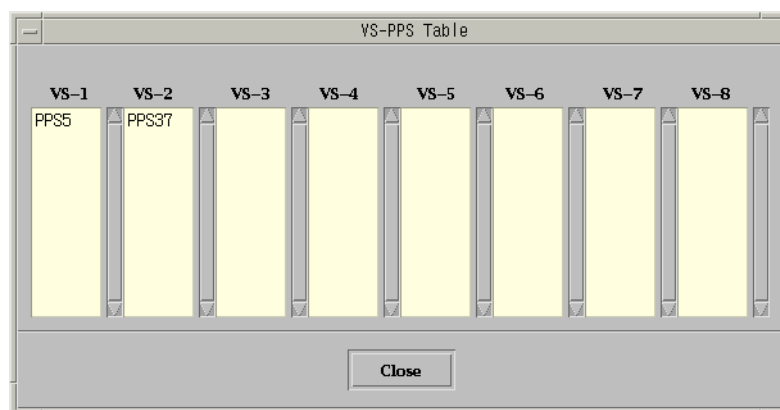
### 4.12. 3 VS-PPS Table Window

When VS-PPS Table... is selected from the View menu, the following window will be displayed.  
 This window shows the PPS numbers assigned to VS of DUTs.  
 To select a DUT, use DUT Select on the tool bar.

 Tip

*The VS-PPS Table window is not displayed when the DUT Select check box is cleared.*

**Figure 4-74 VS-PPS Table Window**



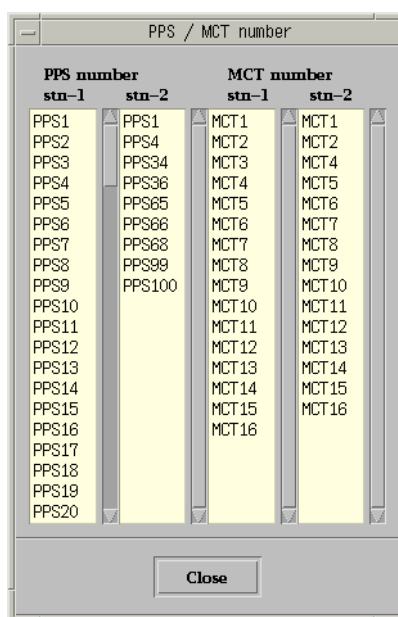
- VS-1 to VS-8 Lists  
These lists indicate the PPS numbers assigned to each VS.
- Close button  
When this button is pressed, the VS-PPS Table window closes.

## 4.12. 4 PPS/MCT Number Window

When PPS/MCT Number... is selected from the View menu, the following window will be displayed.

In this window, the PPS/MCT numbers displayed on the PPS/MCT/VS display in the main window are decimal numbers.

**Figure 4-75 PPS/MCT Number Window**



- PPS Number List  
This list indicates the PPS numbers of each station.
- MCT Number List  
This list indicates the MCT numbers of each station.
- Close button  
When this button is pressed, the PPS/MCT Number window closes.



### 4.12. 5 Highlighting Pin Relay Conditions for Each DUT

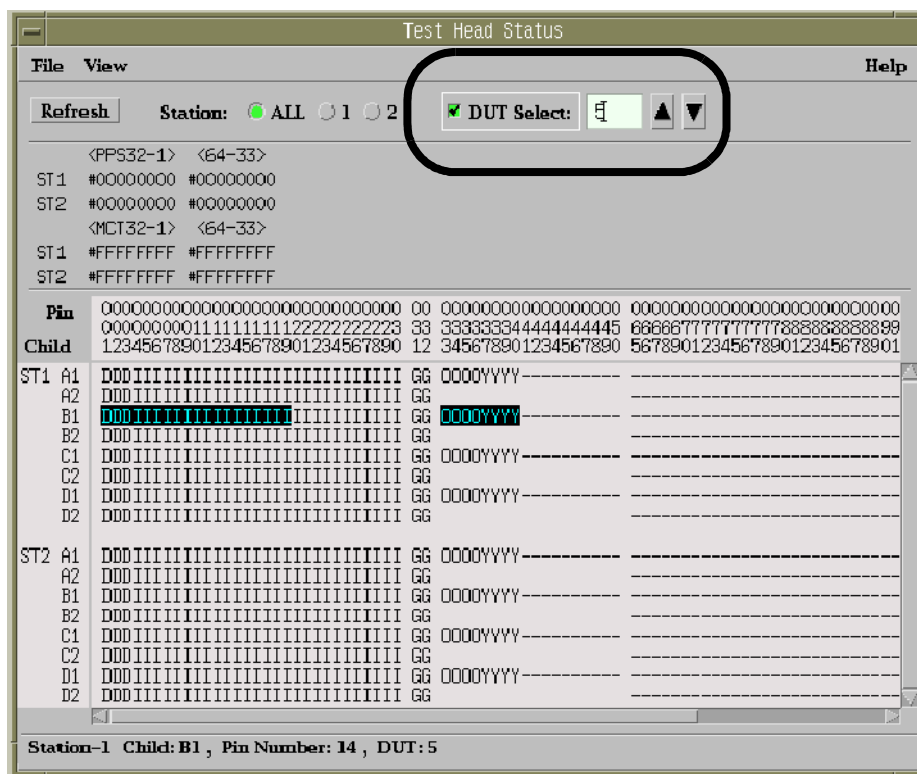
When the DUT Select check box on the tool bar is selected, pin relay conditions set for the DUT (which is selected in the text box) is highlighted. To change the DUT number, click the up or down arrow button or enter the number into the text box directly.

You can specify the DUT number from 1 to the number of devices to be tested in parallel in the program currently running.

When the pin numbers used for a selected DUT are out of view, the display position is automatically adjusted so that entire pin data (relay conditions) can be displayed on the screen.

Clicking the left mouse button on the data display area also allows you to select a target DUT. In this case, when a driver is used for two or more DUTs in parallel, pin relay conditions used for the DUT of the smallest number (of these DUTs) are highlighted.

**Figure 4-76 DUT Select**



## 4.13 Error Message List

Could not get data from Tester.  
Could not obtain data from the tester.

Could not set data to Tester.  
Could not set data to the tester.

No match unit.  
The unit is not correct.

Out of range.  
The setting range is exceeded.

Syntax error.  
Setting is not correct.

Socket file not found.  
Could not find the socket file.

Pattern program file is not sending to Tester.  
No pattern program file is sent to the tester.

File not found.  
A file name is not specified.

File name too long.  
The specified file name is too long.

Please select the Output Monitor.  
Select the monitor for file output.

AFM board is not mounted on the tester.  
AFM board is not mounted on the tester.

Please select a DUT number.

Select the DUT number.

**DUT number error.**

The specified DUT number exceeds the number of devices to be tested in parallel.

**Memory allocate error.**

Memory cannot be allocated.

**Only pin statement is allowed.**

No statements other than the pin statement can be specified.

## 5. Online Manual

The Online Manual is used to search for and display statements, functions, and error codes related to the test system on the workstation in real time.

### Tip

*To use the Online Manual, install Netscape Navigator, Acrobat Reader, and the Online Manual data from the CD-ROM of "PASX51- \*\* 2/2 Online Manual."*

→ For more information, refer to [Online Manual \(HTML\) Reference Manual\(No. : 8700036\)](#) and [Online Manual Reference Manual\(No. : 8431523\)](#).

---

### 5. 1 Main Window

Clicking the Online Manual button in the tool box displays the Online Manual. The Online Manual can also be started from the ASXDebugger Help menu or the TCM Help menu.

### Tip

*If ATLworks is used in Solaris 2.5.1, the PDF version of the Online Manual starts.*

*If Solaris 7 or later is used, the HTML version of the Online Manual starts.*

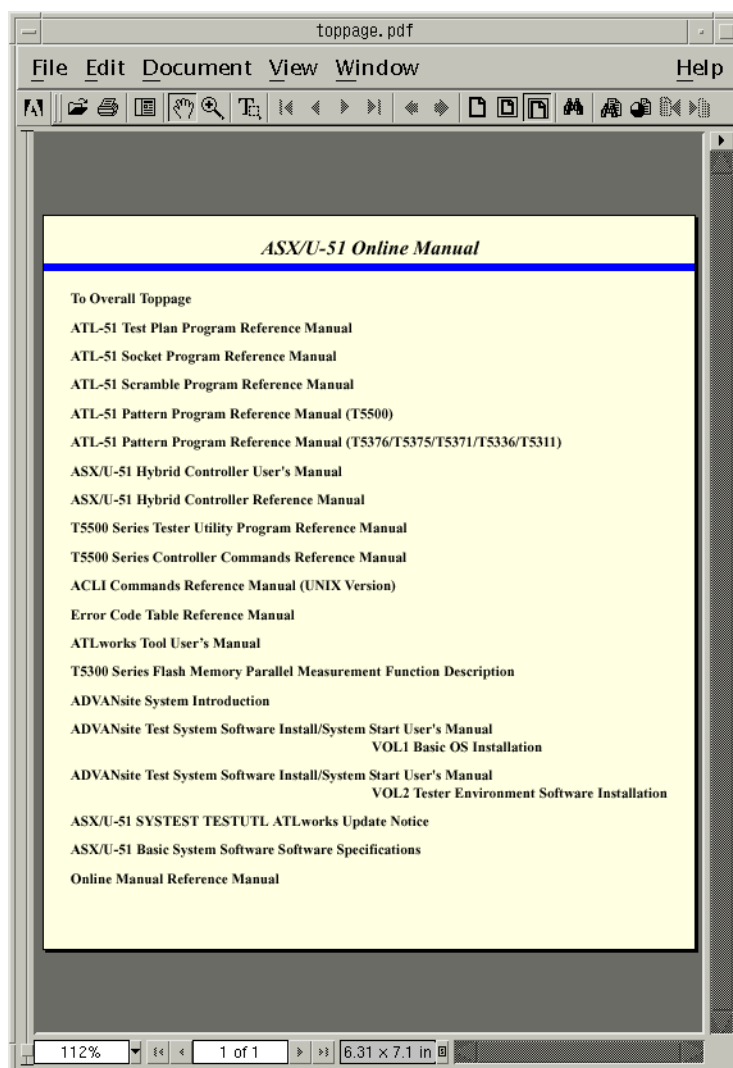
→ For more information, refer to [Online Manual \(HTML\) Reference Manual\(No. : 8700036\)](#) and [Online Manual Reference Manual\(No. : 8431523\)](#).

---

#### 5. 1. 1 Using Solaris 2.5.1

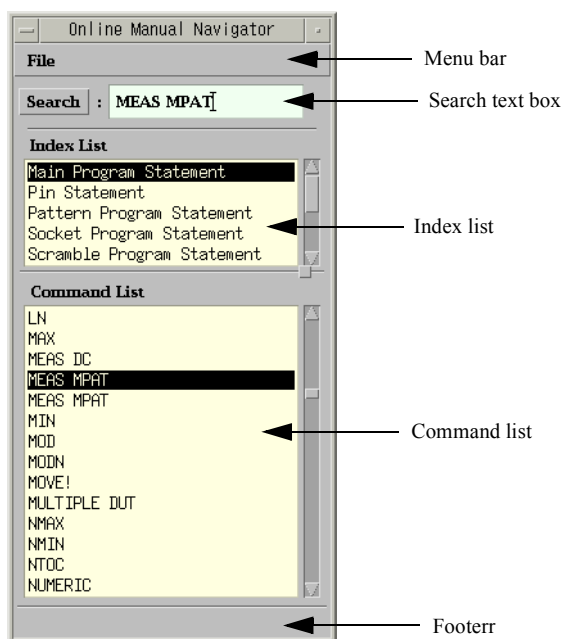
Clicking the Online Manual button in the tool box displays [Figure 5-1](#).

Figure 5-1 Main Window (when Using Solaris 2.5.1)



## 5. 1. 2 Using Solaris 7 or Later

Clicking the Online Manual button in the tool box displays [Figure 5-2](#).

**Figure 5-2 Main Window (when Using Solaris 7 or Later)****Menu bar**

The Menu bar has the File menu.

The File menu includes the following menu command:

- Close button  
Closes the window.

**Search text box**

Enter a command. If the command is found in the specified index list, the Online Manual (HTML version) is displayed. If an Error Code is selected in the index list, enter the error code.

**Search button**

If a character string is selected in another window, clicking this button stores the character string in the search text box and searches for the string.

**Index list**

Displays the title of each manual. Clicking on the list displays the related command list of the selected manual.

**Command list**

Displays the list of each manual items. Clicking on the list displays the Online Manual (HTML version).

**Footer**

Displays the error which occurred during the window operation.

## 5. 2 Example Usage in Solaris 2.5.1

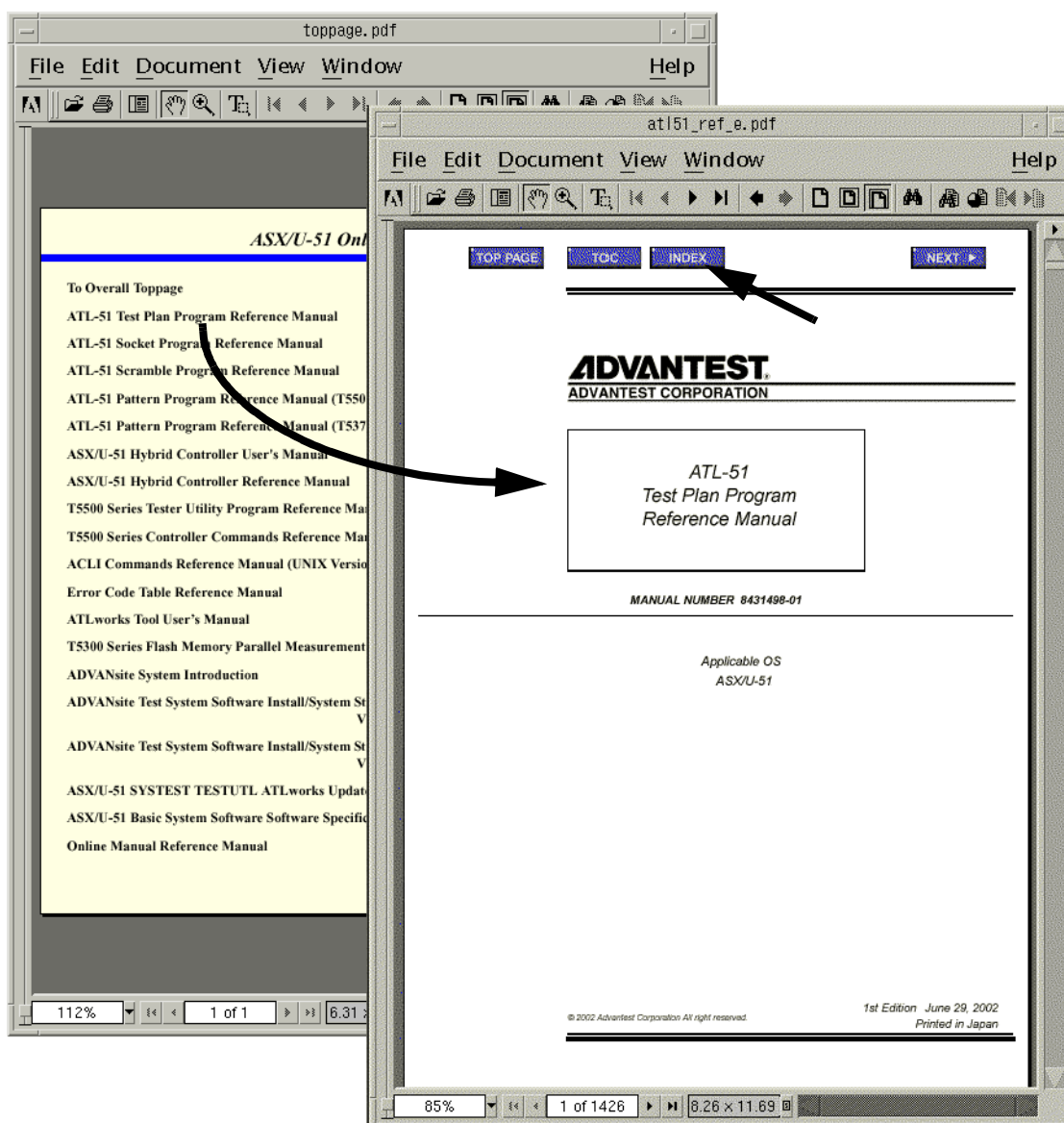
How to refer to the ACLK statement is described below.

### 5. 2. 1 Displaying the Test Plan Program Reference Manual

Clicking the manual name **[ATL-51 Test Plan Program Reference Manual]** in the Main Window displays the top page of the manual.

Clicking the INDEX button on the top page displays the index page.

**Figure 5-3 Displaying the Test Plan Program Reference Manual**

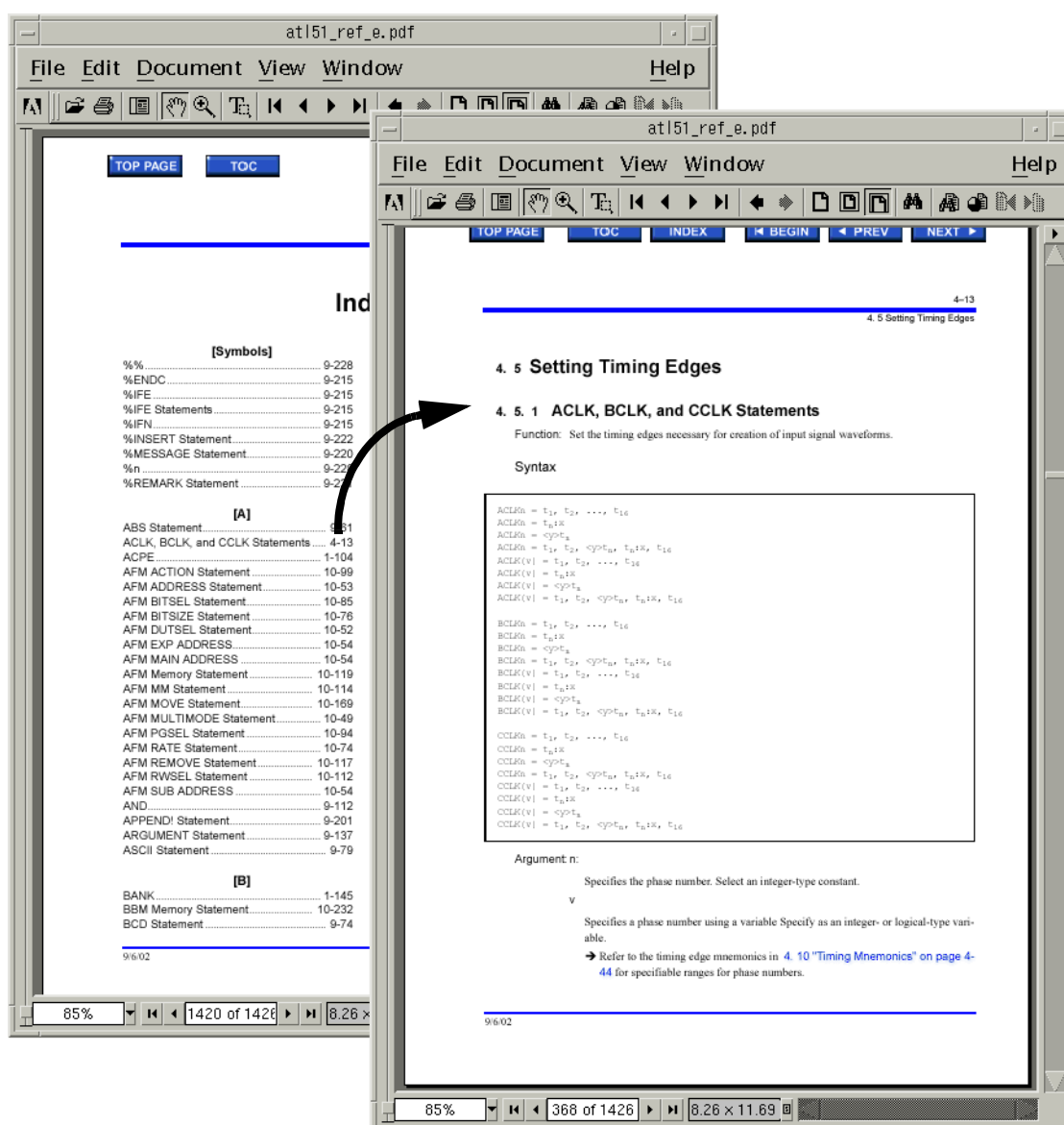


## 5. 2. 2 Displaying the ACLK Statement

Clicking the page number "ACLK..... 4-13" in the index page displays the page related to the ACLK statement.

➔ For more information on other functions, refer to [Online Manual Reference Manual\(No. : 8431523\)](#).

**Figure 5-4 Displaying the Main Body**



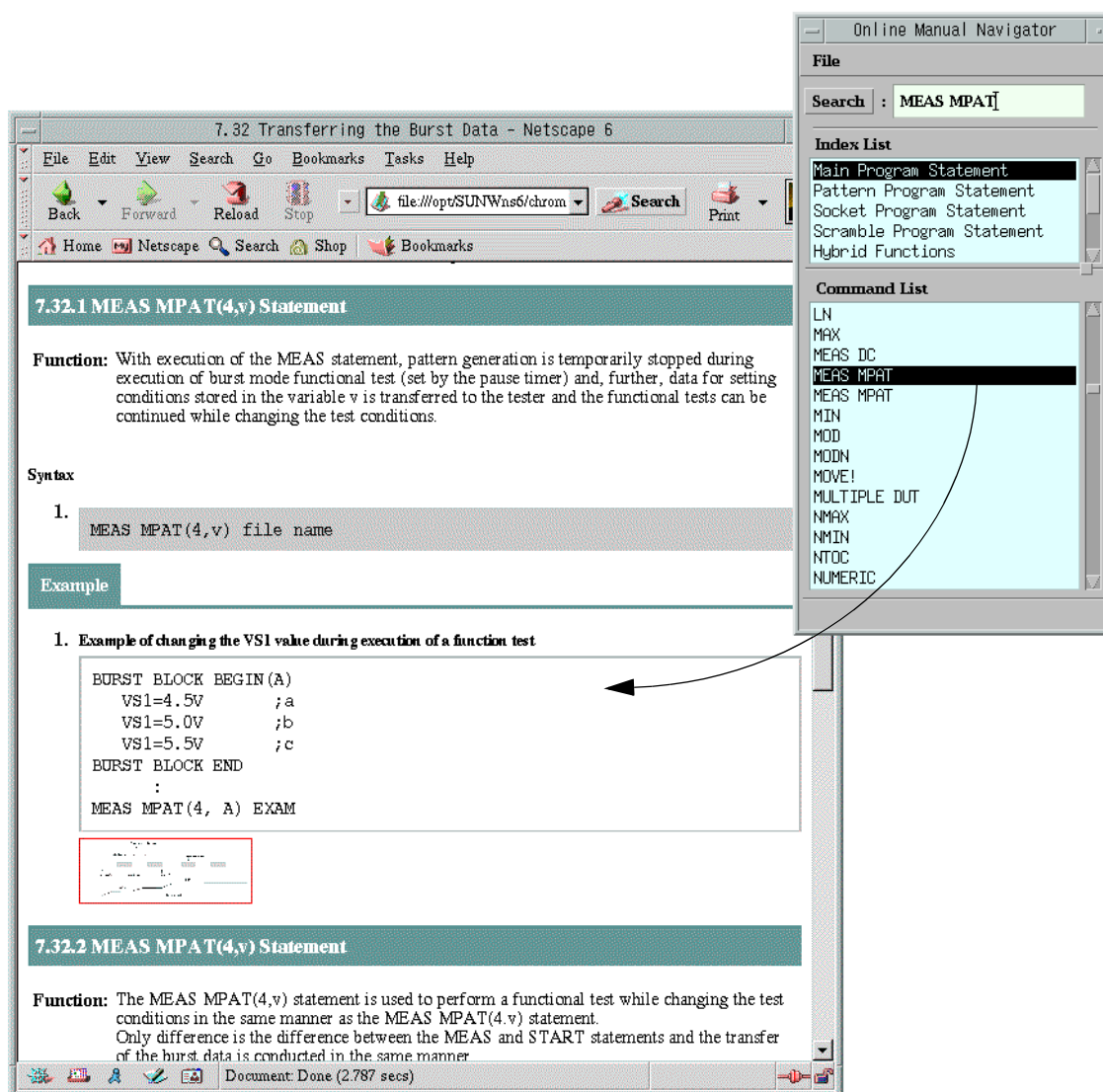


## 5. 3 Example Usage in Solaris 7 or Later

### 5. 3. 1 Displaying MEAS MPAT

- Click on the **[Main Program Statement]** in the index list.
- The command list is displayed. Click **[MEAS MPAT]**.
- Netscape Navigator starts and then the MEAS MPAT page is displayed.

Figure 5-5 MEAS MPAT Display



---

 Tip

*Netscape Navigator's initial settings are not specified. Note the following:*

- *My Sidebar is displayed in the initial state. To display the window as shown in the example, deselect the check button from [View] → [My Sidebar].*

*Only one Netscape Navigator window can be started from the ATLworks. Therefore, clicking on a new command in the command list overwrites the page in the same Netscape Navigator window. To view the previous page, click on the [Back] button.*

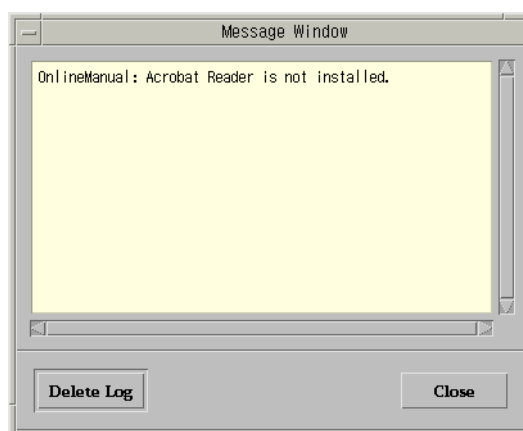
---

## 5. 4 Error Message List

If an error occurs, an error message is displayed in the message window or on the footer depending on the message content.

### 5. 4. 1 Message Displayed in the Message Window

Figure 5-6 Message Window



- OnlineManual: Netscape Navigator is not installed.  
Netscape Navigator is not installed or the search path is not specified.
- OnlineManual: Acrobat Reader is not installed.  
Acrobat Reader is not installed or the search path is not specified.
- OnlineManual: Manual is not installed.  
The Online Manual is not installed.

### 5. 4. 2 Message Displayed on the Footer

- Please select a index list.  
Specify the index list.
- '{key-word}' is not found.  
The specified {key-word} is not found.
- Memory allocate error.  
Memory cannot be allocated.

## 6. Execution Time Measurement Tool(Time Scope)

The execution time measurement tool measures the execution time of each test and the specified tests in the test program.

### Restrictions

- The execution time measurement tool can be used only in the tester on-line mode. It cannot be used when ATLworks is activated in the tester offline mode.
- The execution time measurement tool can perform data collection for up to 10000 lines. Lines exceeding 10000 lines are no displayed.

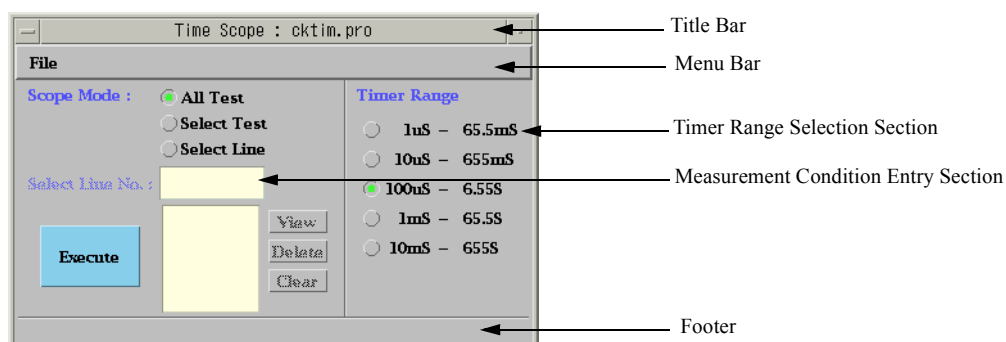
### Tip

- The time for the range between the top of the program and the first TEST statement is the TEST0 execution time.
- The last TEST execution time is the time for up to the end of the program (STOP or END).
- When you use the execution time measurement tool, the timer setting and read operation is performed in the OS and therefore the total execution time of the test program becomes longer to some extent.

### 6. 1 Main Window

When you click the Time Scope button in the tool box, [Figure 6-1](#) is displayed.

**Figure 6-1 Main Window**



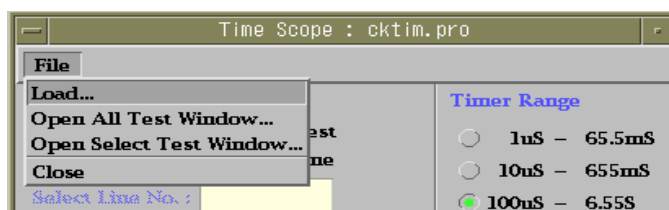
#### Title Bar

Displays the name of the program which is the target of time measurement.

## Menu Bar

Provides the menus you can use with the execution time measurement tool.

**Figure 6-2 Menu Bar**

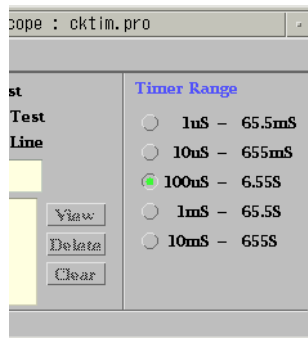


The File menu includes the following four menu commands:

- Load...  
Loads a program which is the target of time measurement.  
➔ For more information, refer to [6. 5 "Loading a Program for Which Measurement is to Be Made \(File Load Window\)" on page 6-10.](#)
- Open All Test Window...  
Displays the all test execution time display window.  
➔ For more information, refer to [6. 3 "All Test Window" on page 6-7.](#)
- Open Select Test Window...  
Displays the execution time display window for the specified test item.  
➔ For more information, refer to [6. 4 "Select Test Window" on page 6-9.](#)
- Close  
Exits the execution time measurement tool.

## Timer Range Selection Section

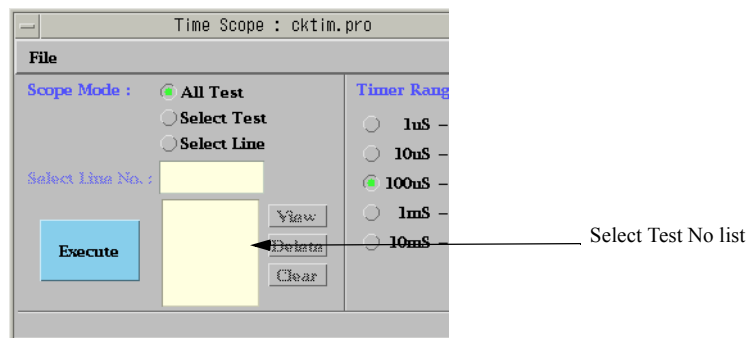
Selects the range for the timer used for time measurement. The resolution to be set differs, depending on the selected range.

**Figure 6-3 Specification of a Resolution**

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| 1 uS -65 ms     | Specifies a resolution of 1 us. Measurement for up to 65 ms is possible.    |
| 10 uS - 655 ms  | Specifies a resolution of 10 us. Measurement for up to 655 ms is possible.  |
| 100 uS - 6.55 S | Specifies a resolution of 100 us. Measurement for up to 6.55 s is possible. |
| 1 mS - 65.5 S   | Specifies a resolution of 1 ms. Measurement for up to 65.5 s is possible.   |
| 10 mS - 655 S   | Specifies a resolution of 10 ms. Measurement for up to 655 s is possible.   |

**Footer**

Displays mainly error messages on a certain operation basis.

**Measurement Condition Entry Section****Figure 6-4 Measurement Condition Entry Section**

Functions of each button in the measurement condition entry section are as follows:

- Scope Mode

Mode selection button. There are three modes as follows:

|             |                                                        |
|-------------|--------------------------------------------------------|
| All Test    | Execution time measurement for all tests               |
| Select Test | Execution time measurement for each test               |
| Select Line | Time measurement for the range between specified lines |

- Select Test No.

The text box into which the test number is entered when the execution time of each test is measured

The entered test number is registered in the Select Test No. list.

In addition, enter line numbers for time measurement for the range between the specified lines. Lines numbers can be entered by concatenating them with a hyphen (-). Specify the third line or after as the measurement start line number.

(Example) To specify the range of time measurement between the 50th and 60th lines, enter 50-60.

- View

After execution of execution time measurement for each test, specify the test number to be displayed.

- Delete

Deletes the test number selected in Select Test No.

- Clear

Clears the entire Select Test No. list.

- Execute

Starts execution time measurement.

## 6. 2 Time Measurement Method

Time measurement can be performed based on test number specification and line number specification.

**test number specification and line number specification.**

To perform execution time measurement for all tests, select the All Test measurement mode, then click the Execute button.

When execution time measurement for all tests is completed, [Figure 6-6](#) is displayed automatically.

In this case, to check the details on the execution time for a certain test, select the Select Test measurement mode and then click the target test number in the test number list in the All Test window to register it in the Select Test No. list of the measurement condition entry section.

➔ For more information, refer to [6. 3. 1 "Selection of a Test Number" on page 6-8](#).

Upon completion of registration to the list, click the Execute button.

When execution time measurement for each test is completed, the Select Test window is displayed automatically and then the execution time for each line in the test is displayed.

In addition, to select the test number to be displayed when multiple tests are registered, select the target test number from the Select Test No. list and then click the View button.

**Execution time measurement with line number specification**

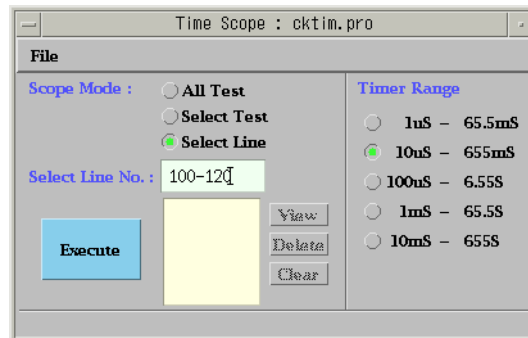
To perform execution time measurement for the range between lines, select the Select Line measurement mode, enter the line number to be measured in the Select Line No. text box, then click the Execute button.

When execution time measurement for the range between lines is completed, the Select Line window is displayed automatically and then the execution time for each specified line is displayed.

The execution time for the lines between the starting line and the thousandth line, or between the starting line and the last line of the program can be measured, when line numbers are specified in the form of "starting line number-\$."

The Select Test window and Select Line window are the same window.



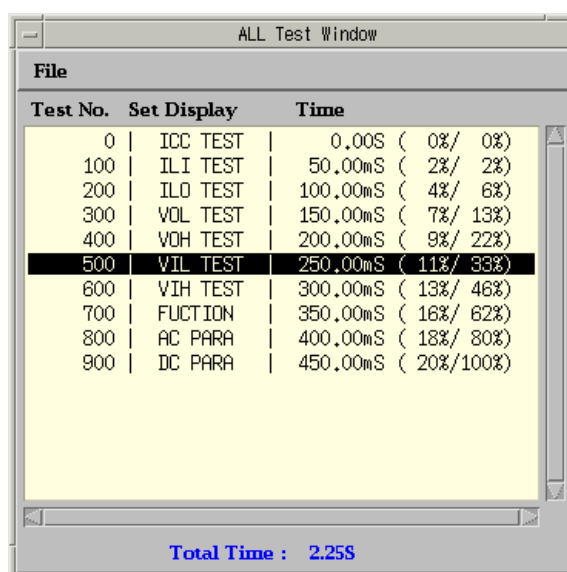
**Figure 6-5 Line Number Entry**

## 6. 3 All Test Window

When you click the Execute button in the All Test mode, execution time measurement for all tests is completed, [Figure 6-6](#) is displayed automatically.

The All Test window displays the test numbers, the execution time of each test, the ratio of each execution time to the total execution time, and the contents of SET DISPLAY.

**Figure 6-6 All Test Window**



| Test No. | Set Display | Time                   |
|----------|-------------|------------------------|
| 0        | ICC TEST    | 0.00S ( 0% / 0%)       |
| 100      | ILI TEST    | 50.00mS ( 2% / 2%)     |
| 200      | ILD TEST    | 100.00mS ( 4% / 6%)    |
| 300      | VOL TEST    | 150.00mS ( 7% / 13%)   |
| 400      | VOH TEST    | 200.00mS ( 9% / 22%)   |
| 500      | VIL TEST    | 250.00mS ( 11% / 33%)  |
| 600      | VIH TEST    | 300.00mS ( 13% / 46%)  |
| 700      | FUCTION     | 350.00mS ( 16% / 62%)  |
| 800      | AC PARA     | 400.00mS ( 18% / 80%)  |
| 900      | DC PARA     | 450.00mS ( 20% / 100%) |

Total Time : 2.25S

The File menu includes the following three menu commands:

### Save...

Saves the result of execution time measurement for all tests in a file.

➔ For more information, refer to [6. 6 "Storing the Measurement Result \(File Save Window\)" on page 6-11.](#)

### Print...

Outputs the result of execution time measurement for all tests to the printer.

➔ For more information, refer to [6. 7 "Printing Out the Measurement Result \(Printing Window\)" on page 6-13.](#)

### Close

Exits the All Test window.

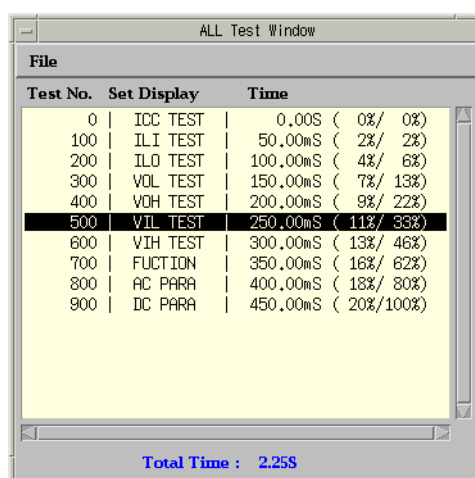
### Tip

*The data exceeding the measurement range is displayed by \*\*\*.  
If there is even one \*\*\*, the total test time and the percentage will also be \*\*\*.*

### 6. 3. 1 Selection of a Test Number

To check the details on the execution time for a certain test after measuring the execution time for all tests, select the Select Test measurement mode and then click the target test number in the test number list in the All Test window to register it in the Select Test No. list of the measurement condition entry section ( [Figure 6-4](#)).

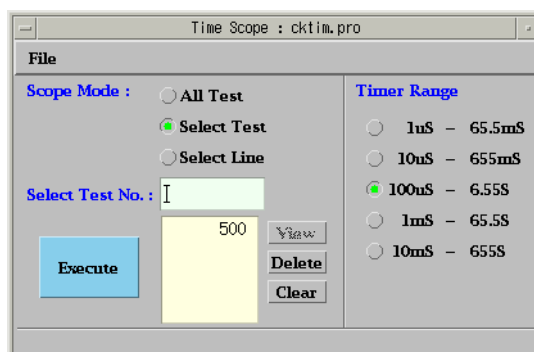
**Figure 6-7 Selection of the All Test Window**



| Test No. | Set Display | Time                   |
|----------|-------------|------------------------|
| 0        | ICC TEST    | 0.00S ( 0% / 0%)       |
| 100      | ILI TEST    | 50.00mS ( 2% / 2%)     |
| 200      | ILO TEST    | 100.00mS ( 4% / 6%)    |
| 300      | VOL TEST    | 150.00mS ( 7% / 13%)   |
| 400      | VOH TEST    | 200.00mS ( 8% / 22%)   |
| 500      | VIL TEST    | 250.00mS ( 11% / 33%)  |
| 600      | VIH TEST    | 300.00mS ( 13% / 46%)  |
| 700      | FUCTION     | 350.00mS ( 16% / 62%)  |
| 800      | AC PARA     | 400.00mS ( 18% / 80%)  |
| 900      | DC PARA     | 450.00mS ( 20% / 100%) |

Total Time : 2.25S

**Figure 6-8 Registration of a Test Number**



Time Scope : cktim.pro

File

Scope Mode : ☐ All Test ☒ Select Test ☐ Select Line

Select Test No. :

Execute

Timer Range

☐ 1uS - 65.5mS

☐ 10uS - 655mS

☒ 100uS - 6.55S

☐ 1mS - 65.5S

☐ 10mS - 655S

View Delete Clear

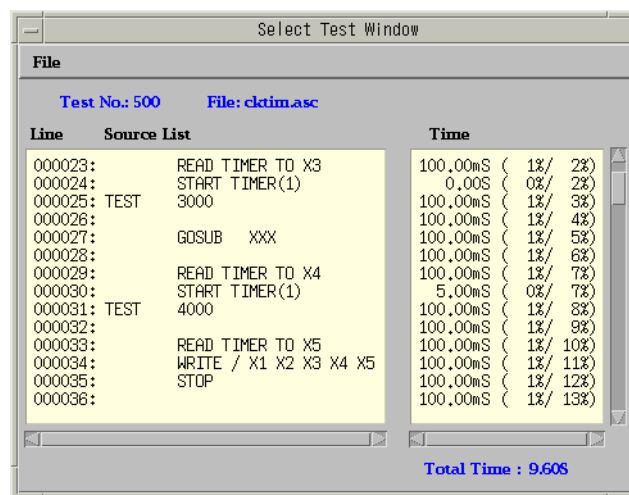
## 6. 4 Select Test Window

Specify the test number to be measured in Select Test mode and then click the Execute button. When execution time measurement is completed, [Figure 6-9](#) is displayed automatically.

The Select Test window displays the execution time for each line in the specified test, the line number, the source list, and the source file name.

By selecting a test number from the Select Test No. list in the measurement condition entry section ([Figure 6-4](#)) and clicking the View button, the details of the test number are displayed.

**Figure 6-9 Select Test Window**



The File menu includes the following three menu commands:

### Save...

Saves the result of execution time measurement for the specified test in a file.

➔ For more information, refer to [6. 6 "Storing the Measurement Result \(File Save Window\)" on page 6-11.](#)

### Print...

Outputs the result of execution time measurement for the specified test to the printer.

➔ For more information, refer to [6. 7 "Printing Out the Measurement Result \(Printing Window\)" on page 6-13.](#)

### Close

Exits the Select Test window.

### Tip

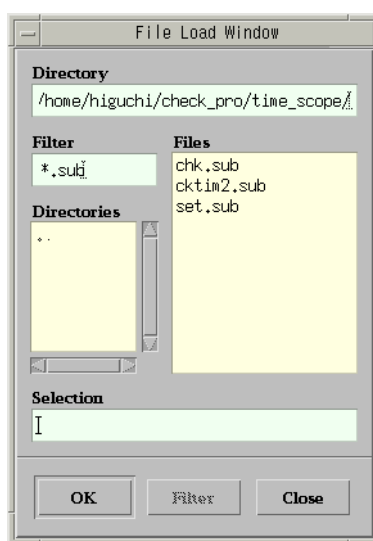
*The data exceeding the measurement range is displayed by \*\*\*.*

*If there is even one \*\*\*, the total test time and the percentage will also be \*\*\*.*

## 6. 5 Loading a Program for Which Measurement is to Be Made (File Load Window)

When you select Load from File from the menu bar in the Main window, [Figure 6-10](#) is displayed. The File Load window is used to load a file for which execution time measurement is to be made.

**Figure 6-10 File Load Window**



Functions of each button are as follows:

### OK

Loads a file for which execution time measurement is to be made.  
The file name is displayed in the title bar.

### Filter

Moves to the directory specified in the Filter text box.

### Close

Exits the File Load window.

### Tip

*The directory for loading a file cannot be changed.*

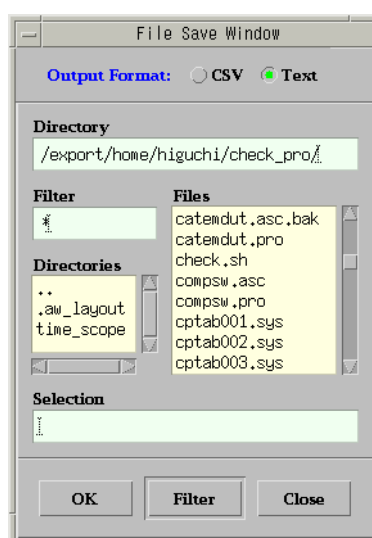
*The program for which measurement is to be made cannot be changed in the All Test mode.*

## 6. 6 Storing the Measurement Result (File Save Window)

When you select Save from the File menu in the All Test window and the Select Test window, [Figure 6-11](#) is displayed.

The File Save window is used to stores the measurement result in the specified file.

**Figure 6-11 File Save Window**



Functions of each button are as follows:

### Output Format

Specifies the file format.

➔ For more information, refer to [6. 6. 1 "Saving in a CSV File" on page 6-12](#) and [6. 6. 2 "Saving in a Text File" on page 6-12](#).

|          |                                      |
|----------|--------------------------------------|
| CSV ...  | Each item is delimited with a comma. |
| Text ... | Each item is delimited with a line.  |

### OK

Stores the measurement result in a file.

### Filter

Moves to the directory specified in the Filter text box.

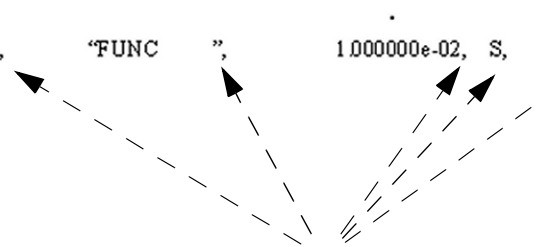
### Close

Exits the File Save window.

### 6. 6. 1 Saving in a CSV File

When you select the CSV format, data is stored in the file with the following format.

```
#Program : demo.pro
#Date : 1998/04/01 12:00
#
#TestNo.      Set Display      Time
#
1000,         "CONTACT",      1.000000e-02, S, 10%, 0%
2000,         "DC",           2.000000e-02, S, 20%, 10%
.
.
10000,        "FUNC",         1.000000e-02, S, 10%, 100%
```



A comma is inserted.

### 6. 6. 2 Saving in a Text File

When you select the Text format, data is stored in the file with the following format.

| Program : demo.pro      |             |                   |
|-------------------------|-------------|-------------------|
| Date : 1998/04/01 12:00 |             |                   |
| Test No.                | Set Display | Time              |
| 1000                    | CONTACT     | 100 mS (10%/ 0%)  |
| 2000                    | DC          | 200 mS (20%/ 10%) |
| .                       |             | .                 |
| 10000                   | FUNC        | 100 mS (10%/100%) |
| Total                   |             | 1.00 S            |

## 6. 7 Printing Out the Measurement Result (Printing Window)

When you select Print from the File menu in the All Test window and the Select Test window, [Figure 6-12](#) is displayed.

The Print window is used to output the measurement result to the specified printer.

The output format is the same as the format in [6. 6. 2 "Saving in a Text File" on page 6-12](#).

**Figure 6-12 Print Window**



Functions of each button are as follows:

**Printer Name**

Specifies the name of the printer to which the measurement result is to be output.

**OK**

Outputs the measurement result to the printer.

**Close**

Quits the Print Window.



## 6. 8 Error Message List

Parameter set error.

An error occurred at the time of parameter setup.

Time data read error.

An error occurred at the time of time data read out.

Execution program not found.

Could not find the program for which execution time measurement is to be made.

During Testing! cannot execution.

Could not make time measurement during the test.

Please input a line number.

Input a line number.

Please select a test number in list.

Select a test number from the list.

Line number error.

There is a mistake in line number specification.

Memory OverFlow.

The available memory area is not sufficient.

Memory allocate error.

A memory allocation error occurred.

Directory Name get error.

Could not get the directory name.

Change directory error.

Could not move the directory.

File open error.

An error occurred at the time of file opening.

File read error.

An error occurred at the time of file reading.

File seek error.

An error occurred at the time of file pointer movement.

Illegal file name.

The attribute of the specified file is inappropriate.

Source text not found.

Could not find the specified source file.

Save data not found.

Could not find the storage data.

Time data not found.

Could not find the time data.

Printer name not found.

Could not find the printer name.

## 7. *DBMtool*

DBMtool is the tool to debug the DBM pattern program.

DBMtool can display and edit the pattern data of the pattern program transferred to the DBM.

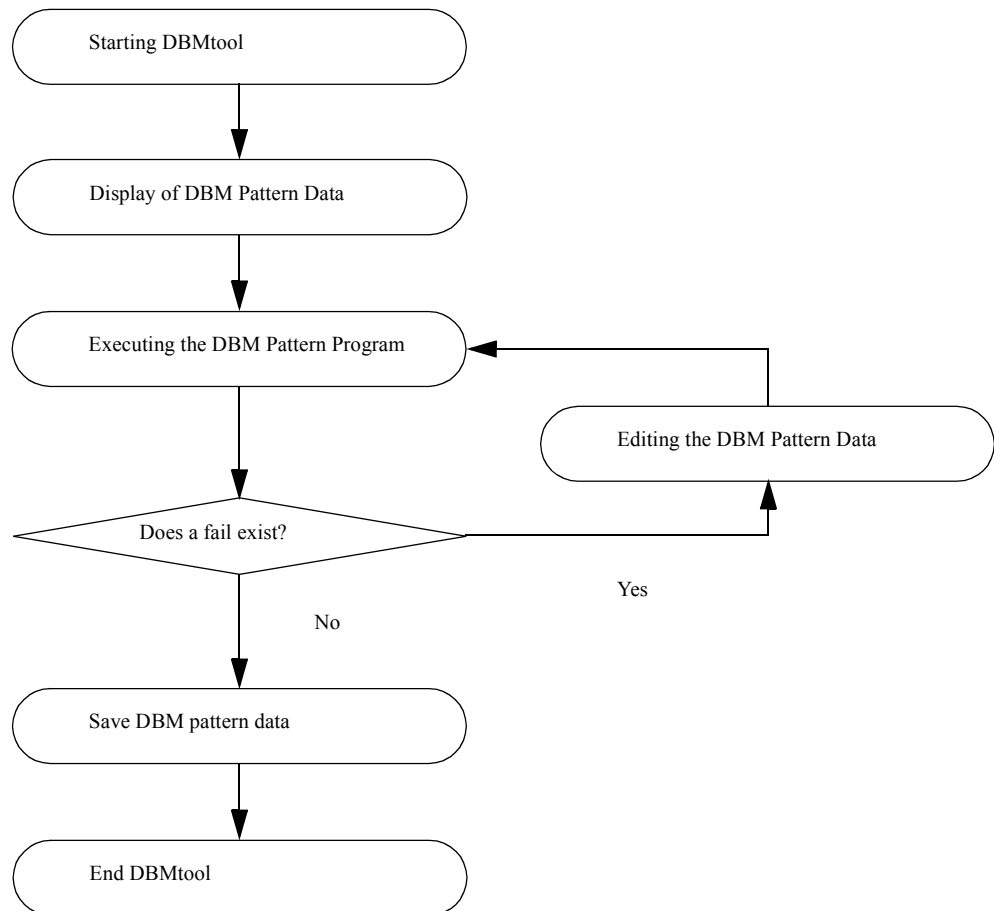
It can also display the fail data by executing the pattern program.

### Restrictions

1. Can be used for the T5593 and T5592 test systems.
2. Can be used for DBM pattern programs (MPAT object files) compiled by the Cross Compiler (PXTL51-00S) REV 6.01 or later.
3. Can be used when ATLworks is in the tester online operation mode.
4. Can be used when other utilities are not operating on the controller window.
5. Must be used with the test mode is set to the Manual or Hold state.
6. Must be used in the Pause test state or while the program is stopped at the breakpoint.
7. If the SEND DBM statement is used, the tool can be used in the DBM pattern program which was transferred most recently.
8. Can be used if the target DBM pattern program (the DBM pattern program that was either transferred most recently or specified in the SELECT DBMFILE statement) file information was not cleared by using the RESET DBM statement.

### 7. 1 Operating Procedure for DBMtool

Use the following procedure to operate DBMtool:

**Figure 7-1 Operating Procedure for DBMtool**

## 7. 2 Starting and Ending DBMtool

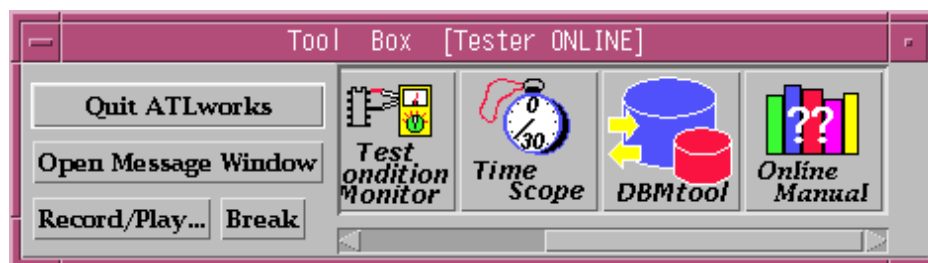
This section describes the starting and ending procedures of DBMtool.

### 7. 2. 1 Starting DBMtool

Click the DBMtool button in the tool box to start DBMtool.

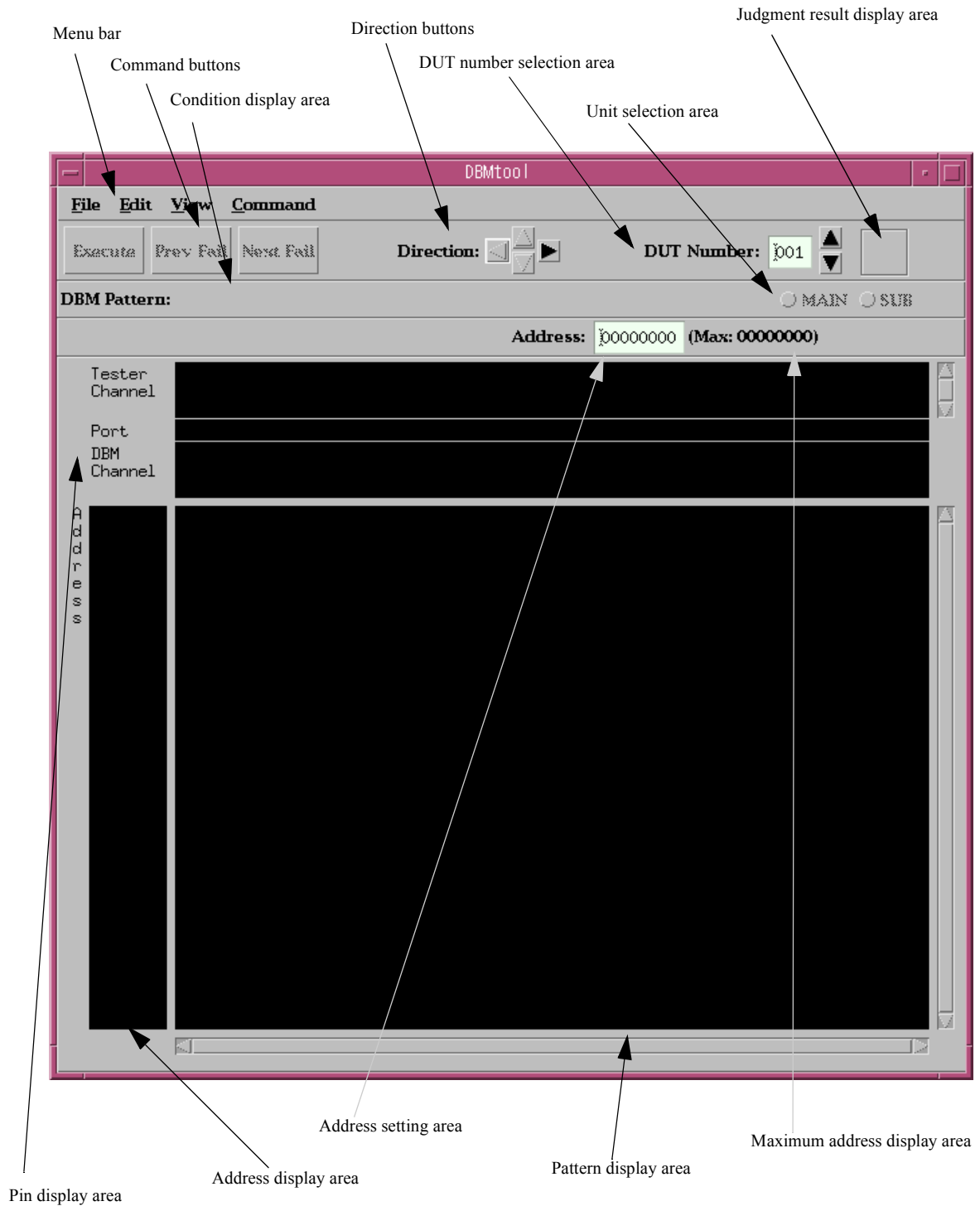
When started, DBMtool displays the main window.

**Figure 7-2 Starting DBMtool**



### 7. 2. 2 Structure of the Main Window

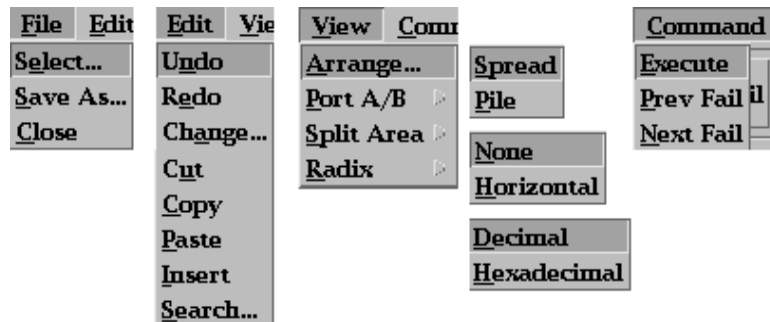
The DBMtool main window contains the following components:

**Figure 7-3 Main Window**

### Menu bar

The menu bar contains a set of menu commands.

**Figure 7-4 Menu Bar**



### Command buttons

The command buttons permit frequently used menu commands to be entered by buttons.

### Direction buttons

These buttons indicate a cursor move direction for key entry and specification of a search direction for pattern retrieval.

### DUT number selection area

Used to select a DUT number.

Fail of the selected DUT number is displayed.

### Judgment result display area

Indicates that a fail occurred.

### Condition display area

Displays the DBM pattern program name and modules (PC values).

If a DBM pattern program is transferred without specifying the PC value, the PC value is not displayed here.

### Unit selection area

Selects the main DBM unit or sub units.

The DBM pattern data of the selected unit is displayed.

### Address setting area

Sets the address of DBM pattern data.

The DBM pattern data of the set address is displayed.

### Maximum address display area

Indicates the maximum address of the DBM pattern data.

### Pin display area

Displays the DBM CHANNEL and the pin name of DBM pattern data.

### Address display area

Indicates the address of the DBM pattern data.

#### Pattern display area

Displays the DBM pattern data.



*If DBM pattern data is not 0, 1, L, H, or X, "-" is displayed.*

---

### 7. 2. 3 Ending DBMtool

Select Close in the File menu to end the DBMtool.

When it ends, the DBMtool window and the dialog box will close.



## 7. 3 Display of DBM Pattern Data

This section describes how to display the DBM pattern data.

### 7. 3. 1 Selecting DBM Pattern Programs

Select DBM pattern programs in the Select dialog box.

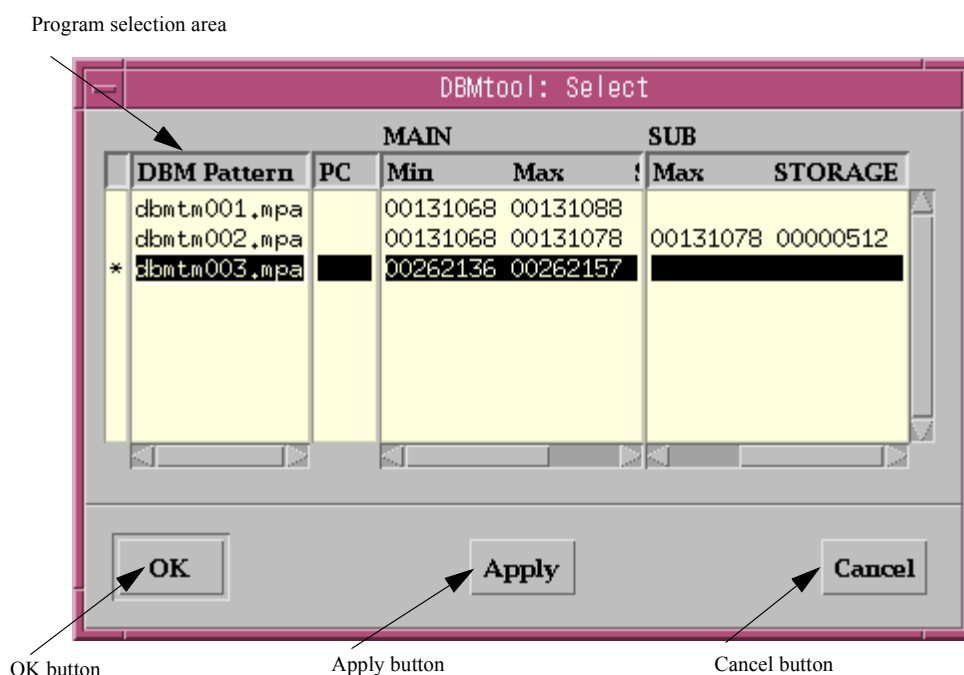
The selected DBM pattern program is displayed.

Selecting Select... from the File menu displays the Select dialog box.



*The Select dialog box can be displayed when the T5593 is used.*

**Figure 7-5 Select Dialog Box**



#### Program selection area

Selects a DBM pattern program to be displayed.

The program selection area displays the following information:

- Transferred DBM pattern program names
- PC values of the transferred modules

If a DBM pattern program is transferred without specifying the PC value, the PC value is not displayed here.

- The Minimum and maximum address data of the main DBM (Min/Max), and the number of addresses which can be inserted by using the Insert edit command (STORAGE).  
If the program is not transferred to the main DBM, Min, Max, and STORAGE are not displayed.
- The Minimum and maximum address data of the sub DBM (Min/Max), and the number of addresses which can be inserted by using the Insert edit command (STORAGE).  
If the program is not transferred to the sub DBM, Min, Max, and STORAGE are not displayed.

**Tip**

*"\*" is displayed at the top of the target DBM pattern program (the DBM pattern program which was either transferred most recently or specified in the SELECT DBMFILE statement).*

**OK button**

Confirms the setting in the dialog box and then closes the dialog box.

**Apply button**

Applies (confirms) the setting in the dialog box.

**Cancel button**

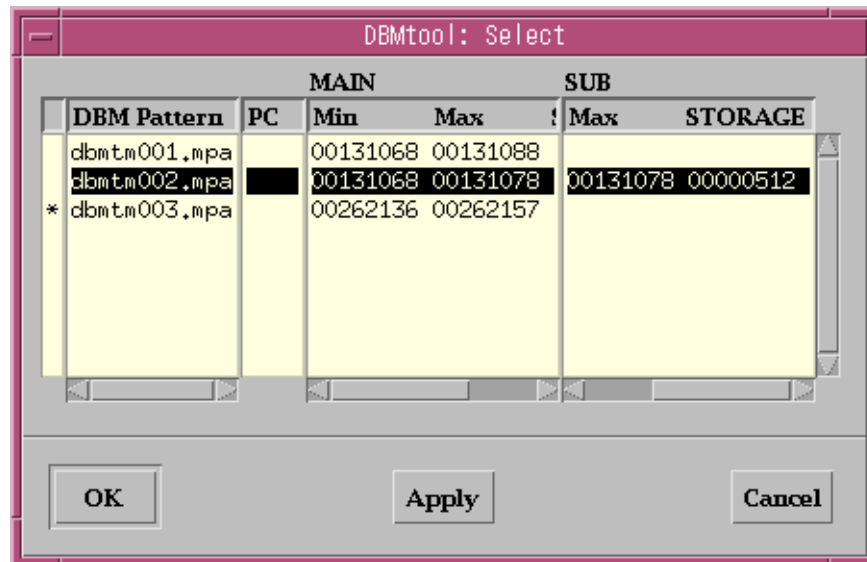
Cancels the setting in the dialog box and then closes the dialog box.

### 7. 3. 1. 1 How to Select DBM Pattern Programs

Use the following procedure to select DBM pattern programs:

- 1) Click the DBM pattern program to be displayed in the program selection area.

Figure 7-6 Selecting DBM Pattern Programs



2) To confirm the setting, click the OK or Apply button.

To cancel the setting, click the Cancel button.

## 7. 3. 2 Selecting DBM Units

Select the main DBM unit or a sub unit in the unit selection area.

The DBM pattern data of the selected unit is displayed.

### 7. 3. 2. 1 Displaying the DBM Pattern Data for the Main Unit

Select MAIN in the unit selection area to display the DBM pattern data for the main unit.

Figure 7-7 Selecting MAIN



### 7. 3. 2. 2 Displaying the DBM Pattern Data for a Sub Unit

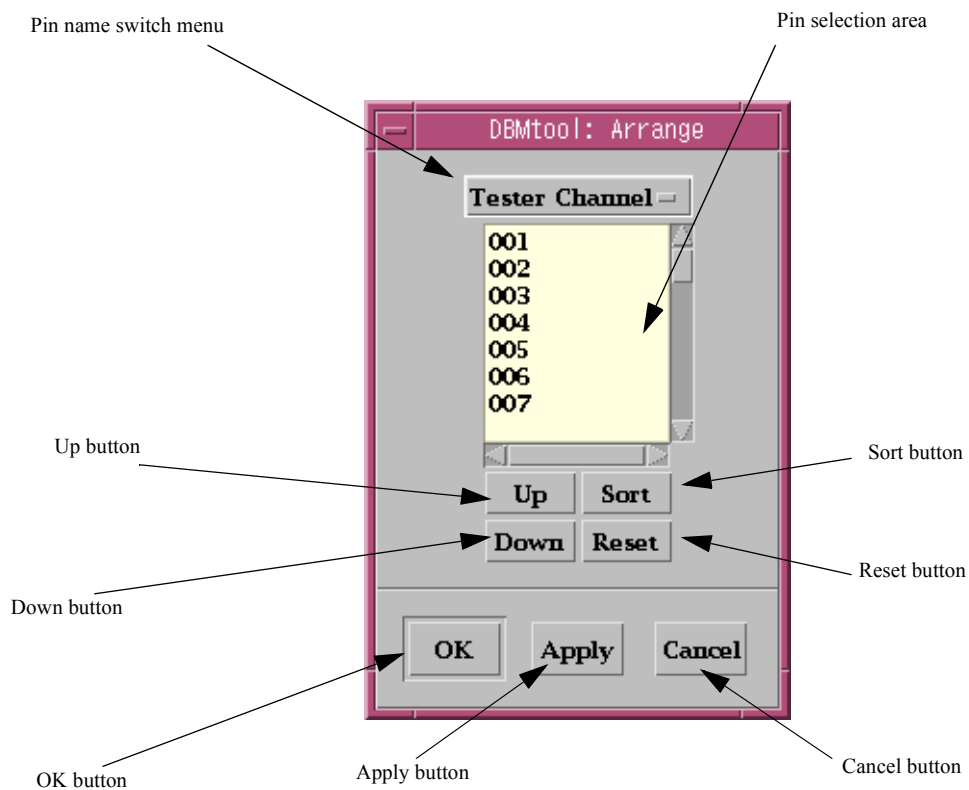
Select SUB in the unit selection area to display the DBM pattern data for a sub unit.

**Figure 7-8 Selecting SUB**

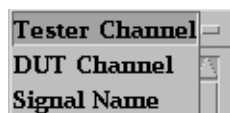
### 7. 3. 3 Changing the Pin Display

Use the Arrange dialog box to change the pin display.

The Arrange dialog box appears when you select Arrange... in the View menu.

**Figure 7-9 Arrange Dialog Box**

Pin name switch menu  
Switches pin names.

**Figure 7-10 Pin Name Switch Menu****Pin selection area**

Selects pins you want to rearrange.

**Up button**

Moves the cursor one pin position upward.

**Down button**

Moves the cursor one pin position downward.

**Sort button**

Rearranges pins by name.

**Reset button**

Rearranges pins in the initial order

**OK button**

Confirms the settings in the dialog box and closes the box.

**Apply button**

Confirms the settings in the dialog box.

**Cancel button**

Cancels the settings in the dialog box and closes the box.

 **Tip**

*The up-to-down pin arrangement in the pin selection area appears to be the left-to-right pin arrangement in the pin display area of the main window.*

**7. 3. 3. 1 Switching a Pin Name to the Tester Channel Number**

Use the following procedure to switch a pin name to the tester channel number:

- 1) Select Tester Channel in the Pin Name Switch menu.
  - 2) Click the OK or Apply button when you complete the procedure.
- Click the Cancel button to cancel.

**Figure 7-11 Displaying the Tester Channel Number**

|                |                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------|
| Tester Channel | 00000000000000000000000000000000<br>000000000111115544333333544444444<br>1234567890123412109876543098765432  |
| Port           | AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA                                                                         |
| DBM Channel    | 00000000000000000000000000000000<br>0000000000111111111122222222223333<br>0123456789012345678901234567890123 |

### 7. 3. 3. 2 Switching a Pin Name to the DUT Pin Number

Use the following procedure to switch a pin name to the DUT pin number:

- 1) Select DUT Channel in the Pin Name Switch menu.
- 2) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

**Figure 7-12 Displaying the DUT Pin Number**

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| DUT Channel | 00000000000000000000000000000000<br>000000000111115544333333544444444<br>1234567890123412109876543098765432  |
| Port        | AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA                                                                         |
| DBM Channel | 00000000000000000000000000000000<br>0000000000111111111122222222223333<br>0123456789012345678901234567890123 |

### 7. 3. 3. 3 Switching a Pin Name to the DUT Pin Name

Use the following procedure to switch a pin name to the DUT pin name:

- 1) Select Signal Name in the Pin Name Switch menu.
- 2) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

**Figure 7-13 Displaying the DUT Pin Name**

|             |                                      |
|-------------|--------------------------------------|
| Signal Name | CCCCCCCCRRRSCSSDDDDDDDDDDDDDDDDDD    |
|             | FFTT00000000CMIQQQQQQQQQQQQQQQQQQ    |
|             | MMMMLLLLWWWK000AAAAAAAABBBBBBBBB     |
|             | N N01234012 01876543210876543210     |
| Port        | AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA |
| DBM         | 00000000000000000000000000000000     |
| Channel     | 000000000011111111122222222223333    |
|             | 0123456789012345678901234567890123   |

**Tip**

*Switching to the DUT pin name is disabled if the DUT pin name is not set in the socket program.*

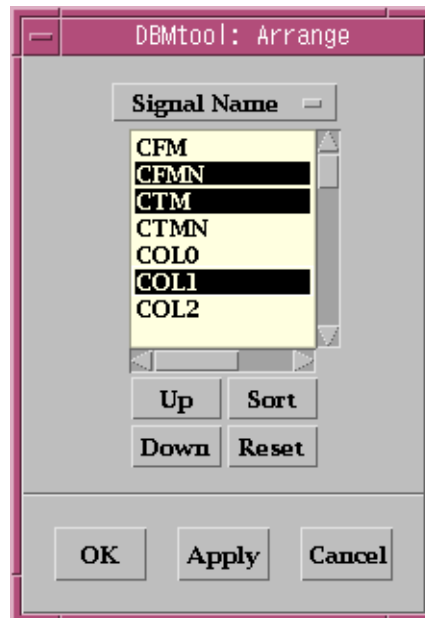
**7. 3. 3. 4 Rearranging Pins Arbitrarily**

Use the following procedure to rearrange pins arbitrarily:

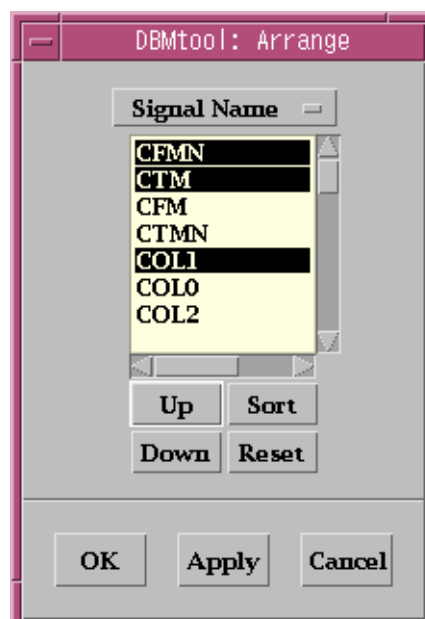
- 1) Click the pin you want to rearrange in the pin selection area.

**Tip**

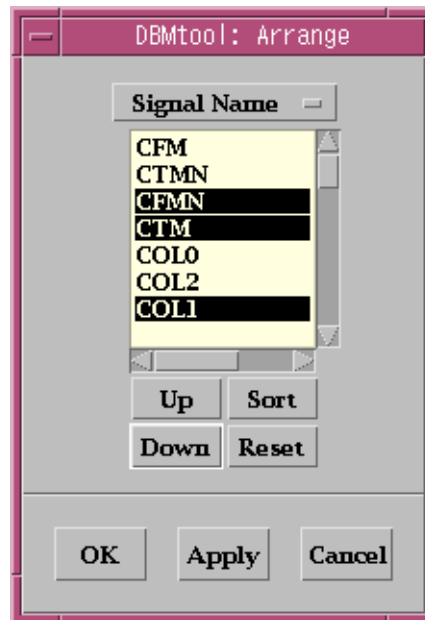
*To select two or more pins at a time, drag them together by the cursor or click them while pressing the Ctrl or Shift key.*

**Figure 7-14 Selecting Pins**

- 2) Specify the target pin position by pressing the Up or Down button.

**Figure 7-15 Rearrangement by the Up Button**



**Figure 7-16 Rearrangement by the Down Button**

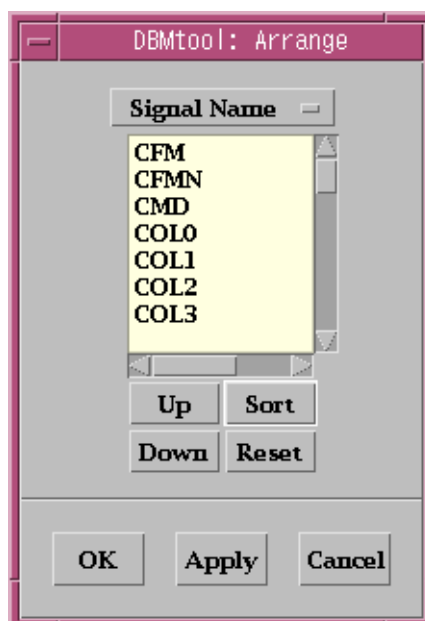
3) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

### 7. 3. 3. 5 Rearranging Pins by Pin Name

Use the following procedure to rearrange pins by pin name:

1) Click the Sort button.

**Figure 7-17 Rearrangement by the Sort Button****Tip**

*Pins are rearranged in the order of pin numbers when you have selected Tester Channel or DUT Channel in the Pin Name Switch menu.*

*Pins are rearranged in the alphabetical order of their name when you have selected Signal Name in the Pin Name Switch menu.*

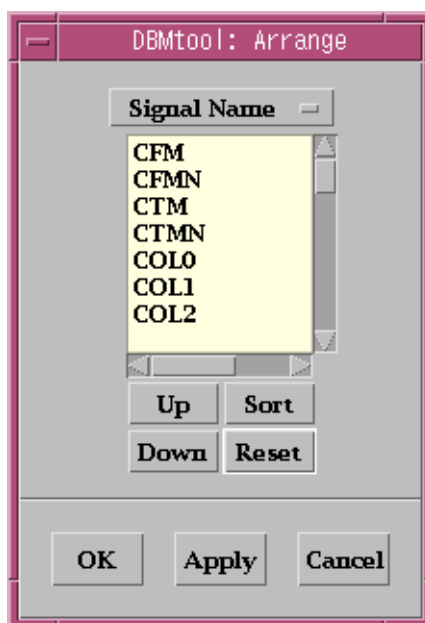
2) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

**7. 3. 3. 6 Rearranging Pins in the Initial Order**

Use the following procedure to rearrange pins in the initial order:

1) Click the Reset button.

**Figure 7-18 Rearrangement by the Reset Button**

Tip

*The initial order means the pin arrangement when DBMtool was started.*

2) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

### 7. 3. 4 Changing the Display of Ports A and B

Use the Port A/B menu in the View menu to change the display of Ports A and B.

Tip

*DBMtool displays Ports A and B even in BITMODE1 or BITMODE3.  
The same DBM CHANNEL is displayed for both Ports A and B.*

#### 7. 3. 4. 1 Displaying Ports A and B Arranged Horizontally

Select Spread in the Port A/B menu in the View menu to display Ports A and B arranged horizontally.



Figure 7-21 Dividing the Pattern Display Area Horizontally into Two Parts

|   |          |                                                                            |
|---|----------|----------------------------------------------------------------------------|
| A | 00000800 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
| d | 00000801 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
| r | 00000802 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
| e | 00000803 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
| s | 00000804 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000805 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000806 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000807 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000808 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000809 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000810 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000811 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000812 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000813 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00000814 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001000 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001001 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001002 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001003 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001004 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001005 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001006 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001007 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001008 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001009 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001010 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001011 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001012 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001013 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |
|   | 00001014 | 010100000000011XLLLLLLLLLLLLLLLLLLLLL101000000000011XLLLLLLLLLLLLLLLLLLLLL |

### 7. 3. 5. 2 Canceling the Division of the Pattern Display Area

Select None in the Split Area menu in the View menu to cancel the division of the pattern display area.

Figure 7-22 Canceling the Division of the Pattern Display Area

|   |          |                  |                      |                  |                      |
|---|----------|------------------|----------------------|------------------|----------------------|
| A | 00000800 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
| d | 00000801 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
| r | 00000802 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
| e | 00000803 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
| s | 00000804 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000805 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000806 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000807 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000808 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000809 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000810 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000811 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000812 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000813 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000814 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000815 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000816 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000817 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000818 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000819 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000820 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000821 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000822 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000823 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000824 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000825 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000826 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000827 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000828 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |
|   | 00000829 | 010100000000011X | LLLLLLLLLLLLLLLLLLLL | 101000000000011X | LLLLLLLLLLLLLLLLLLLL |

### 7. 3. 6 Displaying the DBM Pattern Data of an Optional Address

Enter the address value into the address setting area to display the DBM pattern data of an optional address.

Enter the address value based on the radix (decimal or hexadecimal number).

The address value is confirmed when the Return key is pressed.

#### Tip

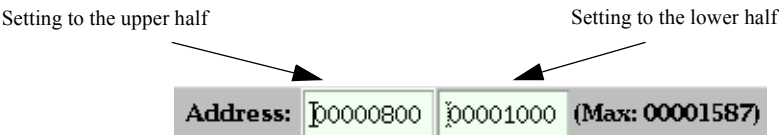
*The address value that can be entered into the address setting area must be within the range of the minimum address to the maximum address (value of the maximum address display area) specified by a DBMA statement.*

Figure 7-23 When the Pattern Display Area is Not Divided

Address: 00000800 (Max: 00001587)

When the pattern display area is divided horizontally into two parts, enter the address value into the left address setting area for the upper half, and similarly, enter the address into the right area of the lower half.

Figure 7-24 When the Pattern Display Area is Divided Horizontally into Two Parts



7. 3. 7 Changing the Radix (Decimal or Hexadecimal Number) for Display and Setting

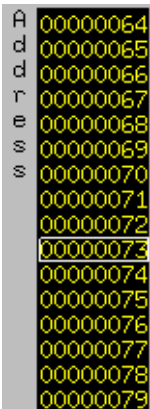
Use the Radix menu in the View menu to change the radix for display and setting.  
The radix for any of the following areas can be changed:

- Address display area of the main window
- Address setting area of the main window
- Largest address display area of the main window
- PC value in the condition display area of the main window
- Address display area in the Fail Information dialog box
- PC value in the Fail Information dialog box

7. 3. 7. 1 Switching to a Decimal Number

Select Decimal in the Radix menu in the View menu to switch to a decimal number.

Figure 7-25 Displaying an Address in Decimal Notation



7. 3. 7. 2 Switching to a Hexadecimal Number

Select Hexadecimal in the Radix menu in the View menu to switch to a hexadecimal number.

Figure 7-26 Displaying an Address in Hexadecimal Notation

|   |          |
|---|----------|
| A | 00000040 |
| d | 00000041 |
| d | 00000042 |
| r | 00000043 |
| e | 00000044 |
| s | 00000045 |
| s | 00000046 |
|   | 00000047 |
|   | 00000048 |
|   | 00000049 |
|   | 0000004A |
|   | 0000004B |
|   | 0000004C |
|   | 0000004D |
|   | 0000004E |
|   | 0000004F |

### 7. 3. 8 Inserting and Deleting Separators

A separator can be inserted or deleted by double-clicking the pin display area.



Tip

*Separators are used for easy viewing of DBM pattern data.*

#### 7. 3. 8. 1 Inserting a Separator

Double-click a pin in the pin display area to insert a separator.

A separator is inserted where you double click.

Figure 7-27 Inserting a Separator

Double click

|          |                  |                                       |
|----------|------------------|---------------------------------------|
| Tester   | 0000000000000000 | 0000000000000000                      |
| Channel  | 0000000000111115 | 544333333354444444                    |
|          | 123456789012341  | 2109876543098765432                   |
| DBM      | 0000000000000000 | 0000000000000000                      |
| Channel  | 000000000011111  | 111112222222223333                    |
| (Port A) | 012345678901234  | 5678901234567890123                   |
| DBM      | 0000000000000000 | 0000000000000000                      |
| Channel  | 3333334444444444 | 455555555556666666                    |
| (Port B) | 456789012345678  | 9012345678901234567                   |
| A        | 00000432         | 010100000000010 XLLLLLLLLLLLLLLLLLLLL |
| d        |                  | 101000000000010 XLLLLLLLLLLLLLLLLLLLL |
| d        | 00000433         | 010100000000010 XLLLLLLLLLLLLLLLLLLLL |
| r        |                  | 101000000000010 XLLLLLLLLLLLLLLLLLLLL |
| e        | 00000434         | 010100000000010 XLLLLLLLLLLLLLLLLLLLL |
| s        |                  | 101000000000010 XLLLLLLLLLLLLLLLLLLLL |
| s        | 00000435         | 010100000000010 XLLLLLLLLLLLLLLLLLLLL |
|          |                  | 101000000000010 XLLLLLLLLLLLLLLLLLLLL |





## 7. 4 Executing the DBM Pattern Program

This section describes how to execute the DBM pattern program.

If a fail occurs during execution of the DBM pattern program, the fail data is displayed.

 **Tip**

*Up to 64 cycles of fail data stored in the DFM are displayed.*

### 7. 4. 1 Selecting a DUT for Which Fail Data is Obtained

Select the DUT number in the DUT number selection area to select a DUT for which you want to obtain the fail data.

The DUT number can be selected by any of the following methods:

**Figure 7-29 DUT Number Selection Area**

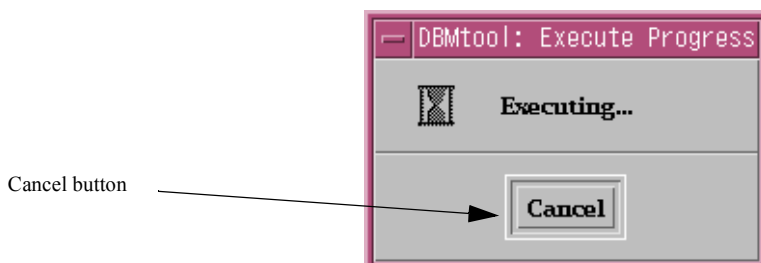


- Keying-in the DUT number  
Key-in the DUT number into the DUT number input area in decimal notation.  
Press the Return key to confirm the DUT number.
- Clicking the DUT number input buttons  
Click the upward arrow button (or downward arrow button) to increment (or decrement) the DUT number in the DUT number input area.

### 7. 4. 2 Executing the DBM Pattern Program

Click the Execute button or select Execute in the Command menu to execute the DBM pattern program.

The Execution Progress dialog box is displayed during execution of the pattern program.

**Figure 7-30 Execution Progress Dialog Box**

Cancel button

Cancels the execution and closes the dialog box.

### 7. 4. 3 Checking to See If a Fail Occurred

Check the following display area to see if a fail occurred:

- Judgment result display area  
Indicates that a fail occurred.

**Figure 7-31 Judgment Result Display Area**

#### Tip

The judgment result display area indicates that a fail occurred even if the fail occurred in a DBM unit other than the selected DBM unit.

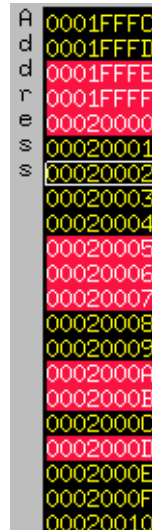
- Pin display area  
Indicates the pin(s) at which the fail occurred in red.

**Figure 7-32 Pin Display Area**

|         |                                  |
|---------|----------------------------------|
| Tester  | 00000000000000000000000000000000 |
| Channel | 44444444443333333344444444333333 |
|         | 87654321098765438765432109876543 |
| Port    | AAAAAAAAAAAAAAAABBBBBBBBBBBBBBBB |
| DBM     | 00000000000000000000000000000000 |
| Channel | 111111000000000033222222221111   |
|         | 54321098765432101098765432109876 |

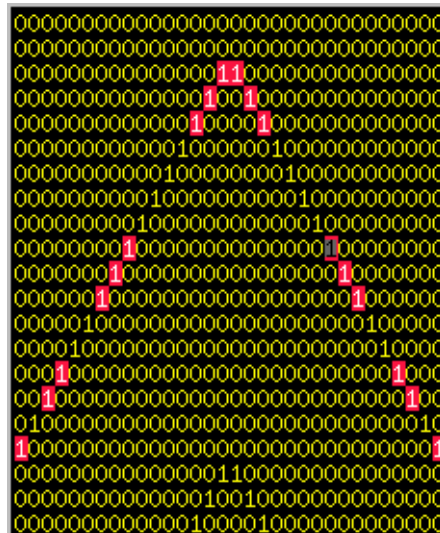
- Address display area  
Indicates the address(es) where the fail occurred in red.

Figure 7-33 Address Display Area



- Pattern display area  
Indicates the pattern data in which the fail occurred in red.

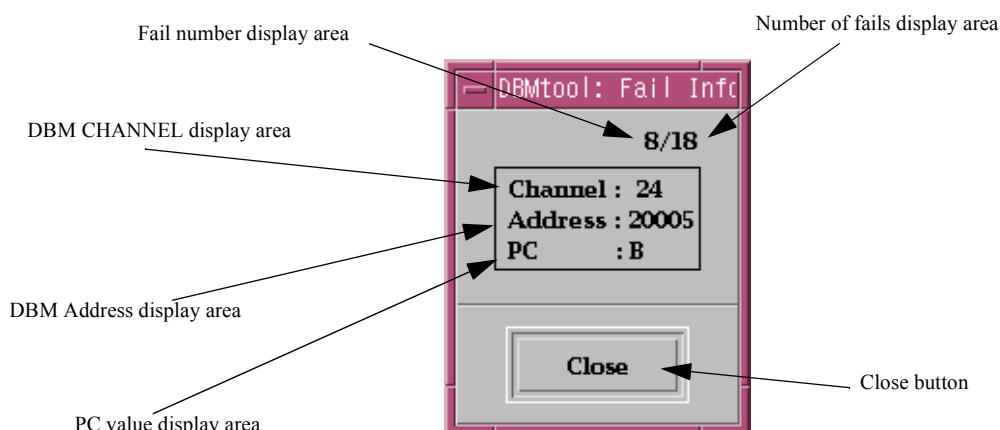
Figure 7-34 Pattern Display Area



#### 7. 4. 4 Checking the Fail Data

Click the pattern data in which a fail occurred to check the fail data.

The Fail Information dialog box is displayed when the pattern data is clicked.

**Figure 7-35 Fail Information Dialog Box****Number of fails display area**

Indicates the total number of fails.

Tip

*The total number of fails refers to the number of fails that occurred in the selected DBM unit.*

**Fail number display area**

Indicates the sequence in which the fails occurred.

**DBM CHANNEL display area**

Indicates the DBM channel in which the fail occurred.

**DBM Address display area**

Indicates the DBM address where the fail occurred.

**PC value display area**

Indicates the PC value of ALPG in which the fail occurred.

**Close button**

Closes the dialog box.

**7. 4. 5 Searching for a Fail**

Click the Prev Fail button or select Prev Fail in the Command menu to search for the fail that occurred just before in the sequence.

Click the Next Fail button or select Next Fail in the Command menu to search for the fail that occurred immediately after in the sequence.

When the fail is found, the Fail Information dialog box is displayed.

 Tip

*When a fail is found by searching, the pattern data is specified in the scope of editing.  
The pattern data can be edited using an Edit command (Change..., Copy, or Paste).*

---

## 7. 5 Editing the DBM Pattern Data

This section describes how to edit the DBM pattern data.

### 7. 5. 1 Editing with Mouse Operation

Use the following procedure for editing with mouse operation:

- 1) Specifying the scope of editing
- 2) Selecting an edit command

#### 7. 5. 1. 1 Specifying the Scope of Editing

Specify the scope of editing in any of the following areas by operating the mouse.

The pattern data specified in the scope of editing is highlighted.

- Pattern display area
- Pin display area
- Address display area

The mouse can be operated in the following ways:

- Dragging with the left button of the mouse
- Clicking the left button of the mouse and then clicking the middle button of the mouse



Tip

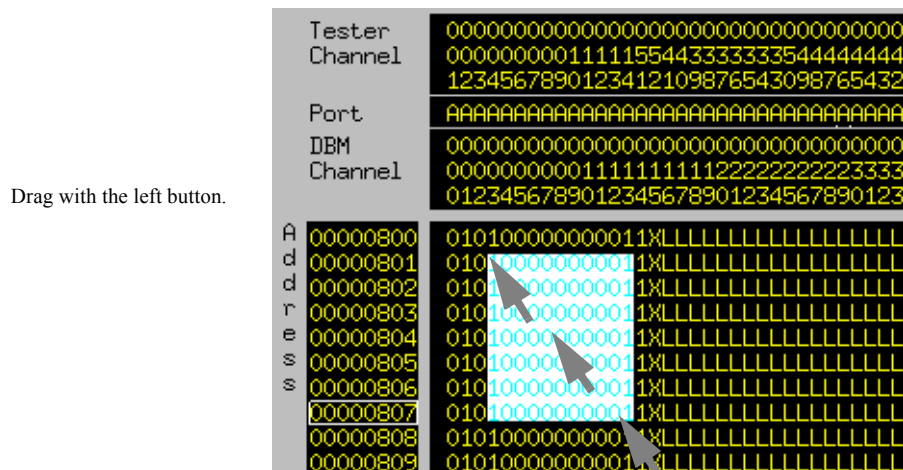
*Click the right button of the mouse to release the specified scope of editing.*

#### 7. 5. 1. 1. 1 Scope of Editing in the Pattern Display Area

The following figures show how to specify an editing area in the pattern display area.

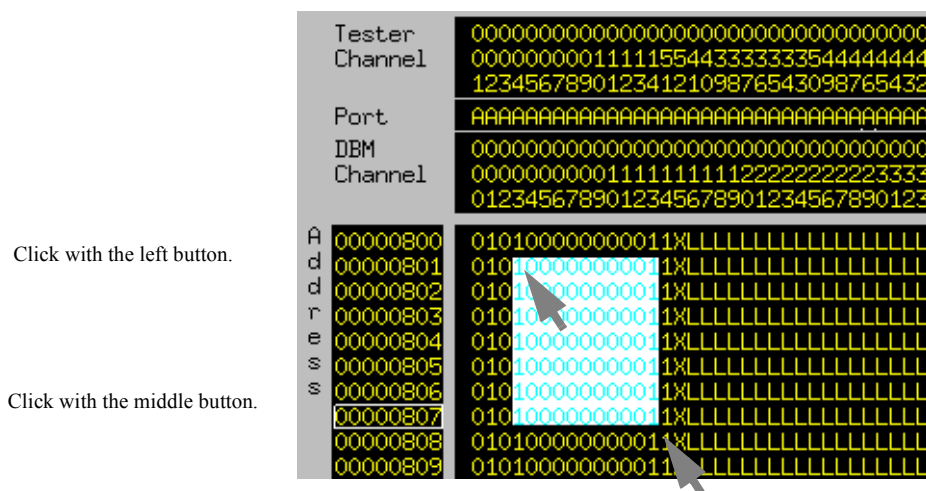
- Dragging with the left button of the mouse

Figure 7-36 Scope of Editing in the Pattern Display Area (1)



- Clicking the left button of the mouse and then clicking the middle button of the mouse

Figure 7-37 Scope of Editing in the Pattern Display Area (2)



### 7. 5. 1. 1. 2 Scope of Editing in the Pin Display Area

The following figures show how to specify an editing area in the pin display area.

In the pin display area, the DBM pattern data of all addresses of the specified pin is the scope of editing.

- Dragging with the left button of the mouse



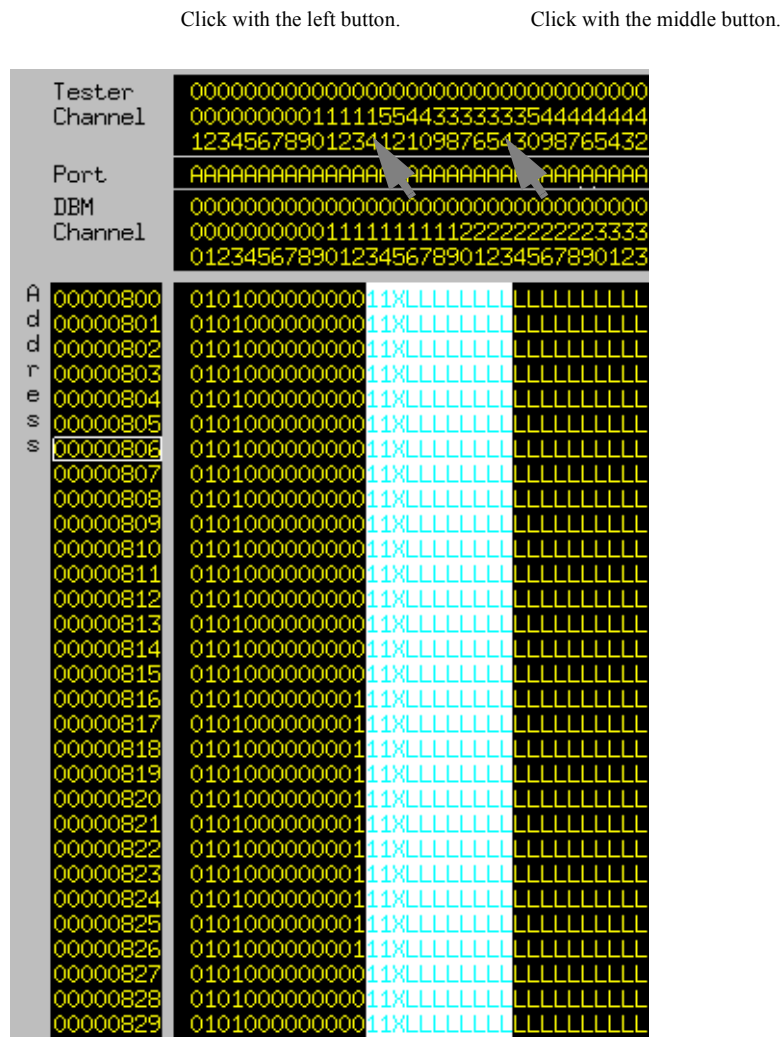
Figure 7-38 Scope of Editing in the Pin Display Area (1)

Drag with the left button.

|                |                                               |
|----------------|-----------------------------------------------|
| Tester Channel | 00000000000000000000000000000000              |
| Port           | 0000000001111155443333333544444444            |
| DBM Channel    | 1234567890123412109876543098765432            |
| Channel        | AAAAAAAAAAAAA AAAA AAAA AAAAAAA               |
|                | 00000000000000000000000000000000              |
|                | 0000000000111111111122222222223333            |
|                | 0123456789012345678901234567890123            |
| Addresses      | 00000800 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000801 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000802 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000803 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000804 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000805 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000806 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000807 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000808 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000809 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000810 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000811 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000812 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000813 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000814 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000815 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000816 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000817 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000818 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000819 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000820 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000821 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000822 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000823 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000824 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000825 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000826 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000827 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000828 010100000000011XLLLLLLLLLLLLLLLLLLLL |
|                | 00000829 010100000000011XLLLLLLLLLLLLLLLLLLLL |

- Clicking the left button of the mouse and then clicking the middle button of the mouse

Figure 7-39 Scope of Editing in the Pin Display Area (2)



### 7. 5. 1. 1. 3 Scope of Editing in the Address Display Area

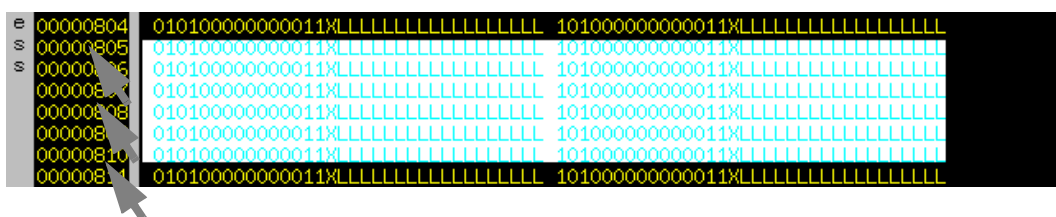
The following figures show how to specify an editing area in the address display area.

In the address display area, the DBM pattern data of all pins of the specified address is the scope of editing.

- Dragging with the left button of the mouse

**Figure 7-40 Scope of Editing in the Address Display Area (1)**

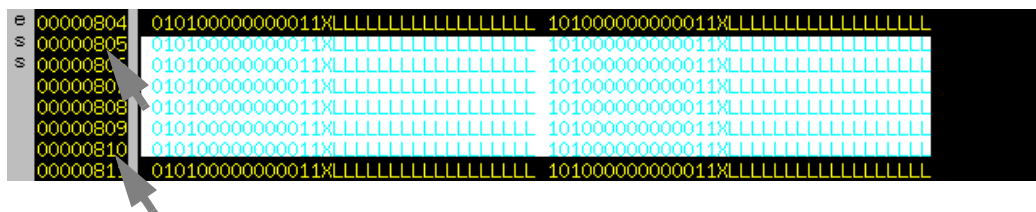
Drag with the left button.



- Clicking the left button of the mouse and then clicking the middle button of the mouse

**Figure 7-41 Scope of Editing in the Address Display Area (2)**

Click with the left button.



Click with the middle button.



#### Limitation

- If Ports A and B are displayed one on the other vertically, you cannot specify the DBM pattern data of only one port as the scope of editing.
- If  $2N$  or  $2N+1$  (where  $N$  is an integer equal to or greater than 0) is specified as the address value when the interleave mode is 2WAY, you cannot specify the DBM pattern data of only one address as the scope of editing.
- If  $4N$ ,  $4N+1$ ,  $4N+2$ , or  $4N+3$  (where  $N$  is an integer equal to or greater than 0) is specified as the address value when the interleave mode is 4WAY, you cannot specify the DBM pattern data of only one address as the scope of editing.

### 7. 5. 1. 2 Selecting an Edit Command

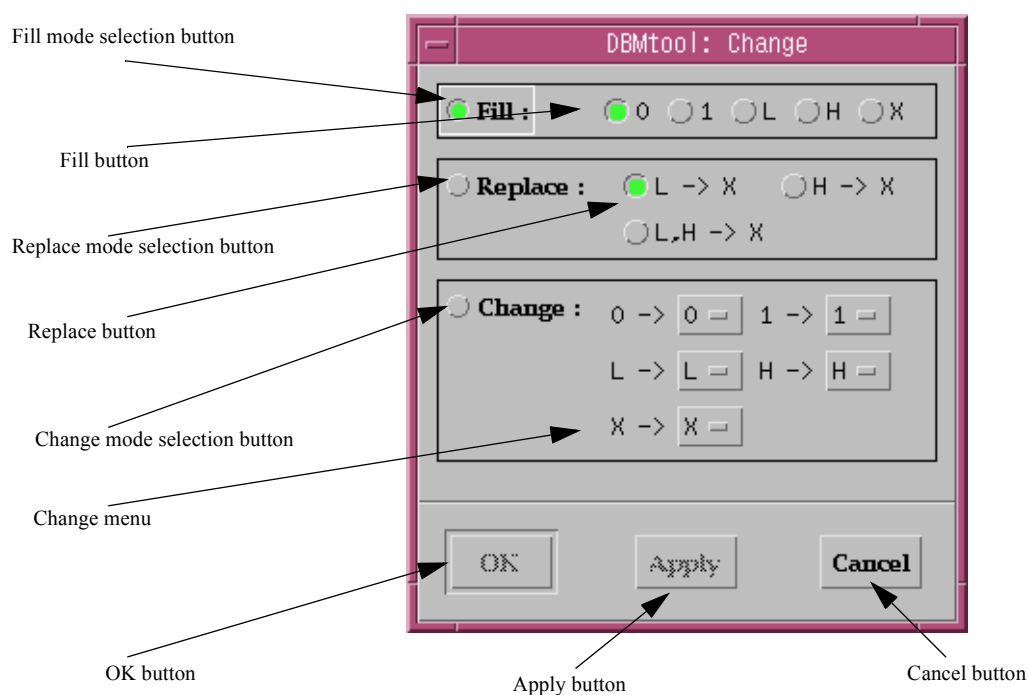
Use the Edit menu for selecting edit commands.

The Edit menu includes the following commands:

- Change...

This command is used to change the pattern data in the scope of editing.

The Change dialog box is displayed.

**Figure 7-42 Change Dialog Box****Fill mode selection button**

Selects the mode that changes the pattern data in the scope of editing into one pattern data element.

**Fill button**

Changes the pattern data into any of 0, 1, L, H, and X.

- 0  
Sets all pattern data to 0.
- 1  
Sets all pattern data to 1.
- L  
Sets all pattern data to L.
- H  
Sets all pattern data to H.
- X  
Sets all pattern data to X.

**Replace mode selection button**

Selects the mode that changes L or H for the scope of editing into X.

**Replace button**

Changes L or H into X.

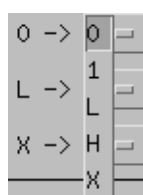
- L -> X  
Sets L to X.
- H -> X  
Sets H to X.
- L,H -> X  
Sets L or H to X.

**Change mode selection button**

Selects the mode that changes the pattern data in the scope of editing arbitrarily.

**Change menu**

Changes 0, 1, L, H, and X into the pattern data selected by the Change menu.

**Figure 7-43 Change Menu****OK button**

Confirms the settings in the dialog box and closes the box.

**Apply button**

Confirms the settings in the dialog box.

**Cancel button**

Cancels the settings in the dialog box and closes the box.

- Cut  
This command is used to store the pattern data in the scope of editing into the copy buffer.  
The pattern data in the scope of editing is deleted.
- Copy  
This command is used to store the pattern data in the scope of editing into the copy buffer.  
The pattern data stored into the copy buffer is displayed enclosed by a frame.
- Paste  
This command is used to write over the pattern data in the scope of editing with that stored in the copy buffer.
- Insert  
This command is used to insert the pattern data stored in the copy buffer into the header address of the pattern data in the scope of editing.

---

**Tip**

The number of addresses that can be inserted using Insert is the sum of the number of addresses specified in the DBM STORAGE statement and the number of addresses deleted using Cut.

→ For more information on the DBM STORAGE statement, refer to [ATL-51 Pattern Program Reference Manual\(No. : 8256292\)](#).

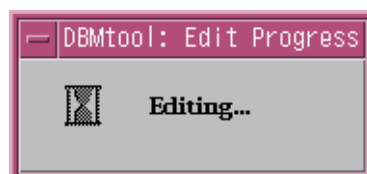
---

**Limitation**

- The pattern data stored into the copy buffer in each area (pattern display, pin display, and address display areas) can be pasted (Paste) when a scope of editing is specified in the area.
  - Cut and Insert can be selected when a scope of editing is specified in the address display area.
- 

The Edit Progress dialog box is displayed while editing is in progress.

**Figure 7-44 Edit Progress Dialog Box**

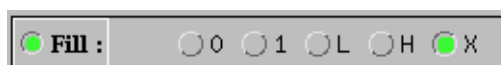


### 7. 5. 1. 3 Editing by Change...

Use the following procedure to perform editing by Change...:

- 1) Specify the scope of editing.
- 2) Select Change... in the Edit menu.
- 3) Select the change mode and change content.
  - For editing in Fill mode
    - Click the Fill mode selection button.
    - Select the change content from the Fill menu.

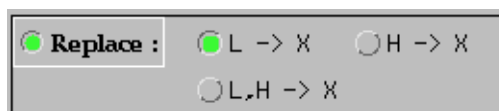
**Figure 7-45 Setting the Pattern Data in the Scope of Editing to X**



- For editing in Replace mode
  - Click the Replace mode selection button.

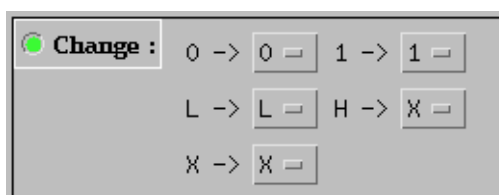
- Select the change content from the Replace menu.

**Figure 7-46 Setting L in the Scope of Editing to X**



- For editing in Change mode
  - Click the Change mode selection button.
  - Select the change content from the Change menu.

**Figure 7-47 Setting H in the Scope of Editing to X**



4) Click the OK or Apply button when you complete the procedure.

Click the Cancel button to cancel.

#### 7. 5. 1. 4 Editing by Cut

Use the following procedure to perform editing by Cut:

- 1) Specify the scope of editing.
- 2) Select Cut in the Edit menu.

#### 7. 5. 1. 5 Editing by Cut and Paste

Use the following procedure to perform editing by Cut and Paste:

- 1) Specify the scope of editing.
- 2) Select Cut in the Edit menu.
- 3) Specify the scope of editing.
- 4) Select Paste in the Edit menu.

### 7. 5. 1. 6 Editing by Cut and Insert

Use the following procedure to perform editing by Cut and Insert:

- 1) Specify the scope of editing.
- 2) Select Cut in the Edit menu.
- 3) Specify the scope of editing.
- 4) Select Insert in the Edit menu.

### 7. 5. 1. 7 Editing by Copy and Paste

Use the following procedure to perform editing by Copy and Paste:

- 1) Specify the scope of editing.
- 2) Select Copy in the Edit menu.
- 3) Specify the scope of editing.
- 4) Select Paste in the Edit menu.

Or use the following procedure:

- 1) Specify the scope of editing.
- 2) Drag the scope of editing using the middle button of the mouse.

### 7. 5. 1. 8 Editing by Copy and Insert

Use the following procedure to perform editing by Copy and Insert:

- 1) Specify the scope of editing.
- 2) Select Copy in the Edit menu.
- 3) Specify the scope of editing.
- 4) Select Insert in the Edit menu.

## 7. 5. 2 Editing by Key Entry

Use the following procedure to perform editing by key entry:

- 1) Use the Direction buttons to indicate the cursor move direction for key entry.
- 2) Double-click the left button of the mouse in the pattern display area to specify the position for key entry.

A cursor is displayed for key entry.

- 3) Key-in any of 0, 1, L, H, or X.



Press the space key to move the cursor.

Tip

*L, H, and X can be entered in lower case.*

4) Press the Return key to terminate editing.

Tip

*Editing by key entry can be terminated by any of the following operations:*

- *Move the mouse pointer outside the pattern display area.*
- *Click a mouse button.*

### 7. 5. 3 Canceling Edit Operations

Select Undo in the Edit menu to cancel edit operations you have made.

Up to five edit operations can be cancelled.

Limitation

If any of the following operations is performed, the edit operations before that cannot be cancelled.

- Editing by Cut
- Editing by Insert
- Changing a DBM unit

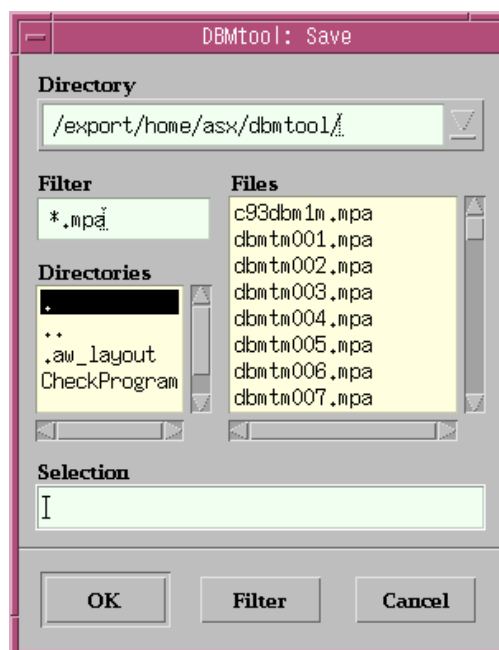
### 7. 5. 4 Reexecuting the Canceled Edit Operations

Select Redo in the Edit menu to reexecute the canceled edit operations.

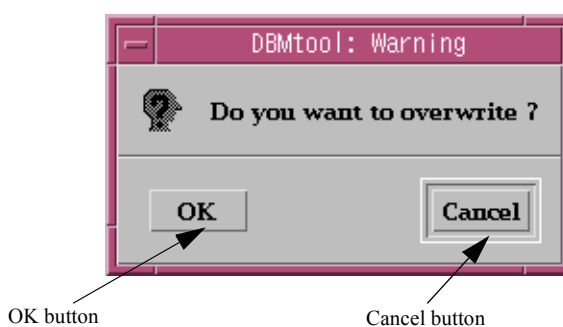
### 7. 5. 5 Saving the Edited DBM Pattern Program

Select Save As... in the File menu to save the edited DBM pattern program.

When Save As... is selected, the File Selection dialog box is displayed.

**Figure 7-48 File Selection Dialog Box**

If an attempt was made to save the program to an existing file, the Warning dialog box is displayed.

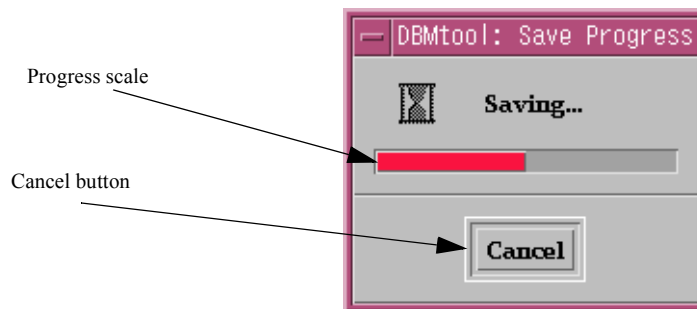
**Figure 7-49 Warning Dialog Box****OK button**

Saves the program to the existing file.

**Cancel button**

Cancels the attempt and closes the dialog box.

The Save Progress dialog box is displayed while saving is in progress.

**Figure 7-50 Save Progress Dialog Box****Progress scale**

Indicates how the saving is progressing.

**Cancel button**

Cancels the attempt and closes the dialog box.

**Tip**

- Suppose one pattern program whose source file and object file are in the same directory is being debugged by DBMtool and ASXDebugger. Then the object file saved by DBMtool will be overwritten, if the source file is compiled afterward by ASXDebugger.
- Use `dbm2asc` to convert the saved DBM pattern program (object file) to the ASCII format.  
➔ For more information, refer to [8. "dbm2asc" on page 8-1](#).

## 7. 6 Searching for DBM Pattern Data

This section describes how to search for the specified DBM pattern data.

Use the following procedure to search for the specified DBM pattern data:

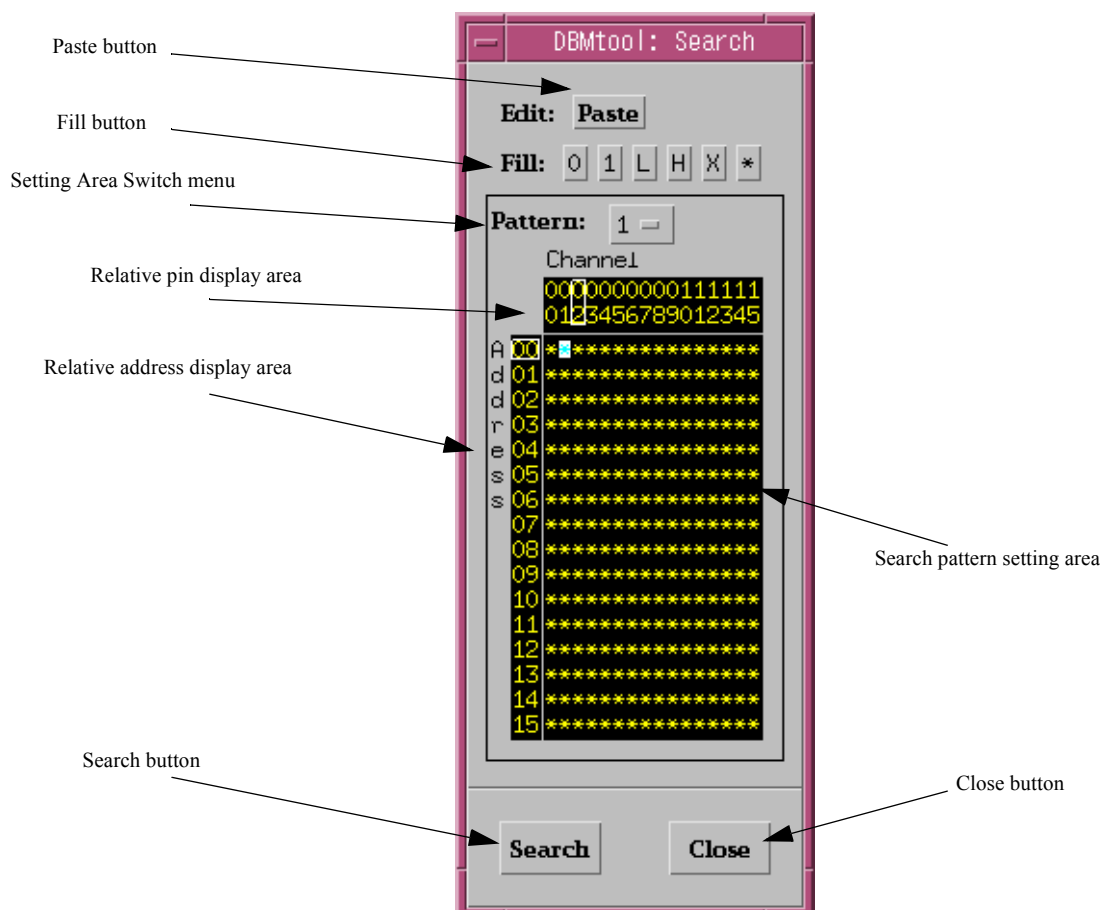
- 1) Display the Search dialog box.
- 2) Set the search target.
- 3) Indicate the search direction
- 4) Specify the search start position.
- 5) Execute the search.

### Tip

*When the desired pattern data is found by searching, the pattern data is specified in the scope of editing.  
The pattern data can be edited using an Edit command (Change..., Copy, or Paste).*

### 7. 6. 1 Displaying the Search Dialog Box

The Search dialog box appears when you select Search... in the Edit menu.

**Figure 7-51 Search Dialog Box****Tip**

An asterisk (\*) displayed in the Search dialog box stands for the pattern data that matches all pattern data, or a wild card.

**Paste button**

This command is used to rewrite the pattern data in the scope of setting with that stored in the copy buffer.

**Fill button**

Changes the pattern data in the scope of setting into any of 0, 1, L, H, X, or \*.

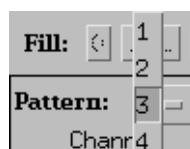
- 0  
Sets all pattern data to 0.
- 1  
Sets all pattern data to 1.

- L  
Sets all pattern data to L.
- H  
Sets all pattern data to H.
- X  
Sets all pattern data to X.
- \*  
Sets all pattern data to \*.

#### Setting Area Switch menu

Switches the search pattern setting area.  
Four search pattern setting areas can be selected.

**Figure 7-52 Setting Area Switch Menu**



#### Relative pin display area

Displays the position of the pattern data in the pin direction.

#### Relative address display area

Displays the position of the pattern data in the address direction.

#### Search pattern setting area

Sets the pattern data of the search target.  
Four types of pattern data can be set by the Setting Area Switch menu.

#### Search button

Executes the search.

#### Close button

Closes the dialog box.

## 7. 6. 2 Setting the Search Target by Mouse Operation

Use the following procedure for setting the search target by mouse operation:

- 1) Specify the scope of setting.
- 2) Select a Set command.

### 7. 6. 2. 1 Specifying the Scope of Setting

Specify the scope of setting by operating the mouse in the search pattern setting area.

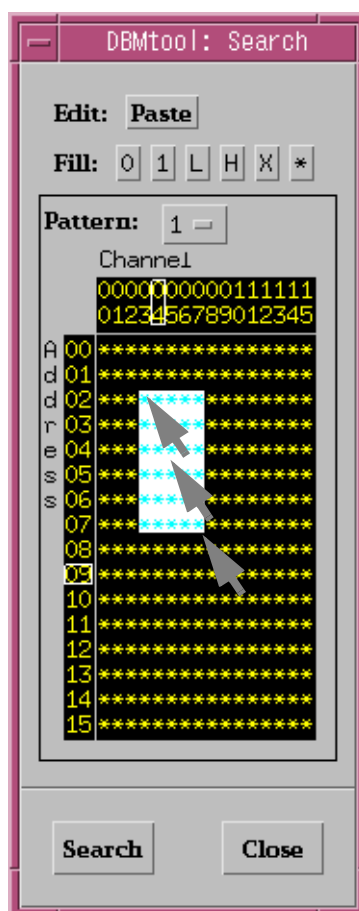
The pattern data specified in the scope of setting is highlighted.

The mouse can be operated in the following ways:

- Dragging with the left button of the mouse

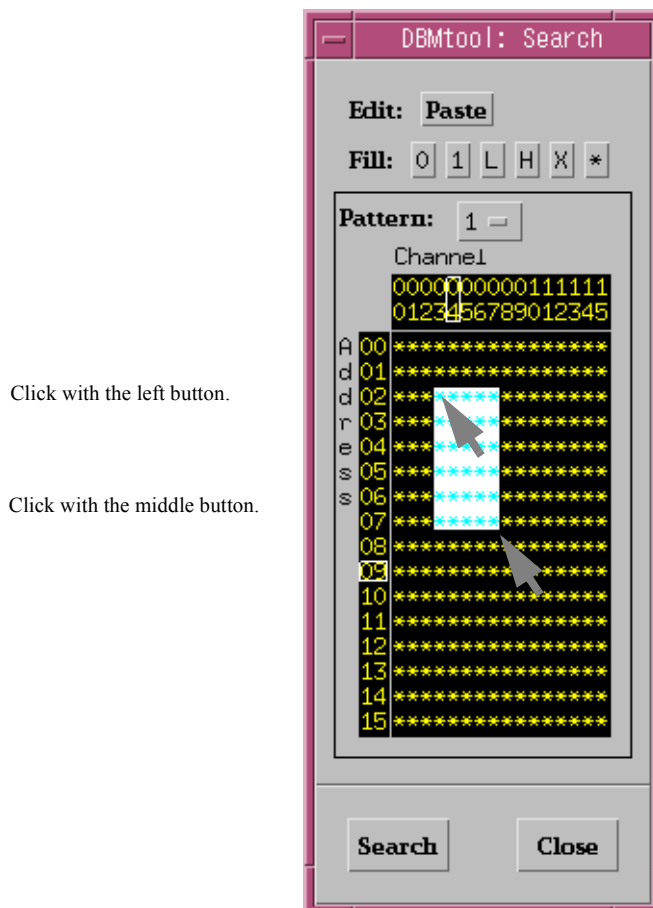
**Figure 7-53 Specifying the Scope of Setting (1)**

Drag with the left button.



- Clicking the left button of the mouse and then clicking the middle button of the mouse

Figure 7-54 Specifying the Scope of Setting (2)



### 7. 6. 2. 2 Selecting a Set Command

Select a Set command using the following buttons:

- Fill button
- Paste button

### 7. 6. 2. 3 Setting the Search Target by Fill

Use the following procedure to set the search target by Fill:

- 1) Specify the scope of setting.
- 2) Select the Fill button.

### 7. 6. 2. 4 Setting the Search Target by Copy and Paste

Use the following procedure to set the search target by Copy and Paste:



- 1) Specify the scope of editing in the pattern display area of the main window.
- 2) Select Copy in the Edit menu.
- 3) Specify the scope of setting.
- 4) Select the Paste button.

### 7. 6. 3 Setting the Search Target by Key Entry

Use the following procedure to set the search target by key entry:

- 1) Use the Direction buttons to indicate the cursor move direction for key entry.
- 2) Double-click the left button of the mouse in the search pattern setting area to specify the position for key entry.

A cursor is displayed for key entry.

- 3) Key-in any of 0, 1, L, H, X, or \*.

Press the space key to move the cursor.

#### Tip

*L, H, and X can be entered in lower case.*

- 4) Press the Return key to terminate setting.

#### Tip

*Setting by key entry can be terminated by any of the following operations:*

- *Move the mouse pointer outside the search pattern setting area.*
- *Click a mouse button.*

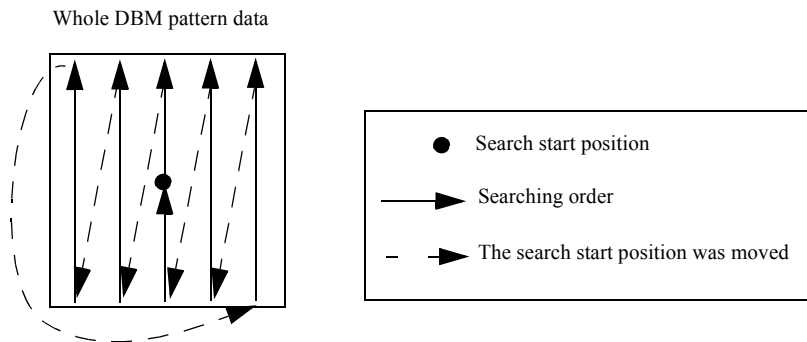
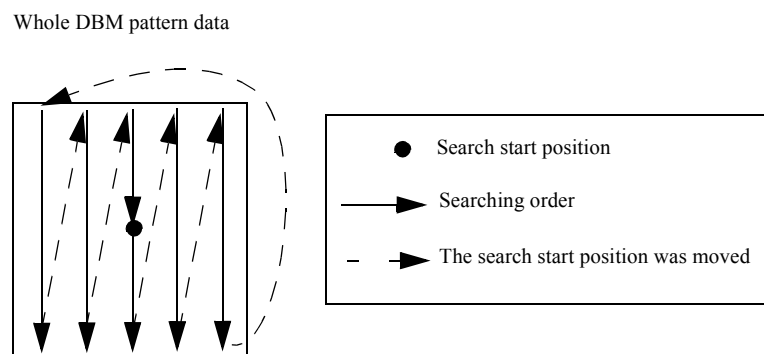
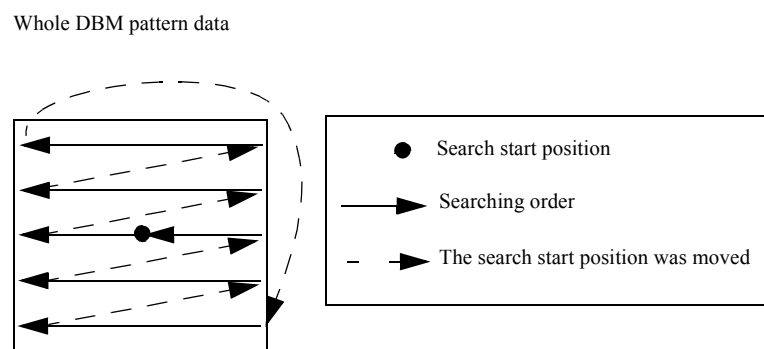
### 7. 6. 4 Specifying the Search Start Position

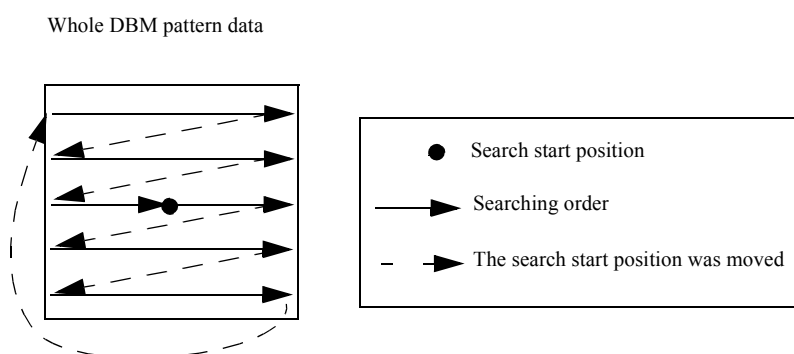
The search start position can be specified by the same procedure as for the scope of editing in the main window.

The search start position is at the upper left of the highlighted pattern data.

### 7. 6. 5 Indicating the Search Direction

Use the Direction buttons to indicate the search direction.

**Figure 7-55 When the Upward Arrow Button Is Selected by Direction Buttons****Figure 7-56 When the Downward Arrow Button Is Selected by Direction Buttons****Figure 7-57 When the Leftward Arrow Button Is Selected by Direction Buttons**

**Figure 7-58 When the Rightward Arrow Button Is Selected by Direction Buttons**

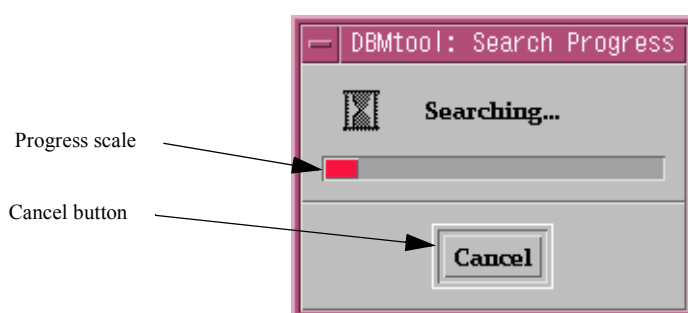
## 7. 6. 6 Executing the Search

Use the Search button to execute the search.

### Tip

*The pattern data displayed in the search pattern setting area is targeted for search.*

The Search Progress dialog box is displayed while searching is in progress.

**Figure 7-59 Search Progress Dialog Box**

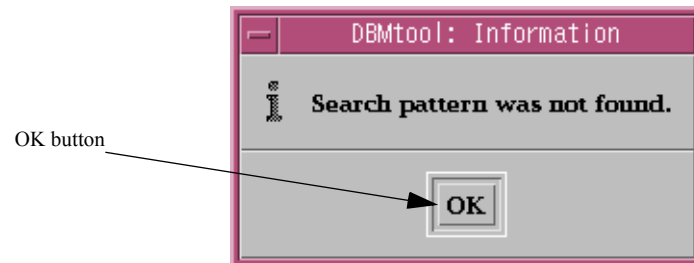
### Progress scale

Indicates how the saving is progressing.

### Cancel button

Cancels the search and closes the dialog box.

If the set pattern data was not found, the Information dialog box is displayed.

**Figure 7-60 Information Dialog Box**

OK button  
Closes the dialog box.

## 7. 7 Error Message List

This section shows DBMtool error messages.

Could not allocate shared memory.

Cut size is equal to the pattern size.

DBM pattern was not found.

File operation error occurred.

Invalid file is specified.

Insert size exceeds DBM STORAGE size.

MPAT object was changed.

Memory allocation error occurred.

Runtime error occurred(*code number*).

The tester type is not supported.

## 8. *dbm2asc*

dbm2asc is a command that is used to convert the DBM pattern data in an MPAT object file to the ASCII format and output the results to a standard output (terminal).

### Restrictions

1. Can be used for the T5593 and T5592 DBM pattern data (CHTYPE13, CHTYPE12, and CHTYPE10).
2. Can be used for MPAT object files compiled by the cross compiler (PXTL51-00S) REV 6.01 or later.

### 8. 1 How to Use dbm2asc

Enter the following command from the command line to use dbm2asc:

```
% dbm2asc [-dbm MAIN|SUB] [-pc PC value] filename.↓
```

#### -dbm MAIN|SUB

Specify the DBM pattern data to be converted.

When MAIN is specified, the pattern data of the main DBM is converted.

When SUB is specified, the pattern data of the sub DBM is converted.

If not specified, the pattern data of the main DBM is converted.

#### -pc PC value

Specify the DBM pattern data to be converted.

The DBM pattern data of the module containing the specified PC value is converted.

The PC value can be specified in either decimal or hexadecimal notation.

For the decimal notation, specify a value within the range of 0 to 2047.

For the hexadecimal notation, specify a value within the range of #0 to #7FF.

If not specified, the DBM pattern data of the common portion is converted.

#### filename

Specify an MPAT object file.

The file name must be specified together with the extension (.mpa).

### 8. 1. 1 Example of Operation

The following example converts the pattern data of the main DBM of the module containing the PC value #100 in the MPAT object file sample.mpa.

```
% dbm2asc -dbm MAIN -pc #100 sample.mpa↵
```

## 8. 2 Ourput Items

The conversion results are output to the standard output (terminal).

Use the output redirection (>) to output the conversion results to a file.

### Tip

*The file to which the conversion results were output cannot be compiled as is.  
Add the required items to compile the file.*

➔ For more information, refer to [ATL-51 Pattern Program Reference Manual\(No. : 8256292\)](#).

The output items are as follows:

#### 1. DBM PATTERN statement

When -dbm MAIN is specified, a MAIN DBM PATTERN statement is output.

When -dbm SUB is specified, a SUB DBM PATTERN statement is output.

If the specification is omitted, a DBM PATTERN statement is output.

#### 2. DBM CHANNEL statement

#### 3. DBM PINSEL statement and DBM PINSEL data

#### 4. DBM STORAGE statement

#### 5. DBMA statement

#### 6. DBM pattern data

### Tip

*When the apply pattern (PPAT), DRE pattern (PDRE), and CPE pattern (PCPE) are set to 0, G is output.*

An output example is shown below:

```
DBM PATTERN
DBM CHANNEL 15-0,31-16
DBM PINSEL(CHTYPE12,BITMODE2)
D0 = P33A
D1 = P34A
...
D31 = P48B
DBM STORAGE #10
DBMA #0
0000 0000 XXXX XXXX@
0000 1111 LLLL HHHH
0000 0101 LHXX LLXX@
1111 0000 LHLH HHHH
...
0000 1010 XXXX XXLL@
0010 0000 XXXX XXXX
```



## 8. 3 Error Message List

This section shows dbm2asc error messages.

Could not open the file.

DBM pattern was not found.

Illegal file format.

Memory allocation error occurred.

Not supported CHTYPE.

## Appendix 1. Keyboard Layout for the Xterm Window

The keyboard layout for the Xterm window is as follows:

Type 4 Keyboard

|                  |                  |                   |                   |                      |       |        |                     |                      |                     |                            |
|------------------|------------------|-------------------|-------------------|----------------------|-------|--------|---------------------|----------------------|---------------------|----------------------------|
| F8<br>↑<br>ESC[A | F9<br>↓<br>ESC[B | F10<br>←<br>ESC[D | F11<br>→<br>ESC[C | F12<br>(~)<br>ESC[Om | <br>_ | Delete | Pause               | Print<br>Screen      | Scroll<br>Lock      | Nm<br>Lock                 |
|                  |                  |                   |                   |                      |       |        | =<br>(PF1)<br>ESCOP | /<br>(PF2)<br>ESC OQ | *<br>(PF3)<br>ESCOR | .<br>(PF4)<br>ESC OS       |
|                  |                  |                   |                   |                      |       |        | 7<br>(7)<br>esc Ow  | 8<br>(8)<br>esc Ox   | 9<br>(9)<br>esc Oy  | +<br>(=)<br>esc Ol         |
|                  |                  |                   |                   |                      |       |        | 4<br>(4)<br>esc Ot  | 5<br>(5)<br>esc Ou   | 6<br>(6)<br>esc Ov  |                            |
|                  |                  |                   |                   |                      |       |        | 1<br>(1)<br>esc Oq  | 2<br>(2)<br>esc Or   | 3<br>(3)<br>esc Os  | Enter<br>(Enter)<br>esc OM |
|                  |                  |                   |                   |                      |       |        | 0<br>(0)<br>esc Op  | .(.)<br>esc On       |                     |                            |



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